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GRADUATE PROGRAM

Advanced Macroeconomics I

Problem Notes and Selected Solutions for Romer 5ed

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Preface

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Chapter 1

Problem 1.1 *Basic properties of growth rates.*

Use the fact that the growth rate of a variable equals the time derivative of its log to show:

(a) The growth rate of the product of two variables equals the sum of their growth rates. That is, if $Z(t) = X(t)\,Y(t)$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} + \frac{\dot{Y}(t)}{Y(t)} \tag{1}$$

(b) The growth rate of the ratio of two variables equals the difference of their growth rates. That is, if $Z(t) = X(t)/Y(t)$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} - \frac{\dot{Y}(t)}{Y(t)} \tag{2}$$

(c) If $Z(t) = a\,X(t)^b$, then

$$\frac{\dot{Z}(t)}{Z(t)} = b \cdot \frac{\dot{X}(t)}{X(t)} \tag{3}$$

Solution

1.1.a For any positive differentiable variable $X(t)$, define its instantaneous growth rate by

$$g_X(t) \equiv \frac{\dot{X}(t)}{X(t)} = \frac{\mathrm{d}\ln X(t)}{\mathrm{d}t}. \tag{4}$$

(a) If $Z(t) = X(t)\,Y(t)$, then

$$\ln Z(t) = \ln X(t) + \ln Y(t),$$
$$\frac{\mathrm{d}\ln Z(t)}{\mathrm{d}t} = \frac{\mathrm{d}\ln X(t)}{\mathrm{d}t} + \frac{\mathrm{d}\ln Y(t)}{\mathrm{d}t}$$

Therefore,

$$\boxed{\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} + \frac{\dot{Y}(t)}{Y(t)}}$$

(b) If $Z(t) = X(t)/Y(t)$, then

$$\begin{aligned} \ln Z(t) &= \ln X(t) - \ln Y(t), \\ \frac{\mathrm{d}\ln Z(t)}{\mathrm{d}t} &= \frac{\mathrm{d}\ln X(t)}{\mathrm{d}t} - \frac{\mathrm{d}\ln Y(t)}{\mathrm{d}t} \end{aligned}$$

Hence,

$$\boxed{\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} - \frac{\dot{Y}(t)}{Y(t)}}$$

(c) Assume $a > 0$ and let $Z(t) = aX(t)^{\alpha}$, where a and α are constants. Taking logs and then differentiating gives

$$\begin{aligned} \ln Z(t) &= \ln a + \alpha \ln X(t), \\ \frac{\mathrm{d}\ln Z(t)}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[\ln a + \alpha \ln X(t)\right] \\ &= 0 + \alpha \frac{\mathrm{d}\ln X(t)}{\mathrm{d}t} \end{aligned}$$

Using the growth-rate identity in (4) for both $Z(t)$ and $X(t)$, we obtain

$$\boxed{\frac{\dot{Z}(t)}{Z(t)} = \alpha \frac{\dot{X}(t)}{X(t)}}$$

Thus a time-invariant multiplicative constant changes the level of a variable, but not its growth rate.

Note **Why logarithms work.** Because instantaneous growth in continuous time is naturally expressed in exponential form, $\lim_{m\to\infty}(1 + g/m)^{mt} = e^{gt}$, the rules above reflect a useful duality: logarithms turn products into sums, while exponentials turn sums into products.

Note **Four growth-rate rules.** For products, add the growth rates. For ratios, subtract them. For powers, multiply the growth rate by the exponent. Multiplying a variable by a constant does not change its growth rate.

Note **Sums and differences.** Sums and differences also have growth-rate formulas, but there is no analogous unweighted rule. For example, the growth rate of $X + Y$ is a level-weighted average of the growth rates of X and Y.

Problem 1.2

Suppose that the growth rate of some variable, X, is constant and equal to $a > 0$ from time 0 to time t_1; drops to 0 at time t_1; rises gradually from 0 to a from time t_1 to time t_2; and is constant and equal to a after time t_2.

(a) Sketch a graph of the growth rate of X as a function of time.

(b) Sketch a graph of $\ln X$ as a function of time.

Solution

Let

$$\gamma_X(t) \equiv \frac{\dot{X}(t)}{X(t)} = \frac{\mathrm{d} \ln X(t)}{\mathrm{d}t}. \tag{1}$$

(a) The question does not specify the exact shape of the gradual increase between t_1 and t_2. For a concrete sketch, suppose it is linear. Then

$$\gamma_X(t) = \begin{cases} a, & 0 \le t < t_1, \\ \dfrac{a(t-t_1)}{t_2-t_1}, & t_1 \le t \le t_2, \\ a, & t > t_2. \end{cases} \tag{2}$$

Thus the growth rate is initially horizontal at a, jumps down to zero at t_1, rises to a, and is horizontal again after t_2.

(b) Since the slope of $\ln X(t)$ equals $\gamma_X(t)$,

$$\ln X(t) - \ln X(0) = \int_0^t \gamma_X(\tau)\,\mathrm{d}\tau. \tag{3}$$

Substituting the linear path in (2) into (3) gives

$$\ln X(t) = \begin{cases} \ln X(0) + at, & 0 \le t \le t_1, \\ \ln X(0) + at_1 + \dfrac{a(t-t_1)^2}{2(t_2-t_1)}, & t_1 \le t \le t_2, \\ \ln X(0) + at_1 + \dfrac{a(t_2-t_1)}{2} + a(t-t_2), & t \ge t_2. \end{cases} \tag{4}$$

Hence $\ln X(t)$ is linear with slope a before t_1. It remains continuous at t_1, but its slope drops to zero. Between t_1 and t_2 it is increasing and convex, and after t_2 it is again linear with slope a.

The two paths derived above are collected in the figure below. The left panel shows the assumed growth-rate path, and the right panel shows its implication for the level of $\ln X(t)$.

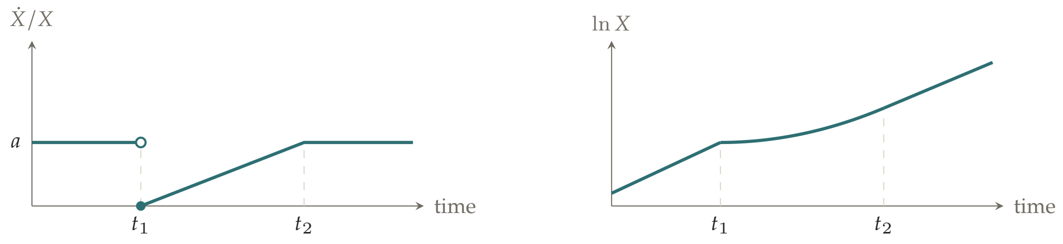

Figure 1.1. The growth rate of X and the corresponding path of $\ln X$. The straight portions of the right-hand panel have slope a.

Note **Reading a log-level graph.** A jump in a growth rate changes the slope of the log-level path, not its level. Thus $\ln X(t)$ has a kink at t_1, not a jump or an inflection point.

Problem 1.3

Describe how, if at all, each of the following developments affects the break-even and actual investment lines in our basic diagram for the Solow model:

(a) The rate of depreciation falls.

(b) The rate of technological progress rises.

(c) The production function is Cobb–Douglas, $f(k) = k^\alpha$, and capital's share, α, rises.

(d) Workers exert more effort, so that output per unit of effective labor for a given value of capital per unit of effective labor is higher than before.

Solution

The law of motion for capital per unit of effective labor is

$$\dot{k} = sf(k) - (n + g + \delta)k. \tag{1}$$

The balanced-growth-path value k^* is determined by

$$sf(k^*) = (n + g + \delta)k^*. \tag{2}$$

(a) By (1), a fall in δ reduces $n+g+\delta$, so the break-even investment line becomes flatter. The actual investment curve $sf(k)$ is unchanged. At the old k^*, actual investment now exceeds break-even investment, and therefore

$$\boxed{k^*_{\text{new}} > k^*_{\text{old}}}$$

(b) A rise in g increases $n + g + \delta$, so the break-even investment line becomes steeper. The actual investment curve is unchanged. At the old k^*, break-even investment now exceeds actual investment, and therefore

$$\boxed{k^*_{\text{new}} < k^*_{\text{old}}}$$

(c) With $f(k) = k^{\alpha}$, actual investment is sk^{α}, while the break-even line is unaffected. For a fixed k,

$$\frac{\partial(sk^{\alpha})}{\partial\alpha} = sk^{\alpha}\ln k. \tag{3}$$

Thus an increase in α lowers actual investment when $0 < k < 1$, leaves it unchanged at $k = 1$, and raises it when $k > 1$. The actual investment curve therefore rotates around $k = 1$; it does not shift uniformly upward or downward.

Solving the steady-state condition (2) for the Cobb–Douglas case gives

$$k^* = \left(\frac{s}{n+g+\delta}\right)^{1/(1-\alpha)}. \tag{4}$$

Taking logs and differentiating with respect to α gives

$$\frac{\partial \ln k^*}{\partial\alpha} = \frac{\ln\big(s/(n+g+\delta)\big)}{(1-\alpha)^2}. \tag{5}$$

Consequently,

$$\boxed{\begin{aligned} s > n+g+\delta &\implies k^* \text{ rises,} \\ s < n+g+\delta &\implies k^* \text{ falls,} \\ s = n+g+\delta &\implies k^* = 1 \text{ and is unchanged.} \end{aligned}}$$

Without additional information, the effect on k^* is ambiguous.

(d) Let the higher effort level raise output at every relevant k, so that the new production function satisfies $\widetilde{f}(k) > f(k)$. Actual investment shifts upward from $sf(k)$ to $s\widetilde{f}(k)$, while the break-even line is unchanged. Hence

$$\boxed{k^*_{\text{new}} > k^*_{\text{old}}}$$

The four comparative-static cases are collected in the figure below. In each panel, the vertical guides mark the old and new balanced-growth-path values of capital per unit of effective labor.

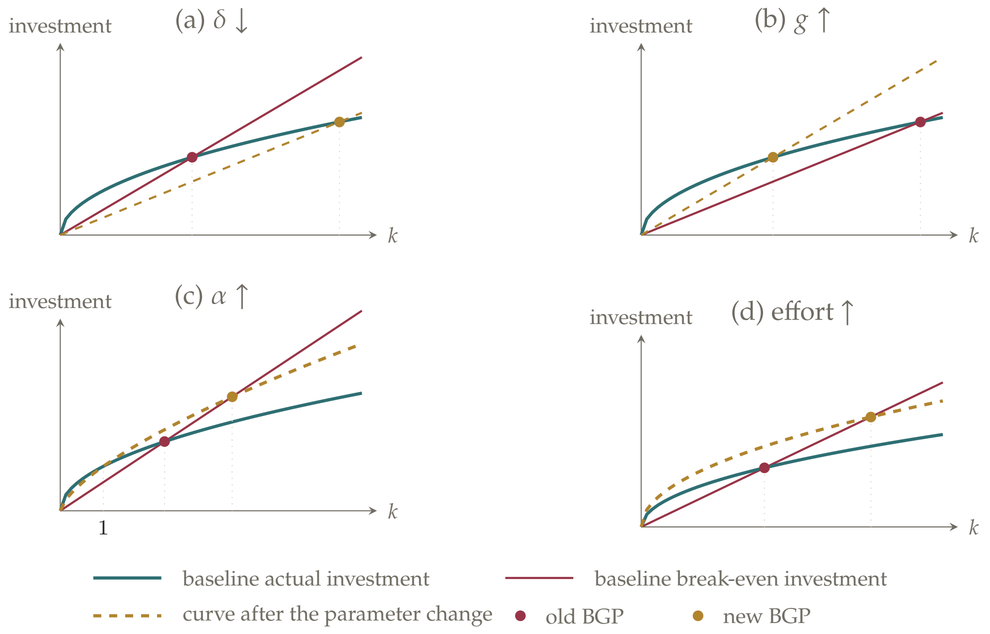

Figure 1.2. Comparative statics in the Solow diagram. Vertical guides locate the old and new balanced-growth-path capital stocks.

Note **Diagram strategy.** First identify which curve changes. Parameters n, g, and δ rotate the break-even line; productivity or effort changes the actual-investment curve.

Note **Changing an exponent.** In part (c), a higher α rotates rather than uniformly shifts sk^{α}: it lowers the curve when $0 < k < 1$, leaves it unchanged at $k = 1$, and raises it when $k > 1$.

Problem 1.4

Consider an economy with technological progress but without population growth that is on its balanced growth path. Now suppose there is a one-time jump in the number of workers.

(a) At the time of the jump, does output per unit of effective labor rise, fall, or stay the same? Why?

(b) After the initial change (if any) in output per unit of effective labor when the new workers appear, is there any further change in output per unit of effective labor? If so, does it rise or fall? Why?

(c) Once the economy has again reached a balanced growth path, is output per unit of effective labor higher, lower, or the same as it was before the new workers appeared? Why?

Solution

Let the one-time increase in labor occur at t_0. Immediately before the shock, the economy is on a balanced growth path, and

$$k(t_0^-) = \frac{K(t_0^-)}{A(t_0^-)L(t_0^-)} = k^*. \tag{1}$$

Suppose $L(t_0^+) = mL(t_0^-)$, where $m > 1$. Capital K and technology A cannot jump at that instant, so $K(t_0^+) = K(t_0^-)$ and $A(t_0^+) = A(t_0^-)$.

(a) Immediately after the increase in workers,

$$\begin{aligned} k(t_0^+) &= \frac{K(t_0^+)}{A(t_0^+)L(t_0^+)} \\ &= \frac{K(t_0^-)}{A(t_0^-)mL(t_0^-)} \\ &= \frac{k(t_0^-)}{m} = \frac{k^*}{m} < k^*. \end{aligned} \tag{2}$$

Since $y = f(k)$ and $f'(k) > 0$,

$$y(t_0^+) = f\left(k(t_0^+)\right) < f\left(k(t_0^-)\right) = y(t_0^-) = y^*. \tag{3}$$

Thus output per unit of effective labor falls immediately: the existing capital stock is spread across more effective labor.

(b) Because there is no population growth, the post-shock law of motion remains

$$\dot{k} = sf(k) - (g + \delta)k. \tag{4}$$

The one-time change in the level of L does not alter either curve in the Solow diagram. Since (2) shows that $k(t_0^+) < k^*$,

$$sf\left(k(t_0^+)\right) > (g + \delta)k(t_0^+),$$

so $\dot{k} > 0$. Capital per unit of effective labor rises toward k^*, and $y = f(k)$ rises with it.

(c) The balanced-growth-path condition is still

$$sf(k^*) = (g + \delta)k^*. \tag{5}$$

Since none of the parameters in this equation has changed, the long-run values of

k^* and $y^* = f(k^*)$ are unchanged. Therefore,

$$\boxed{y^*_{\text{new}} = y^*_{\text{old}}}$$

The transition after the one-time labor increase is summarized in the figure below. A single long arrow represents the instantaneous jump, whereas successive arrowheads represent the economy's gradual movement under the unchanged law of motion.

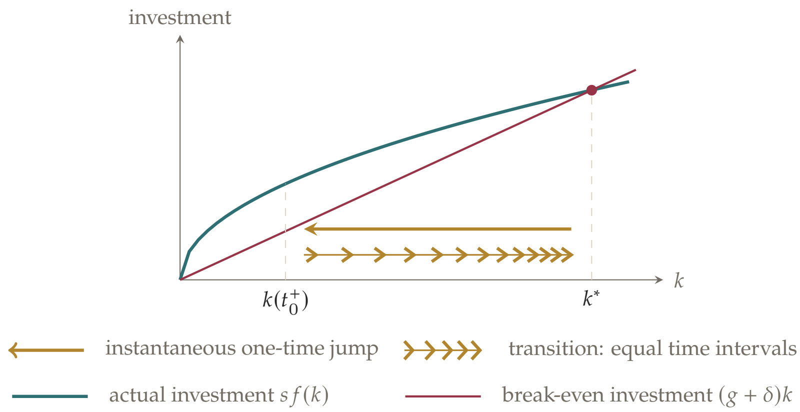

Figure 1.3. A one-time increase in L moves the economy left of k^*, but does not shift either curve. The shrinking gaps between successive rightward arrowheads indicate that the transition slows as k approaches k^*.

NOTE AGGREGATION. Do not confuse total output Y with output per unit of effective labor y. More workers generally raise Y on impact even though $y = Y/(AL)$ falls.

NOTE LEVEL VERSUS GROWTH SHOCK. This is a one-time level shock to L, not a permanent change in population growth. It moves the economy away from its old balanced growth path without shifting either curve in the Solow diagram.

PROBLEM 1.5

Suppose that the production function is Cobb–Douglas.

(a) Find expressions for k^*, y^*, and c^* as functions of the parameters of the model, s, n, δ, g, and α.

(b) What is the golden-rule value of k?

(c) What saving rate is needed to yield the golden-rule capital stock?

SOLUTION

Let the intensive-form production function be

$$f(k) = k^{\alpha}, \qquad 0 < \alpha < 1. \tag{1}$$

The law of motion for capital per unit of effective labor is

$$\dot{k} = sk^\alpha - (n+g+\delta)k. \tag{2}$$

(a) On the balanced growth path, $\dot{k} = 0$. Thus

$$s(k^*)^\alpha = (n+g+\delta)k^*. \tag{3}$$

For the positive steady state, divide by k^* and rearrange:

$$(k^*)^{1-\alpha} = \frac{s}{n+g+\delta}$$

Therefore,

$$\boxed{k^* = \left(\frac{s}{n+g+\delta}\right)^{1/(1-\alpha)}}. \tag{4}$$

Since $y = f(k) = k^\alpha$,

$$\boxed{y^* = \left(\frac{s}{n+g+\delta}\right)^{\alpha/(1-\alpha)}}. \tag{5}$$

Finally, $c = (1-s)y$, and hence

$$\boxed{c^* = (1-s)\left(\frac{s}{n+g+\delta}\right)^{\alpha/(1-\alpha)}}. \tag{6}$$

(b) On a balanced growth path, investment equals break-even investment. Consumption per unit of effective labor can therefore be written as

$$c(k) = f(k) - (n+g+\delta)k = k^\alpha - (n+g+\delta)k. \tag{7}$$

The golden-rule capital stock maximizes consumption in (7). The first-order condition is

$$\frac{\mathrm{d}c(k)}{\mathrm{d}k} = \alpha k^{\alpha-1} - (n+g+\delta) = 0,$$

or equivalently

$$f'(k_{\mathrm{GR}}) = n+g+\delta. \tag{8}$$

Solving for k_{GR} gives

$$\boxed{k_{\mathrm{GR}} = \left(\frac{\alpha}{n+g+\delta}\right)^{1/(1-\alpha)}}. \tag{9}$$

Moreover,

$$\frac{\mathrm{d}^2 c(k)}{\mathrm{d}k^2} = \alpha(\alpha-1)k^{\alpha-2} < 0,$$

so this critical point is the unique maximum.

(c) A saving rate s_{GR} supports the golden-rule capital stock if

$$s_{\mathrm{GR}} f(k_{\mathrm{GR}}) = (n+g+\delta)k_{\mathrm{GR}}. \tag{10}$$

Since $f(k) = k^{\alpha}$,

$$s_{\mathrm{GR}} = (n+g+\delta)k_{\mathrm{GR}}^{1-\alpha}$$

For Cobb–Douglas production, (8) implies $k_{\mathrm{GR}}^{1-\alpha} = \alpha/(n+g+\delta)$. Hence

$$\boxed{s_{\mathrm{GR}} = \alpha}. \tag{11}$$

Under competitive factor pricing, α is also capital's gross share of output.

Note **Steady state versus golden rule.** The steady-state condition $sf(k^*) = (n+g+\delta)k^*$ determines the capital stock generated by a given saving rate. The golden-rule condition $f'(k_{\mathrm{GR}}) = n+g+\delta$ instead identifies the capital stock that maximizes steady-state consumption.

Note **Cobb–Douglas shortcut.** The result $s_{\mathrm{GR}} = \alpha$ is specific to Cobb–Douglas production. Under competitive factor pricing, it says that the golden-rule saving rate equals capital's gross share of output.

Problem 1.6

Consider a Solow economy that is on its balanced growth path. Assume for simplicity that there is no technological progress. Now suppose that the rate of population growth falls.

(a) What happens to the balanced-growth-path values of capital per worker, output per worker, and consumption per worker? Sketch the paths of these variables as the economy moves to its new balanced growth path.

(b) Describe the effect of the fall in population growth on the path of output (that is, total output, not output per worker).

Solution

Because there is no technological progress, A is constant. We can normalize it to one and write

$$k \equiv \frac{K}{L}, \qquad y \equiv \frac{Y}{L} = f(k), \qquad c \equiv \frac{C}{L} = (1-s)f(k). \tag{1}$$

Suppose the population growth rate falls permanently at t_0 from n_{old} to $n_{\text{new}} < n_{\text{old}}$.

(a) Capital per worker evolves according to

$$\dot{k} = sf(k) - (n+\delta)k. \tag{2}$$

Before the change, the economy is on its balanced growth path, so

$$sf\left(k^*_{\text{old}}\right) = (n_{\text{old}}+\delta)k^*_{\text{old}}. \tag{3}$$

Equation (2) shows that the fall in n rotates the break-even investment line downward from $(n_{\text{old}}+\delta)k$ to $(n_{\text{new}}+\delta)k$. It does not change $sf(k)$. Therefore, at the old steady state,

$$\begin{aligned}\dot{k}(t_0^+) &= sf\left(k^*_{\text{old}}\right) - (n_{\text{new}}+\delta)k^*_{\text{old}} \\ &= (n_{\text{old}}-n_{\text{new}})k^*_{\text{old}} > 0. \end{aligned} \tag{4}$$

Thus capital per worker rises monotonically until it reaches the new balanced-growth-path value:

$$\boxed{k^*_{\text{new}} > k^*_{\text{old}}}$$

A change in a growth rate does not create an instantaneous level jump in K or L. Hence

$$k(t_0^+) = k(t_0^-) = k^*_{\text{old}}. \tag{5}$$

After t_0, k rises gradually. Since $f'(k) > 0$ and s is unchanged, output and consumption per worker also rise gradually:

$$\boxed{y^*_{\text{new}} > y^*_{\text{old}}, \qquad c^*_{\text{new}} > c^*_{\text{old}}}$$

(b) Total output satisfies $Y = Ly$, so

$$\frac{\dot{Y}}{Y} = \frac{\dot{L}}{L} + \frac{\dot{y}}{y} = n + \frac{\dot{y}}{y}. \tag{6}$$

On the initial balanced growth path, $\dot{y} = 0$, so (6) becomes

$$\left.\frac{\dot{Y}}{Y}\right|_{\text{old BGP}} = n_{\text{old}}. \tag{7}$$

On the new balanced growth path, $\dot{y} = 0$ again, and therefore

$$\left.\frac{\dot{Y}}{Y}\right|_{\text{new BGP}} = n_{\text{new}}. \tag{8}$$

There is no jump in Y at t_0. Immediately after the change, K is still growing at its old balanced-growth-path rate n_{old}, while labor grows at n_{new}. Growth accounting therefore gives

$$\frac{\dot{Y}(t_0^+)}{Y(t_0^+)} = \alpha_K\left(k^*_{\text{old}}\right) n_{\text{old}} + \left[1 - \alpha_K\left(k^*_{\text{old}}\right)\right] n_{\text{new}}, \tag{9}$$

which lies strictly between n_{new} and n_{old}. During the transition, output continues to grow, but its growth rate gradually approaches n_{new}. Hence total output falls increasingly below the path it would have followed if population growth had remained at n_{old}, even though output per worker is higher.

The steady-state comparison and the transition paths from part (a) are collected in the first figure below. The second figure separates the actual path of total output in part (b) from the no-change counterfactual.

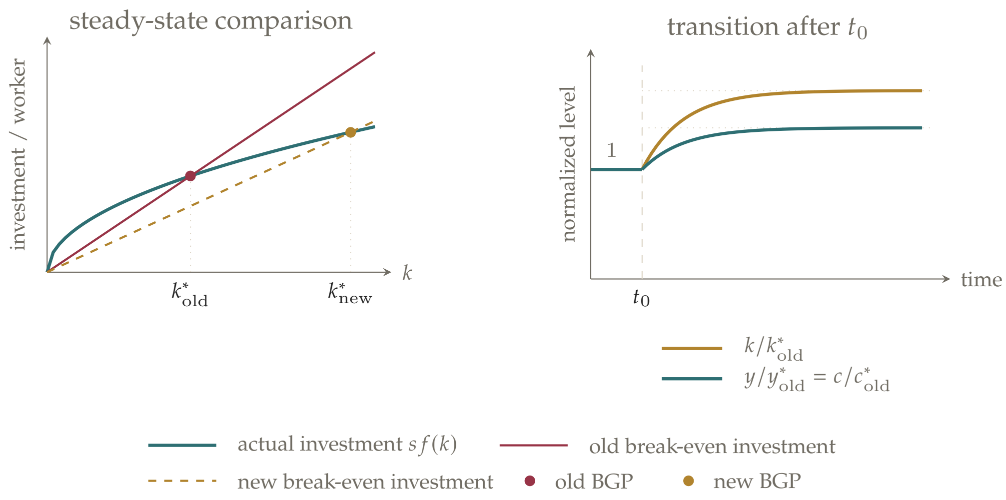

Figure 1.4. A permanent fall in population growth raises capital, output, and consumption per worker. These variables do not jump at t_0; they rise gradually toward their new balanced-growth-path levels.

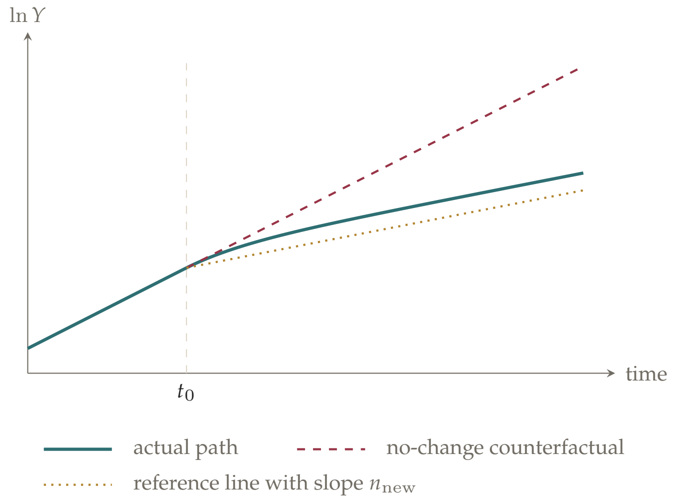

Figure 1.5. Total output is continuous at t_0. Its growth rate falls below n_{old} immediately and converges gradually to n_{new}.

NOTE CONTINUITY. A fall in the *growth rate* of labor does not reduce the level of labor on impact. Per-worker variables are therefore continuous at t_0, unlike the one-time level increase in L studied in Problem 1.4.

PROBLEM 1.7

Find the elasticity of output per unit of effective labor on the balanced growth path, y^*, with respect to the rate of population growth, n. If $\alpha_K(k^*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise y^*?

SOLUTION

The balanced-growth-path conditions are

$$y^* = f(k^*), \tag{1}$$
$$sf(k^*) = (n + g + \delta)k^*. \tag{2}$$

Differentiate the steady-state condition (2) with respect to n:

$$sf'(k^*)\frac{\partial k^*}{\partial n} = k^* + (n + g + \delta)\frac{\partial k^*}{\partial n}.$$

Collecting the terms that contain $\partial k^*/\partial n$ gives

$$\left[sf'(k^*) - (n + g + \delta)\right]\frac{\partial k^*}{\partial n} = k^*,$$

and hence

$$\frac{\partial k^*}{\partial n} = \frac{k^*}{sf'(k^*) - (n + g + \delta)}.$$

Use the steady-state condition to write

$$s = \frac{(n + g + \delta)k^*}{f(k^*)}.$$

Then

$$\begin{aligned} sf'(k^*) &= (n + g + \delta)\frac{k^* f'(k^*)}{f(k^*)} \\ &= (n + g + \delta)\alpha_K(k^*), \end{aligned}$$

where

$$\alpha_K(k^*) \equiv \frac{k^* f'(k^*)}{f(k^*)}. \tag{3}$$

Therefore,

$$\frac{\partial k^*}{\partial n} = -\frac{k^*}{(n + g + \delta)\,[1 - \alpha_K(k^*)]} < 0. \tag{4}$$

Since $y^* = f(k^*)$, the elasticity of y^* with respect to n is

$$\begin{aligned} \varepsilon_{y^*,n} &\equiv \frac{n}{y^*}\frac{\partial y^*}{\partial n} \\ &= \frac{n}{f(k^*)} f'(k^*)\frac{\partial k^*}{\partial n} \\ &= \boxed{-\frac{n}{n + g + \delta}\frac{\alpha_K(k^*)}{1 - \alpha_K(k^*)}}. \end{aligned} \tag{5}$$

For a rough calculation, evaluate (5) at the midpoint $\bar{n} = 1.5\%$:

$$\begin{aligned} \varepsilon_{y^*,n} &\approx -\frac{0.015}{0.015 + 0.02 + 0.03}\frac{1/3}{1 - 1/3} \\ &= -0.115 \approx -0.12. \end{aligned}$$

The fall from 2% to 1% is a 50% fall relative to the initial rate. The conventional textbook-style elasticity calculation therefore gives

$$\frac{\Delta y^*}{y^*} \approx (-0.12)(-0.50) = 0.06.$$

Thus the standard rough answer is

$$\boxed{y^* \text{ rises by about } 6\%.}$$

There is a small numerical subtlety because this is a large change in n. If $\alpha_K = 1/3$ is treated as constant over the interval, a base-consistent log-change calculation gives

$$\begin{aligned}\Delta \ln y^* &\approx -\frac{1/3}{1-1/3} \ln\left(\frac{0.01+0.02+0.03}{0.02+0.02+0.03}\right) \\ &= \frac{1}{2} \ln\left(\frac{0.07}{0.06}\right) \approx 0.077,\end{aligned}$$

which corresponds to an increase of about 8%. The difference reflects the crudeness of applying a local elasticity to a 50% change.

Note **Local versus finite changes.** For grading against the standard solution, the expected back-of-the-envelope answer is about 6%. A consistent finite-change calculation is closer to 8% because halving n is not a small change.

Note **Economic intuition.** Population growth enters through $n + g + \delta$, so a change in n alters only part of the total break-even-investment rate. Plausible differences in population growth therefore have only modest level effects on y^*.

Problem 1.8

Suppose that investment as a fraction of output in the United States rises permanently from 0.15 to 0.18. Assume that capital's share is $1/3$.

(a) By about how much does output eventually rise relative to what it would have been without the rise in investment?

(b) By about how much does consumption rise relative to what it would have been without the rise in investment?

(c) What is the immediate effect of the rise in investment on consumption? About how long does it take for consumption to return to what it would have been without the rise in investment?

Solution

Let $s_{\text{old}} = 0.15$ and $s_{\text{new}} = 0.18$. The proportional increase in the investment rate is

$$\frac{s_{\text{new}} - s_{\text{old}}}{s_{\text{old}}} = \frac{0.03}{0.15} = 0.20.$$

Thus the saving rate rises by 20%, not by 3%; the latter number is the change in percentage points.

(a) From equation (1.28) in the text, the balanced-growth-path elasticity of output with respect to the saving rate is

$$\varepsilon_{y^*,s} \equiv \frac{s}{y^*}\frac{\partial y^*}{\partial s} = \frac{\alpha_K(k^*)}{1-\alpha_K(k^*)}. \tag{1}$$

With $\alpha_K = 1/3$, (1) becomes

$$\varepsilon_{y^*,s} = \frac{1/3}{1-1/3} = \frac{1}{2}$$

Hence the elasticity approximation gives

$$\frac{\Delta y^*}{y^*} \approx \frac{1}{2}(0.20) = 0.10$$

Therefore,

$$\boxed{\text{Long-run output is about } 10\% \text{ higher.}}$$

If a Cobb–Douglas production function with capital share $1/3$ is used to calculate the finite change exactly, then

$$\frac{y^*_{\text{new}}}{y^*_{\text{old}}} = \left(\frac{s_{\text{new}}}{s_{\text{old}}}\right)^{(1/3)/(2/3)} = \sqrt{1.2} \approx 1.095,$$

so the exact increase is 9.5%, which confirms the approximation.

(b) Balanced-growth-path consumption per unit of effective labor is

$$c^* = (1-s)y^*. \tag{2}$$

Therefore, using (2),

$$\begin{aligned}\frac{c^*_{\text{new}}}{c^*_{\text{old}}} &= \frac{1-s_{\text{new}}}{1-s_{\text{old}}}\frac{y^*_{\text{new}}}{y^*_{\text{old}}} \\ &= \frac{0.82}{0.85}\sqrt{1.2} \\ &\approx 1.057\end{aligned}$$

Thus

$$\boxed{\text{Long-run consumption is about } 6\% \text{ higher.}}$$

Consumption rises less than output because the economy devotes a larger fraction of its higher output to investment.

(c) At the instant the saving rate rises, K, A, L, and hence y, cannot jump. Consumption does jump because the consumed fraction of output falls immediately:

$$\begin{aligned}\frac{c(t_0^+)}{c(t_0^-)} &= \frac{(1-s_{\text{new}})y(t_0)}{(1-s_{\text{old}})y(t_0)}\\ &= \frac{0.82}{0.85} \approx 0.965\end{aligned}$$

Hence

$$\boxed{\text{Consumption falls immediately by about } 3.5\%.}$$

To estimate the recovery time, use the text's benchmark $n+g+\delta \approx 0.06$. The local convergence rate is

$$\begin{aligned}\lambda &= [1-\alpha_K(k^*)]\,(n+g+\delta)\\ &= \left(1-\frac{1}{3}\right)(0.06) = 0.04.\end{aligned} \tag{3}$$

Normalize consumption on the old path to one, and define

$$\begin{aligned}x_0 &\equiv \frac{c(t_0^+)}{c^*_{\text{old}}} = \frac{0.82}{0.85} \approx 0.965,\\ x^*_{\text{new}} &\equiv \frac{c^*_{\text{new}}}{c^*_{\text{old}}} \approx 1.057\end{aligned}$$

The linearized transition is

$$x(t) - x^*_{\text{new}} \approx e^{-\lambda(t-t_0)}\left(x_0 - x^*_{\text{new}}\right). \tag{4}$$

Consumption has returned to its old path when $x(t) = 1$. Substituting this condition into (4) gives

$$\begin{aligned}e^{-\lambda(t-t_0)} &= \frac{x^*_{\text{new}} - 1}{x^*_{\text{new}} - x_0}\\ &\approx \frac{1.057-1}{1.057-0.965} \approx 0.62\end{aligned}$$

Taking logs gives

$$\begin{aligned}t - t_0 &\approx -\frac{\ln(0.62)}{0.04}\\ &\approx 12 \text{ years}\end{aligned}$$

Therefore,

$$\boxed{\text{Consumption returns to its old path after roughly } 12 \text{ years.}}$$

The impact effect and the gradual recovery of consumption in part (c) are summarized in the figure below. The horizontal benchmark is the consumption path that would have prevailed without the increase in investment.

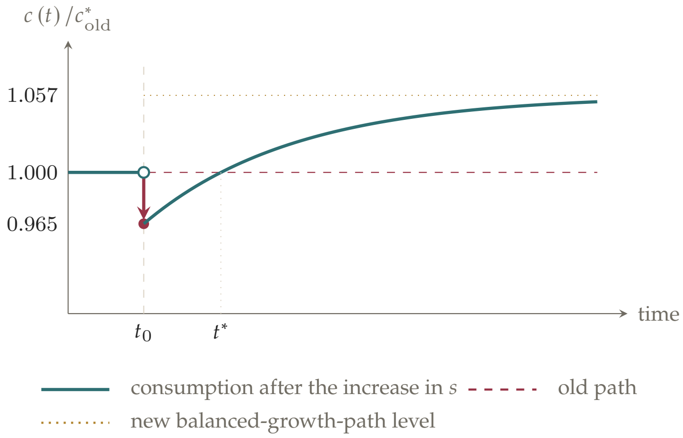

Figure 1.6. Consumption falls on impact, crosses its old path at $t^* \approx t_0 + 12$ years, and then approaches a level about 6% above the old path.

Problem 1.9 *Factor payments in the Solow model.*

Assume that both labor and capital are paid their marginal products. Let w denote $\partial F(K, AL)/\partial L$ and r denote $[\partial F(K, AL)/\partial K] - \delta$.

(a) Show that the marginal product of labor, w, is $A\left[f(k) - kf'(k)\right]$.

(b) Show that if both capital and labor are paid their marginal products, constant returns to scale imply that the total amount paid to the factors of production equals total net output. That is, show that under constant returns, $wL + rK = F(K, AL) - \delta K$.

(c) The return to capital (r) is roughly constant over time, as are the shares of output going to capital and to labor. Does a Solow economy on a balanced growth path exhibit these properties? What are the growth rates of w and r on a balanced growth path?

(d) Suppose the economy begins with a level of k less than k^*. As k moves toward k^*, is w growing at a rate greater than, less than, or equal to its growth rate on the balanced growth path? What about r?

Solution

Write the constant-returns production function in intensive form:

$$Y = F(K, AL) = ALf(k), \qquad k \equiv \frac{K}{AL}. \tag{1}$$

(a) The wage is the marginal product of labor, holding K and A fixed. First,

$$\frac{\partial k}{\partial L} = \frac{\partial}{\partial L}\left(\frac{K}{AL}\right) = -\frac{K}{AL^2} = -\frac{k}{L}$$

Apply the product and chain rules to $Y = ALf(k)$:

$$\begin{aligned} w \equiv \frac{\partial Y}{\partial L} &= Af(k) + ALf'(k)\frac{\partial k}{\partial L} \\ &= Af(k) + ALf'(k)\left(-\frac{k}{L}\right) \\ &= \boxed{A\left[f(k) - kf'(k)\right]}. \end{aligned} \tag{2}$$

(b) First derive the marginal product of capital. Since

$$\frac{\partial k}{\partial K} = \frac{1}{AL},$$

we have

$$\frac{\partial Y}{\partial K} = ALf'(k)\frac{1}{AL} = f'(k). \tag{3}$$

Therefore the net return to capital is

$$r = f'(k) - \delta. \tag{4}$$

Total labor income is

$$\begin{aligned} wL &= AL\left[f(k) - kf'(k)\right] \\ &= ALf(k) - ALkf'(k) \\ &= Y - Kf'(k), \end{aligned}$$

where $ALk = K$. Total net capital income is

$$rK = Kf'(k) - \delta K$$

Adding the two payments gives

$$\begin{aligned} wL + rK &= Y - Kf'(k) + Kf'(k) - \delta K \\ &= \boxed{F(K, AL) - \delta K}. \end{aligned} \tag{5}$$

Equation (5) is Euler's theorem for a constant-returns production function, with depreciation deducted from capital's payment.

(c) On a balanced growth path, $k = k^*$ is constant. Hence

$$r = f'(k^*) - \delta \tag{6}$$

is constant, so

$$\boxed{\frac{\dot{r}}{r} = 0}. \tag{7}$$

Evaluating the wage formula (2) at k^* gives

$$w = A\left[f(k^*) - k^* f'(k^*)\right]. \tag{8}$$

The expression in brackets is constant on the balanced growth path, while A grows at rate g. Therefore,

$$\boxed{\frac{\dot{w}}{w} = g}. \tag{9}$$

The factor shares are constant as well. For example, r is constant, and both K and Y grow at $n + g$, so

$$\frac{\mathrm{d}}{\mathrm{d}t} \ln\left(\frac{rK}{Y}\right) = 0 + (n + g) - (n + g) = 0. \tag{10}$$

Likewise, w grows at g, L grows at n, and Y grows at $n + g$, so wL/Y is constant. The shares in net output are also constant because $Y - \delta K$, wL, and rK all grow at $n + g$ on the balanced growth path.

(d) Let

$$h(k) \equiv f(k) - kf'(k), \tag{11}$$

so that $w = Ah(k)$. Its derivative is

$$h'(k) = f'(k) - \left[f'(k) + kf''(k)\right] = -kf''(k)$$

Thus

$$\begin{aligned} \frac{\dot{w}}{w} &= \frac{\dot{A}}{A} + \frac{h'(k)\dot{k}}{h(k)} \\ &= g - \frac{kf''(k)\dot{k}}{f(k) - kf'(k)}. \end{aligned} \tag{12}$$

When $k < k^*$, stability implies $\dot{k} > 0$. With diminishing marginal returns, $f''(k) <$

0, and labor's gross marginal product satisfies $f(k) - kf'(k) > 0$. Therefore,

$$\boxed{\frac{\dot{w}}{w} > g}$$

The wage grows faster than on the balanced growth path because workers benefit from both technological progress and capital deepening.

For the net return to capital,

$$r = f'(k) - \delta, \qquad \dot{r} = f''(k)\dot{k} < 0. \tag{13}$$

If $r > 0$, its proportional growth rate is

$$\frac{\dot{r}}{r} = \frac{f''(k)\dot{k}}{f'(k) - \delta} < 0. \tag{14}$$

Hence

$$\boxed{r \text{ falls during the transition, so its growth rate is below } 0.}$$

It approaches the constant balanced-growth-path return from above.

Note **Net versus gross return.** The problem defines $r = f'(k) - \delta$, the *net* return. Therefore $\dot{r}/r = f''(k)\dot{k}/[f'(k) - \delta]$; using $f'(k)$ in the denominator would instead be the growth rate of the gross marginal product of capital.

Problem 1.10

(Piketty, 2014; Piketty and Zucman, 2014; Rognlie, 2015) This problem asks you to use a Solow-style model to investigate some ideas that have been discussed in the context of Thomas Piketty's recent work. Consider an economy described by the assumptions of the Solow model, except that factors are paid their marginal products (as in the previous problem), and all labor income is consumed and all other income is saved. Thus, $C(t) = L(t)[\partial Y(t)/\partial L(t)]$.

(a) Show that the properties of the production function and our assumptions about the behavior of L and A imply that the capital-output ratio, K/Y, is rising if and only if the growth rate of K is greater than $n + g$–that is, if and only if k is rising.

(b) Assume that the initial conditions are such that $\partial Y/\partial K$ at $t = 0$ is strictly greater than $n + g + \delta$. Describe the qualitative behavior of the capital-output ratio over time. (For example, does it grow or fall without bound? Gradually approach some constant level from above or below? Something else?) Explain your reasoning.

(c) Many popular summaries of Piketty's work describe his thesis as: Since the return to capital exceeds the growth rate of the economy, the capital-output ratio tends to grow without bound. By the assumptions in part (b), this economy starts in a situation where the return to capital exceeds the economy's growth rate. If you found in (b) that K/Y grows without bound, explain intuitively whether the driving force of this unbounded growth is that the return to capital exceeds the economy's growth rate. Alternatively, if you found in (b) that K/Y does not grow without bound, explain intuitively what is wrong with the statement that the return to capital exceeding the economy's growth rate tends to cause K/Y to grow without bound.

(d) Suppose $F(\cdot)$ is Cobb–Douglas and that the initial situation is as in part (b). Describe the qualitative behavior over time of the share of net capital income (that is, $K(t)[\partial Y(t)/\partial K(t) - \delta]$) in net output (that is, $Y(t) - \delta K(t)$). Explain your reasoning. Is the common statement that an excess of the return to capital over the economy's growth rate causes capital's share to rise over time correct in this case?

Solution

Let $F_K \equiv \partial F(K, AL)/\partial K$, and define capital's gross output elasticity by

$$\alpha_K(t) \equiv \frac{F_K(t)\,K(t)}{Y(t)}. \tag{1}$$

Because the production function has constant returns to scale, labor income plus gross capital income exhausts output. Since all labor income is consumed,

$$Y - C = F_K K. \tag{2}$$

The capital accumulation equation is therefore

$$\begin{aligned}\dot{K} &= Y - C - \delta K \\ &= (F_K - \delta)\,K.\end{aligned} \tag{3}$$

Thus the growth rate of capital equals the net return to capital:

$$\frac{\dot{K}}{K} = F_K - \delta. \tag{4}$$

(a) Differentiate $Y = F(K, AL)$ with respect to time:

$$\dot{Y} = F_K\dot{K} + F_{AL}\,(\dot{AL})$$

Divide by Y, multiply and divide the first term by K, and multiply and divide the second term by AL:

$$\frac{\dot{Y}}{Y} = \frac{F_K K}{Y}\frac{\dot{K}}{K} + \frac{F_{AL}(AL)}{Y}\frac{(\dot{AL})}{AL}$$

Constant returns imply

$$\frac{F_K K}{Y} + \frac{F_{AL}(AL)}{Y} = 1, \tag{5}$$

and AL grows at $n + g$. Hence

$$\frac{\dot{Y}}{Y} = \alpha_K \frac{\dot{K}}{K} + (1 - \alpha_K)(n + g). \tag{6}$$

The growth rate of the capital-output ratio is therefore

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \ln\left(\frac{K}{Y}\right) &= \frac{\dot{K}}{K} - \frac{\dot{Y}}{Y} \\ &= (1 - \alpha_K) \left[\frac{\dot{K}}{K} - (n + g)\right]. \end{aligned} \tag{7}$$

With $0 < \alpha_K < 1$, this expression is positive if and only if

$$\frac{\dot{K}}{K} > n + g$$

Finally, since $k = K/(AL)$,

$$\frac{\dot{k}}{k} = \frac{\dot{K}}{K} - (n + g). \tag{8}$$

Thus

$$\boxed{\frac{K}{Y} \text{ rises} \quad \Longleftrightarrow \quad \frac{\dot{K}}{K} > n + g \quad \Longleftrightarrow \quad \dot{k} > 0.}$$

(b) In intensive form, $F_K = f'(k)$. Using the accumulation equation (3),

$$\begin{aligned} \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - (n + g) \\ &= f'(k) - (n + g + \delta). \end{aligned} \tag{9}$$

The assumption in the question is

$$f'(k(0)) > n + g + \delta,$$

so $\dot{k}(0) > 0$. Under the usual Solow assumptions, $f''(k) < 0$, so the marginal product $f'(k)$ falls as k rises. Let $\bar{k}$ denote the unique value satisfying

$$f'(\bar{k}) = n + g + \delta. \tag{10}$$

Then

$$\begin{aligned} k < \bar{k} &\Longrightarrow \dot{k} > 0, \\ k > \bar{k} &\Longrightarrow \dot{k} < 0 \end{aligned}$$

Since the initial condition implies $k(0) < \bar{k}$, capital per unit of effective labor rises monotonically toward $\bar{k}$.

Moreover,

$$\frac{K}{Y} = \frac{k}{f(k)}, \tag{11}$$

and

$$\frac{\mathrm{d}\ln[k/f(k)]}{\mathrm{d}\ln k} = 1 - \frac{kf'(k)}{f(k)} = 1 - \alpha_K(k) > 0. \tag{12}$$

Therefore K/Y rises whenever k rises and converges to the finite value $\bar{k}/f(\bar{k})$. Hence

$$\boxed{\frac{K}{Y} \text{ approaches a finite constant from below.}}$$

(c) At the start of the transition,

$$r \equiv F_K - \delta > n + g. \tag{13}$$

Since $\dot{K}/K = r$, capital initially grows faster than effective labor, and K/Y rises. But the return r is not fixed: as k rises, diminishing marginal returns reduce $F_K = f'(k)$, and hence reduce r. The process stops when

$$r = n + g, \tag{14}$$

equivalently when $f'(k) = n + g + \delta$. At that point k and K/Y are constant.

Thus the statement that r exceeding the economy's growth rate causes K/Y to grow without bound incorrectly treats the return to capital as exogenous. In this model,

$$\boxed{\text{capital accumulation lowers } r \text{ until the excess return disappears.}}$$

(d) With Cobb–Douglas production,

$$Y = K^{\alpha}(AL)^{1-\alpha}, \qquad 0 < \alpha < 1, \tag{15}$$

so

$$F_K = \alpha\frac{Y}{K}. \tag{16}$$

Net capital income is

$$K(F_K - \delta) = \alpha Y - \delta K, \tag{17}$$

while net output is $Y - \delta K$. Let

$$\beta \equiv \frac{K}{Y}. \tag{18}$$

The share of net capital income in net output is

$$\begin{aligned} \theta_K^{\text{net}} &\equiv \frac{\alpha Y - \delta K}{Y - \delta K} \\ &= \frac{\alpha - \delta\beta}{1 - \delta\beta}. \end{aligned} \tag{19}$$

Differentiate with respect to β:

$$\frac{\mathrm{d}\,\theta_K^{\text{net}}}{\mathrm{d}\beta} = \frac{-\delta(1-\delta\beta)-(\alpha-\delta\beta)(-\delta)}{(1-\delta\beta)^2}$$
$$= \frac{\delta(\alpha-1)}{(1-\delta\beta)^2} < 0. \tag{20}$$

Equations (11) and (12) show that $\beta = K/Y$ rises monotonically. Therefore,

$$\boxed{\theta_K^{\text{net}} \text{ falls monotonically toward a constant.}}$$

Thus, in the Cobb–Douglas case, an initial excess of the return to capital over the economy's growth rate is associated with a *fall*, not a rise, in capital's share of net output.

The qualitative paths established in parts (b)–(d) are collected in the figure below. Each horizontal guide marks the corresponding long-run value.

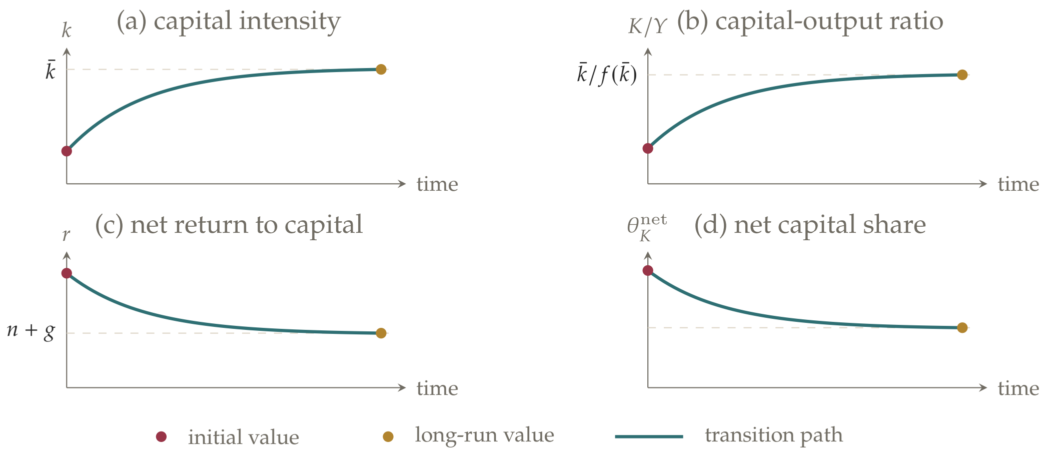

Figure 1.7. With $f'(k(0)) > n+g+\delta$, capital intensity and the capital-output ratio rise, while the net return falls to $n+g$. Under Cobb–Douglas production, capital's share of net output also falls.

Note **Transition, not a parameter restriction.** The comparison $r > g_Y$ is not a permanent parameter restriction here. It is a transitional inequality: capital deepening lowers the marginal product of capital until $r = n+g$ on the balanced growth path.

Problem 1.11

Consider Problem 1.10. Suppose there is a marginal increase in K.

(a) Derive an expression (in terms of K/Y, δ, the marginal product of capital F_K, and the elasticity of substitution between capital and effective labor in the gross production function $F(\cdot)$) that determines whether a marginal increase in K increases, reduces, or has no effect on the share of net capital income in net output.

(b) Suppose the capital-output ratio is 3, $\delta = 3\%$, and the rate of return on capital

$(F_K - \delta) = 5\%$. How large must the elasticity of substitution in the gross production function be for the share of net capital income in net output to rise when K rises?

Solution

Let

$$\theta_K^{\text{net}} \equiv \frac{K(F_K - \delta)}{Y - \delta K} = \frac{F_K - \delta}{Y/K - \delta} \tag{1}$$

denote capital's share of net output. We maintain the usual assumptions that $F_K > \delta$, $Y/K > \delta$, and $F_{KK} < 0$.

(a) Along a constant-returns-to-scale production function, the elasticity of substitution between capital and effective labor can be written as

$$\sigma = -\frac{\partial \ln(K/Y)}{\partial \ln F_K} = \frac{\partial \ln(Y/K)}{\partial \ln F_K}. \tag{2}$$

Hence

$$\frac{\partial(Y/K)}{\partial F_K} = \sigma \frac{Y/K}{F_K}. \tag{3}$$

Differentiate (1) with respect to F_K:

$$\begin{aligned} \frac{\partial \theta_K^{\text{net}}}{\partial F_K} &= \frac{1}{Y/K - \delta} - \frac{F_K - \delta}{(Y/K - \delta)^2} \frac{\partial(Y/K)}{\partial F_K} \\ &= \frac{(Y/K - \delta) - \sigma(F_K - \delta)(Y/K)/F_K}{(Y/K - \delta)^2}. \end{aligned} \tag{4}$$

Since F_K falls when K rises, the sign of the response to K is the opposite of the sign in (4). Therefore,

$$\boxed{\begin{aligned} \theta_K^{\text{net}} \text{ rises with } K &\iff \sigma > \frac{F_K[1 - \delta(K/Y)]}{F_K - \delta}, \\ \theta_K^{\text{net}} \text{ is locally unchanged} &\iff \sigma = \frac{F_K[1 - \delta(K/Y)]}{F_K - \delta}, \\ \theta_K^{\text{net}} \text{ falls with } K &\iff \sigma < \frac{F_K[1 - \delta(K/Y)]}{F_K - \delta}. \end{aligned}} \tag{5}$$

(b) The net return is $F_K - \delta = 0.05$, so $F_K = 0.08$. Substituting $K/Y = 3$ and $\delta = 0.03$ into (5) gives

$$\sigma > \frac{0.08[1 - (0.03)(3)]}{0.08 - 0.03} = \frac{0.0728}{0.05} = 1.456$$

Thus

$$\boxed{\sigma > 1.456}$$

is required for a marginal increase in K to raise capital's share of net output.

Note **Sign check.** The derivative in (4) is taken with respect to F_K, not K. Diminishing marginal returns imply $\partial F_K/\partial K < 0$, so the final inequality reverses.

Note **Gross versus net.** F_K is the gross marginal product, whereas $F_K - \delta$ is the net return. Likewise, the relevant denominator is net output $Y - \delta K$, not gross output Y.

Problem 1.12

Consider the same setup as at the start of Problem 1.10: the economy is described by the assumptions of the Solow model, except that factors are paid their marginal products and all labor income is consumed and all other income is saved. Show that the economy converges to a balanced growth path, and that the balanced-growth-path level of k equals the golden-rule level of k. What is the intuition for this result?

Solution

Constant returns to scale and marginal-product factor pricing imply that gross capital income is $F_K K$. Since all labor income is consumed and all other income is saved, capital accumulation satisfies

$$\dot{K} = (F_K - \delta)K. \tag{1}$$

Let $k = K/(AL)$. Because AL grows at rate $n + g$ and $F_K = f'(k)$, equation (1) yields

$$\dot{k} = \left[f'(k) - (n + g + \delta)\right]k. \tag{2}$$

Apart from the zero-capital steady state, the balanced-growth-path value $k^* > 0$ is determined by

$$f'(k^*) = n + g + \delta. \tag{3}$$

Since $f''(k) < 0$,

$$\begin{aligned} k < k^* &\Longrightarrow f'(k) > n + g + \delta \Longrightarrow \dot{k} > 0, \\ k > k^* &\Longrightarrow f'(k) < n + g + \delta \Longrightarrow \dot{k} < 0. \end{aligned}$$

Thus every path with positive initial capital converges to k^*. Once k is constant, both K and Y grow at rate $n + g$, so the limiting path is a balanced growth path.

Consumption per unit of effective labor at any constant k is

$$c(k) = f(k) - (n + g + \delta)k.$$

The golden-rule capital stock maximizes $c(k)$, and its first-order condition is

$$f'(k_{\text{GR}}) = n + g + \delta. \tag{4}$$

Equations (3) and (4) are identical; hence

$$\boxed{k^* = k_{\text{GR}}.}$$

The intuition follows directly from saving behavior. Saving per unit of effective labor is capital's gross income, $f'(k)k$, while break-even investment is $(n+g+\delta)k$. Capital rises when the former exceeds the latter and falls when it is smaller. The adjustment ends precisely where the marginal product of capital equals the resource cost of maintaining one more unit of k, which is the golden-rule condition.

Note **Interpretation.** The saving rate is endogenous in this model: households save all capital income and consume all labor income. It should not be treated as the fixed s of the basic Solow model.

Problem 1.13

Go through steps analogous to those in equations (1.29)–(1.32) to find how quickly y converges to y^* in the vicinity of the balanced growth path.

Hint: Since $y = f(k)$, we can write $k = g(y)$, where $g(\cdot) = f^{-1}(\cdot)$.

Solution

Let $k = g(y) = f^{-1}(y)$. Differentiating $y = f(k)$ and using the Solow law of motion gives

$$\begin{aligned} \dot{y} &= f'(k)\dot{k} \\ &= f'(k)\big[sf(k) - (n + g + \delta)k\big] \equiv H(y). \end{aligned} \tag{1}$$

At $y^* = f(k^*)$, $H(y^*) = 0$. A first-order approximation around y^* is therefore

$$\dot{y}(t) \simeq H'(y^*)[y(t) - y^*] \equiv -\lambda_y[y(t) - y^*]. \tag{2}$$

The solution to the linearized equation is

$$y(t) - y^* \simeq e^{-\lambda_y t}[y(0) - y^*]. \tag{3}$$

It remains to calculate $H'(y^*)$. By the chain rule,

$$
\begin{aligned}
H'(y^*) &= \frac{\partial \dot{y}}{\partial k}\frac{\partial k}{\partial y}\bigg|_{k=k^*} \\
&= \left\{f''(k)[sf(k)-(n+g+\delta)k] + f'(k)[sf'(k)-(n+g+\delta)]\right\}\frac{1}{f'(k)}\bigg|_{k=k^*} \\
&= sf'(k^*) - (n+g+\delta), \qquad (4)
\end{aligned}
$$

where the steady-state condition $sf(k^*) = (n+g+\delta)k^*$ eliminates the first term.

Consequently,

$$
\begin{aligned}
\lambda_y &= (n+g+\delta) - sf'(k^*) \\
&= (n+g+\delta)\left[1 - \frac{k^* f'(k^*)}{f(k^*)}\right] \\
&= \left[1 - \alpha_K(k^*)\right](n+g+\delta). \qquad (5)
\end{aligned}
$$

Thus

$$
\boxed{\lambda_y = \lambda_k = \left[1 - \alpha_K(k^*)\right](n+g+\delta).}
$$

Output per unit of effective labor therefore converges locally at exactly the same rate as capital per unit of effective labor.

Note **Economic intuition.** Near the balanced growth path, $y - y^* \simeq f'(k^*)(k - k^*)$. This local proportionality is why the two gaps shrink at the same exponential rate even though y and k have different units.

Problem 1.14 *Embodied technological progress.*

(Solow, 1960; Sato, 1966) One view of technological progress is that the productivity of capital goods built at t depends on the state of technology at t and is unaffected by subsequent technological progress. This is known as *embodied technological progress* (technological progress must be "embodied" in new capital before it can raise output). This problem asks you to investigate its effects.

(a) As a preliminary, let us modify the basic Solow model to make technological progress capital-augmenting rather than labor-augmenting. So that a balanced growth path exists, assume that the production function is Cobb–Douglas:

$$
Y(t) = [A(t)K(t)]^{\alpha} L(t)^{1-\alpha}
$$

Assume that A grows at rate μ: $\dot{A}(t) = \mu A(t)$. Show that the economy converges to a balanced growth path, and find the growth rates of Y and K on the balanced growth path.

Hint: Show that we can write $Y/(A^{\phi}L)$ as a function of $K/(A^{\phi}L)$, where $\phi = \alpha/(1-\alpha)$. Then analyze the dynamics of $K/(A^{\phi}L)$.

(b) Now consider embodied technological progress. Specifically, let the production function be $Y(t) = J(t)^{\alpha} L(t)^{1-\alpha}$, where $J(t)$ is the effective capital stock. The dynamics of $J(t)$ are given by $\dot{J}(t) = sA(t)Y(t) - \delta J(t)$. The presence of the $A(t)$ term in this expression means that the productivity of investment at t depends on the technology at t. Show that the economy converges to a balanced growth path. What are the growth rates of Y and J on the balanced growth path?

Hint: Let $\bar{J}(t) \equiv J(t)/A(t)$. Then use the same approach as in (a), focusing on $\bar{J}/(A^{\phi}L)$ instead of $K/(A^{\phi}L)$.

(c) What is the elasticity of output on the balanced growth path with respect to s?

(d) In the vicinity of the balanced growth path, how rapidly does the economy converge to the balanced growth path?

(e) Compare your results for (c) and (d) with the corresponding results in the text for the basic Solow model.

Solution

Let

$$\phi \equiv \frac{\alpha}{1-\alpha}.$$

(a) Define capital and output relative to $A^{\phi}L$:

$$\kappa(t) \equiv \frac{K(t)}{A(t)^{\phi} L(t)}, \qquad y(t) \equiv \frac{Y(t)}{A(t)^{\phi} L(t)}.$$

Since $1+\phi = 1/(1-\alpha)$, dividing the production function by $A^{\phi}L$ gives

$$y(t) = \kappa(t)^{\alpha}. \tag{1}$$

The normalizing factor $A^{\phi}L$ grows at rate $\phi\mu + n$. Using $\dot{K} = sY - \delta K$, the law of motion for κ is

$$\dot{\kappa} = s\kappa^{\alpha} - (n + \phi\mu + \delta)\kappa. \tag{2}$$

The unique positive steady state is

$$\kappa^{*} = \left(\frac{s}{n+\phi\mu+\delta}\right)^{1/(1-\alpha)}. \tag{3}$$

For $0 < \kappa < \kappa^{*}$, actual investment exceeds break-even investment and $\dot{\kappa} > 0$; for $\kappa > \kappa^{*}$, the reverse holds. Thus $\kappa \to \kappa^{*}$ and, by (1), $y \to y^{*}$.

Since $K = A^{\phi}L\kappa$ and $Y = A^{\phi}Ly$, the balanced-growth-path growth rates are

$$\boxed{g_K^{\text{BGP}} = g_Y^{\text{BGP}} = n + \phi\mu.}$$

(b) Under embodied technological progress, define $\bar{J} \equiv J/A$. Its law of motion is

$$\dot{\bar{J}} = \frac{\dot{J}}{A} - \mu\bar{J} = sY - (\delta + \mu)\bar{J}. \tag{4}$$

The production function becomes $Y = (A\bar{J})^{\alpha}L^{1-\alpha}$, which has the same form as in part (a). Define

$$j(t) \equiv \frac{\bar{J}(t)}{A(t)^{\phi}L(t)} = \frac{J(t)}{A(t)^{1+\phi}L(t)}.$$

Then

$$y(t) = j(t)^{\alpha}, \qquad y(t) \equiv \frac{Y(t)}{A(t)^{\phi}L(t)}. \tag{5}$$

Combining (4) with the growth rate of $A^{\phi}L$ yields

$$\dot{j} = sj^{\alpha} - \left[n + \delta + (1+\phi)\mu\right]j. \tag{6}$$

Thus the unique positive steady state is

$$j^* = \left(\frac{s}{n + \delta + (1+\phi)\mu}\right)^{1/(1-\alpha)}, \tag{7}$$

and the same sign argument as in part (a) shows that $j \to j^*$.

On the balanced growth path, y and j are constant. Hence

$$\boxed{g_Y^{\text{BGP}} = n + \phi\mu,} \qquad \boxed{g_J^{\text{BGP}} = n + (1+\phi)\mu.}$$

The growth rate of J is higher because $J = A^{1+\phi}Lj$, whereas output is $Y = A^{\phi}Ly$.

(c) From (5) and (7),

$$y^* = \left(\frac{s}{n + \delta + (1+\phi)\mu}\right)^{\alpha/(1-\alpha)}.$$

Since $Y = A^{\phi}Ly$, the elasticity of balanced-growth-path output with respect to s is

$$\boxed{\frac{\partial \ln Y^{\text{BGP}}}{\partial \ln s} = \frac{\partial \ln y^*}{\partial \ln s} = \frac{\alpha}{1-\alpha}.} \tag{8}$$

(d) Let $B \equiv n + \delta + (1+\phi)\mu$. Linearizing (6) around j^* gives

$$\dot{j} \simeq \left[s\alpha(j^*)^{\alpha-1} - B\right](j - j^*).$$

The steady-state condition implies $s(j^*)^{\alpha-1} = B$. Therefore,

$$\boxed{\lambda_{\text{emb}} = (1-\alpha)\big[n + \delta + (1+\phi)\mu\big].} \tag{9}$$

Locally,

$$j(t) - j^* \simeq e^{-\lambda_{\text{emb}} t}[j(0) - j^*].$$

Because $y = j^\alpha$ is smooth and strictly increasing, y converges at the same rate.

(e) For Cobb–Douglas production, the basic Solow model gives

$$\frac{\partial \ln y^*}{\partial \ln s} = \frac{\alpha}{1-\alpha}, \qquad \lambda_{\text{basic}} = (1-\alpha)(n + \mu + \delta).$$

Thus embodiment leaves the saving elasticity unchanged. It increases the convergence rate, since

$$\lambda_{\text{emb}} - \lambda_{\text{basic}} = \alpha\mu > 0.$$

The two convergence mechanisms are summarized in the figure below. Both use the same concave actual-investment schedule; embodiment adds the extra $\mu\bar{J}$ dilution term visible in (4), making the break-even line steeper in normalized units.

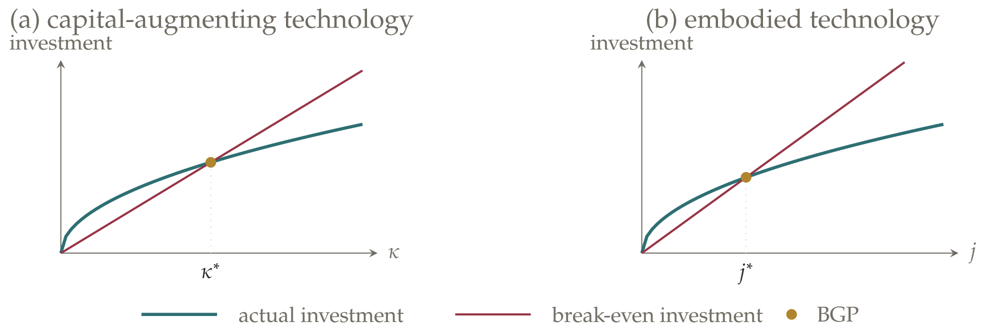

Figure 1.8. Normalized dynamics under capital-augmenting and embodied technological progress. The break-even schedules are $(n + \phi\mu + \delta)\kappa$ and $[n + \delta + (1+\phi)\mu]j$, respectively.

Note **Notation.** J is the effective capital stock, $\bar{J} = J/A$ removes the productivity of the investment vintage, and $j = J/(A^{1+\phi}L)$ is the stationary normalized variable. Mixing these three objects leads to incorrect growth rates.

Note **Economic intuition.** As the frontier A rises, existing embodied capital loses ground relative to newly produced capital goods. This adds μ to normalized break-even investment and makes convergence faster than in the basic model.

Problem 1.15

Consider a Solow economy on its balanced growth path. Suppose the growth-accounting techniques described in Section 1.7 are applied to this economy.

(a) What fraction of growth in output per worker does growth accounting attribute to growth in capital per worker? What fraction does it attribute to technological progress?

(b) How can you reconcile your results in (a) with the fact that the Solow model implies that the growth rate of output per worker on the balanced growth path is determined solely by the rate of technological progress?

Solution

Let $\alpha_K(t) \equiv F_K(t) K(t) / Y(t)$ denote capital's output elasticity. Growth accounting for output per worker gives

$$\frac{\dot{Y}}{Y} - \frac{\dot{L}}{L} = \alpha_K(t)\left(\frac{\dot{K}}{K} - \frac{\dot{L}}{L}\right) + R(t), \tag{1}$$

where $R(t)$ is the Solow residual.

(a) On the Solow balanced growth path,

$$\frac{\dot{Y}}{Y} - \frac{\dot{L}}{L} = \frac{\dot{K}}{K} - \frac{\dot{L}}{L} = g. \tag{2}$$

Substituting (2) into (1) gives

$$g = \alpha_K(k^*)g + R, \qquad R = [1 - \alpha_K(k^*)]g. \tag{3}$$

Thus growth accounting attributes

$$\boxed{\alpha_K(k^*) \text{ of growth to capital deepening}}$$

and

$$\boxed{1 - \alpha_K(k^*) \text{ of growth to technological progress.}}$$

With the common calibration $\alpha_K = 1/3$, the shares are one-third and two-thirds, respectively.

(b) There is no contradiction because growth accounting records *proximate* contributions, whereas the Solow model identifies the *underlying* source of sustained growth. Growth in A raises output per worker directly. For a fixed saving rate, it also induces capital per worker to grow at rate g, so part of technology's total effect operates through capital deepening. Growth accounting assigns this induced channel to capital rather than to the technological change that caused it.

Hence the accounting decomposition is correct, but it should not be read as a causal decomposition of the ultimate sources of balanced-growth-path growth.

NOTE **ACCOUNTING VERSUS CAUSALITY.** The term $\alpha_K g$ answers how much current capital deepening contributes mechanically to output growth. It does not answer why capital deepening occurs.

NOTE **SOLOW RESIDUAL.** Labor-augmenting technology grows at rate g, but the residual in (3) is $(1 - \alpha_K)g$. The remaining $\alpha_K g$ appears in the capital-deepening term.

PROBLEM 1.16

(a) In the model of convergence and measurement error in equations (1.39) and (1.40), suppose the true value of b is -1. Does a regression of $\ln(Y/N)_{1979} - \ln(Y/N)_{1870}$ on a constant and $\ln(Y/N)_{1870}$ yield a biased estimate of b? Explain.

(b) Suppose there is measurement error in measured 1979 income per capita but not in 1870 income per capita. Does a regression of $\ln(Y/N)_{1979} - \ln(Y/N)_{1870}$ on a constant and $\ln(Y/N)_{1870}$ yield a biased estimate of b? Explain.

SOLUTION

Write true log income per capita in 1870 as x_i^*, measured log income as x_i, and true 1979 log income as z_i^*.

(a) Equations (1.39) and (1.40) are

$$z_i^* - x_i^* = a + bx_i^* + \varepsilon_i, \tag{1}$$
$$x_i = x_i^* + u_i, \tag{2}$$

where u_i is uncorrelated with x_i^* and ε_i. The regression uses measured growth $z_i^* - x_i$. Substituting $x_i^* = x_i - u_i$ into (1) gives

$$z_i^* - x_i = a + bx_i + \varepsilon_i - (1+b)u_i. \tag{3}$$

In general, $x_i = x_i^* + u_i$ is correlated with the composite error. Indeed,

$$\mathrm{Cov}\,(x_i, \varepsilon_i - (1+b)u_i) = -(1+b)\,\mathrm{Var}(u_i). \tag{4}$$

When the true $b = -1$, however, the measurement-error term disappears from (3); the regression error is simply ε_i, which is uncorrelated with x_i. Therefore,

$$\boxed{b = -1 \Longrightarrow \text{the OLS slope is unbiased.}}$$

(b) Now suppose 1870 income is measured without error, so $x_i = x_i^*$, but measured 1979 income is $z_i = z_i^* + v_i$. Measured growth satisfies

$$z_i - x_i = a + bx_i + \varepsilon_i + v_i. \tag{5}$$

Under classical measurement error, $\mathrm{E}[v_i \mid x_i] = 0$. Thus the new composite error $\varepsilon_i + v_i$ remains uncorrelated with the regressor, and

$$\boxed{\text{measurement error only in 1979 income does not bias } \hat{b}.}$$

It raises the variance of the regression error and therefore reduces precision.

Note **Why initial-income error is special.** Measured 1870 income enters both sides with opposite signs. A positive error makes a country look richer initially and slower growing, mechanically favoring convergence unless $b = -1$.

Note **Classical-error qualification.** Part (b) requires mean-zero 1979 error uncorrelated with initial income and the structural disturbance. Systematic error can still bias OLS.

Problem 1.17

Derive equation (1.50).

Hint: Follow steps analogous to those in equations [1.47] and [1.48].

Solution

The hypothetical economy removes resource and land limitations by assuming that resources, land, and labor all grow at rate n:

$$g_R = g_T = g_L = n, \qquad g_A = g.$$

The capital accumulation equation gives $g_K = s(Y/K) - \delta$. A constant balanced-growth-path value of g_K therefore requires Y/K to be constant, and hence

$$g_K^{\mathrm{BGP}} = g_Y^{\mathrm{BGP}}. \tag{1}$$

Log-differentiating the production function gives

$$g_Y = \alpha g_K + \beta g_R + \gamma g_T + (1 - \alpha - \beta - \gamma)(g_A + g_L). \tag{2}$$

Substitute $g_R = g_T = g_L = n$ and $g_A = g$:

$$\begin{aligned} g_Y &= \alpha g_K + (\beta + \gamma)n + (1 - \alpha - \beta - \gamma)(n + g) \\ &= \alpha g_K + (1 - \alpha)n + (1 - \alpha - \beta - \gamma)g. \end{aligned} \tag{3}$$

Imposing (1) yields

$$\begin{aligned} (1 - \alpha)\widetilde{g}_Y^{\mathrm{BGP}} &= (1 - \alpha)n + (1 - \alpha - \beta - \gamma)g, \\ \widetilde{g}_Y^{\mathrm{BGP}} &= n + \frac{1 - \alpha - \beta - \gamma}{1 - \alpha} g. \end{aligned} \tag{4}$$

The tilde distinguishes the hypothetical economy from the economy with fixed land and declining resource use. Output per worker grows at total-output growth minus population growth. Therefore,

$$\boxed{\widetilde{g}_{Y/L}^{\mathrm{BGP}} = \widetilde{g}_{Y}^{\mathrm{BGP}} - n = \frac{1-\alpha-\beta-\gamma}{1-\alpha} g.} \tag{5}$$

This is equation (1.50).

Chapter 2

Problem 2.1

Consider N firms each with the constant-returns-to-scale production function $Y = F(K, AL)$, or (using the intensive form) $Y = ALf(k)$. Assume $f'(\cdot) > 0$, $f''(\cdot) < 0$. Assume that all firms can hire labor at wage wA and rent capital at cost r, and that all firms have the same value of A.

(a) Consider the problem of a firm trying to produce Y units of output at minimum cost. Show that the cost-minimizing level of k is uniquely defined and is independent of Y, and that all firms therefore choose the same value of k.

(b) Show that the total output of the N cost-minimizing firms equals the output that a single firm with the same production function has if it uses all the labor and capital used by the N firms.

Solution

It is useful to treat effective labor, AL, as one input. Let $X \equiv AL$, so that $k = K/X$ and output is $Y = Xf(k)$.

(a) For any proposed capital intensity k, producing Y requires

$$X = \frac{Y}{f(k)}, \qquad K = kX = \frac{kY}{f(k)}.$$

The resulting total cost is therefore

$$\text{Cost}(Y, k) = wX + rK = Y\frac{w + rk}{f(k)}. \tag{1}$$

Thus Y only multiplies unit cost; it cannot affect the value of k that minimizes cost.

For an interior optimum, differentiate the last term in (1) with respect to k:

$$\frac{\mathrm{d}}{\mathrm{d}k}\left(\frac{w + rk}{f(k)}\right) = \frac{rf(k) - (w + rk)f'(k)}{[f(k)]^2} = 0.$$

Rearranging gives the usual tangency condition

$$\frac{r}{w} = \frac{f'(k)}{f(k) - kf'(k)}. \tag{2}$$

The numerator on the right is the marginal product of capital; the denominator is the marginal product of effective labor.

To verify uniqueness rather than merely state the first-order condition, differentiate the right-hand side of (2):

$$\frac{\mathrm{d}}{\mathrm{d}k}\left[\frac{f'(k)}{f(k) - kf'(k)}\right] = \frac{f''(k)f(k)}{[f(k) - kf'(k)]^2} < 0. \tag{3}$$

Hence the marginal-product ratio is strictly decreasing in k, so a given positive factor-price ratio r/w can be matched by at most one value, denoted $\bar{k}$. Under the chapter's standard interiority and Inada assumptions, such a value exists. It is independent of Y, and because every firm faces the same r, w, and A, every firm chooses the same $\bar{k}$.

(b) Let K_i, L_i, and Y_i denote firm i's choices and output. Since $K_i/(AL_i) = \bar{k}$, aggregation gives

$$\begin{aligned}\sum_{i=1}^{N} Y_i &= \sum_{i=1}^{N} AL_i f(\bar{k}) = A\left(\sum_{i=1}^{N} L_i\right) f(\bar{k}) \\ &= ALf\left(\frac{K}{AL}\right) = F(K, AL), \qquad K \equiv \sum_{i=1}^{N} K_i, \quad L \equiv \sum_{i=1}^{N} L_i.\end{aligned} \tag{4}$$

The second line uses $K/(AL) = \sum_i K_i/[A\sum_i L_i] = \bar{k}$. Thus the N cost-minimizing firms produce exactly what one firm would produce with their combined capital and labor.

Note **Why output drops out.** Constant returns to scale separates the firm's decision into a scale choice and an input-ratio choice. Output Y scales total cost in (1), while relative factor prices determine capital intensity.

Note **When aggregation works.** The result uses common technology, a common value of A, common factor prices, and constant returns. If firms chose different capital intensities, the final equality in (4) would generally fail.

Problem 2.2 *The elasticity of substitution with constant-relative-risk-aversion utility.*

Consider an individual who lives for two periods and whose utility is given by equation (2.43). Let P_1 and P_2 denote the prices of consumption in the two periods, and let W denote the value of the individual's lifetime income; thus the budget constraint is $P_1C_1 + P_2C_2 = W$.

(a) What are the individual's utility-maximizing choices of C_1 and C_2, given P_1, P_2, and W?

(b) The elasticity of substitution between consumption in the two periods is

$$-\frac{P_1/P_2}{C_1/C_2}\frac{\partial\,(C_1/C_2)}{\partial\,(P_1/P_2)} \quad \text{or} \quad -\frac{\partial \ln{(C_1/C_2)}}{\partial \ln{(P_1/P_2)}} \tag{1}$$

Show that with the utility function (2.43), the elasticity of substitution between C_1 and C_2 is $1/\theta$.

Solution

The utility function in equation (2.43) is

$$U(C_1, C_2) = \frac{C_1^{1-\theta}}{1-\theta} + \frac{1}{1+\rho}\frac{C_2^{1-\theta}}{1-\theta}.$$

(a) Attach multiplier λ to the lifetime budget constraint. The two first-order conditions are

$$C_1^{-\theta} = \lambda P_1, \qquad \frac{1}{1+\rho}C_2^{-\theta} = \lambda P_2. \tag{2}$$

Dividing the first condition in (2) by the second and solving for the consumption ratio yields

$$\frac{C_1}{C_2} = \phi, \qquad \phi \equiv (1+\rho)^{1/\theta}\left(\frac{P_2}{P_1}\right)^{1/\theta}. \tag{3}$$

Substituting $C_1 = \phi C_2$ into $P_1C_1 + P_2C_2 = W$ gives the Marshallian demands

$$\boxed{C_1^* = \frac{\phi W}{P_1\phi + P_2}, \qquad C_2^* = \frac{W}{P_1\phi + P_2}.} \tag{4}$$

Both choices are positive when $P_1, P_2, W > 0$, and strict concavity of utility makes this interior optimum unique.

(b) Take logs of (3) and write the relative price as P_1/P_2:

$$\ln\left(\frac{C_1}{C_2}\right) = \frac{1}{\theta}\ln(1+\rho) - \frac{1}{\theta}\ln\left(\frac{P_1}{P_2}\right).$$

Therefore,

$$\boxed{-\frac{\partial \ln(C_1/C_2)}{\partial \ln(P_1/P_2)} = \frac{1}{\theta}.}$$

Thus a larger θ means that relative consumption responds less to a relative-price change. The formulas also apply to the logarithmic limiting case $\theta = 1$.

Note Income versus substitution. Lifetime wealth W scales both consumption choices in (4); it does not affect C_1/C_2. Relative prices determine the intertemporal allocation, and $1/\theta$ measures the strength of that response.

Problem 2.3

(a) Suppose it is known in advance that at some time t_0 the government will confiscate half of whatever wealth each household holds at that time. Does consumption change discontinuously at time t_0? If so, why (and what is the condition relating consumption immediately before t_0 to consumption immediately after)? If not, why not?

(b) Suppose it is known in advance that at t_0 the government will confiscate from each household an amount of wealth equal to half of the wealth of the average household at that time. Does consumption change discontinuously at time t_0? If so, why (and what is the condition relating consumption immediately before t_0 to consumption immediately after)? If not, why not?

Solution

Write

$$C_{\text{before}} \equiv C(t_0^-), \qquad C_{\text{after}} \equiv C(t_0^+).$$

The distinction between the two policies is whether an additional dollar saved by one household changes the amount taken from that household.

(a) Choose a small number $\varepsilon > 0$, and compare the two dates $t_0 - \varepsilon$ and $t_0 + \varepsilon$. Let $\Delta > 0$ denote a small amount by which the household reduces its total consumption at $t_0 - \varepsilon$. The household invests this amount and consumes the proceeds at $t_0 + \varepsilon$.

Because half of the household's wealth is confiscated at t_0, only one-half of this marginal saving survives. Including discounting and the return earned over the short interval, optimality requires

$$e^{-\rho(t_0-\varepsilon)} u'(C(t_0 - \varepsilon))\,\Delta = \frac{1}{2} e^{-\rho(t_0+\varepsilon)} u'(C(t_0 + \varepsilon))\, e^{R(t_0+\varepsilon)-R(t_0-\varepsilon)} \Delta.$$

Here Δ is the small amount of consumption shifted between the two dates. ε measures how close the two comparison dates are to t_0, and $R(t) = \int_0^t r(v)\,\mathrm{d}v$.

As ε approaches zero, the two dates approach t_0 from opposite sides, and the discounting and interest factors over the vanishing interval approach one. Canceling

Δ then gives the jump condition

$$u'(C_{\text{before}}) = \frac{1}{2} u'(C_{\text{after}}). \tag{1}$$

Since u' is decreasing, (1) implies $C_{\text{after}} < C_{\text{before}}$: consumption jumps down at t_0. With CRRA utility, $u'(C) = C^{-\theta}$, so the size of the jump is

$$\boxed{\frac{C_{\text{after}}}{C_{\text{before}}} = 2^{-1/\theta} < 1.}$$

Anticipation raises consumption before the confiscation because saving across t_0 earns an exceptionally poor after-confiscation return.

(b) Each household now takes the confiscated amount, one-half of average wealth, as fixed. An extra dollar saved by this household does not raise its tax bill, so the marginal dollar survives in full. Repeating the same perturbation therefore gives

$$u'(C_{\text{before}}) = u'(C_{\text{after}}).$$

Strict concavity implies

$$\boxed{C_{\text{after}} = C_{\text{before}}.}$$

Consumption is continuous at t_0, even though the policy causes a discrete loss of wealth. The anticipated loss affects the entire planned consumption path, not the marginal return exactly at the confiscation date.

Note **Average versus own wealth.** In a symmetric equilibrium the two policies confiscate the same amount from each household. Their marginal incentives differ: under part (a), more own wealth raises confiscation; under part (b), one household treats average wealth as given.

Note **What anticipation rules out.** A known discrete wealth loss need not make consumption discontinuous. A jump is optimal only when the intertemporal marginal return itself has a discrete wedge, as in part (a).

Problem 2.4

Assume that the instantaneous utility function $u(C)$ in equation (2.2) is $\ln C$. Consider the problem of a household maximizing (2.2) subject to (2.7). Find an expression for C at each time as a function of initial wealth plus the present value of labor income, the path of $r(t)$, and the parameters of the utility function.

Solution

Let

$$R(t) \equiv \int_0^t r(v)\, dv$$

and use $\mathcal{W}$ for the household's lifetime resources (to avoid confusing this object with Romer's wage $W(t)$):

$$\mathcal{W} \equiv \frac{K(0)}{H} + \int_0^\infty e^{-R(t)} W(t) \frac{L(t)}{H} \, \mathrm{d}t. \tag{1}$$

Because utility is increasing in consumption, the present-value budget constraint holds with equality:

$$\int_0^\infty e^{-R(t)} C(t) \frac{L(t)}{H} \, \mathrm{d}t = \mathcal{W}.$$

Attach multiplier λ to this constraint. Differentiating the Lagrangian with respect to consumption at date t gives

$$e^{-\rho t} \frac{1}{C(t)} = \lambda e^{-R(t)}. \tag{2}$$

The common factor $L(t)/H$ cancels from the first-order condition. Equation (2) implies

$$C(t) = \lambda^{-1} e^{R(t) - \rho t}.$$

Use $L(t) = L(0)e^{nt}$ and substitute this path into the budget constraint:

$$\mathcal{W} = \lambda^{-1} \frac{L(0)}{H} \int_0^\infty e^{-(\rho - n)t} \, \mathrm{d}t = \lambda^{-1} \frac{L(0)/H}{\rho - n}.$$

The integral is finite because the log-utility case of Romer's parameter restriction is $\rho > n$. Solving for λ gives the complete consumption path:

$$\boxed{C(t) = C(0)e^{R(t) - \rho t}, \qquad C(0) = (\rho - n)\frac{\mathcal{W}}{L(0)/H}.}$$

Thus initial consumption is the marginal propensity $\rho - n$ times lifetime wealth per initial household member. The interest-rate path affects consumption after date 0 through $R(t)$, but it does not affect this initial propensity to consume.

NOTE **WHY n ENTERS.** $C(t)$ is consumption per person, whereas $\mathcal{W}$ is wealth per household. A household has $L(t)/H$ members, so population growth makes its total consumption grow faster than its per-person consumption and changes the propensity from ρ to $\rho - n$.

PROBLEM 2.5

Consider a household with utility given by (2.2)–(2.3). Assume that the real interest rate is constant, and let W denote the household's initial wealth plus the present value of its lifetime labor income (the right-hand side of [2.7]). Find the utility-maximizing path of C, given r, W, and the parameters of the utility function.

Solution

Here W is the household's initial wealth plus the present value of its lifetime labor income. Since the real interest rate is constant, the binding budget constraint is

$$\int_0^\infty e^{-rt} C(t) \frac{L(t)}{H} \, \mathrm{d}t = W.$$

The first-order condition for consumption at date t is

$$e^{-\rho t} C(t)^{-\theta} = \lambda e^{-rt}. \tag{1}$$

Taking logs of (1) and differentiating with respect to time gives the Euler equation. Integrating it gives the corresponding level path:

$$\frac{\dot{C}(t)}{C(t)} = \frac{r-\rho}{\theta}, \qquad C(t) = C(0) e^{[(r-\rho)/\theta]t}. \tag{2}$$

It remains to determine $C(0)$. Since $L(t) = L(0)e^{nt}$, substitution into the budget constraint gives

$$W = C(0) \frac{L(0)}{H} \int_0^\infty e^{-\{r-n-(r-\rho)/\theta\}t} \, \mathrm{d}t.$$

For a finite interior solution, the convergence term must be positive. Define

$$\eta \equiv r - n - \frac{r-\rho}{\theta} = \frac{\rho - r}{\theta} + r - n > 0. \tag{3}$$

The integral is $1/\eta$, so $C(0) = \eta W / [L(0)/H]$. Therefore the utility-maximizing path is

$$\boxed{C(t) = \frac{W}{L(0)/H} \left[\frac{\rho - r}{\theta} + r - n \right] e^{[(r-\rho)/\theta]t}.} \tag{4}$$

When $r > \rho$, consumption per person rises; when $r < \rho$, it falls. For $\theta = 1$, the coefficient becomes $\rho - n$, so (4) reduces to the constant-interest-rate case of Problem 2.4.

Problem 2.6 *Growth, saving, and $r - g$.*

Piketty (2014) argues that a fall in the growth rate of the economy is likely to lead to an increase in the difference between the real interest rate and the growth rate. This problem asks you to investigate this issue in the context of the Ramsey–Cass–Koopmans model. Specifically, consider a Ramsey–Cass–Koopmans economy that is on its balanced growth path, and suppose there is a permanent fall in g.

(a) How, if at all, does this affect the $\dot{k} = 0$ curve?

(b) How, if at all, does this affect the $\dot{c} = 0$ curve?

(c) At the time of the change, does c rise, fall, or stay the same, or is it not possible to tell?

(d) At the time of the change, does $r-g$ rise, fall, or stay the same, or is it not possible to tell?

(e) In the long run, does $r-g$ rise, fall, or stay the same, or is it not possible to tell?

(f) Find an expression for the impact of a marginal change in g on the fraction of output that is saved on the balanced growth path. Can one tell whether this expression is positive or negative?

(g) For the case where the production function is Cobb–Douglas, $f(k)=k^{\alpha}$, rewrite your answer to part (f) in terms of ρ, n, g, θ, and α.

Hint: Use the fact that $f'(k^*)=\rho+\theta g$.

Solution

Let g_{old} and g_{new} denote the growth rates before and after the permanent change, with $g_{\text{new}} < g_{\text{old}}$.

(a) Capital per unit of effective labor obeys

$$\dot{k}=f(k)-c-(n+g)k, \qquad \dot{k}=0 \iff c=f(k)-(n+g)k. \tag{1}$$

For every $k>0$, a fall in g reduces break-even investment $(n+g)k$. Hence the $\dot{k}=0$ locus shifts upward, by more at larger values of k. It remains anchored at the origin.

(b) The Euler equation for consumption per unit of effective labor is

$$\frac{\dot{c}}{c}=\frac{f'(k)-\rho-\theta g}{\theta}, \qquad \dot{c}=0 \iff f'(k^*)=\rho+\theta g. \tag{2}$$

A fall in g lowers the right-hand side of the steady-state condition. Since $f''(k)<0$, the marginal product is lower only at a larger capital intensity. Thus the vertical $\dot{c}=0$ locus shifts right: $k^*_{\text{new}} > k^*_{\text{old}}$.

(c) Capital intensity is a state variable, so it cannot jump: $k(t_0^+)=k(t_0^-)=k^*_{\text{old}}$. Consumption is a control variable and must jump to the new saddle path. The new saddle path may pass above, below, or exactly through the old balanced-growth point. Therefore the impact change in c is ambiguous: c may rise, fall, or remain unchanged at t_0.

After the impact, both k and c rise along the new saddle path. To verify that the new long-run value of c is higher, differentiate $c^* = f(k^*) - (n+g)k^*$ and use (2):

$$\begin{aligned}\frac{\partial c^*}{\partial g} &= \left[\rho - n - (1-\theta)g\right]\frac{\partial k^*}{\partial g} - k^* \\ &= \left[\rho - n - (1-\theta)g\right]\frac{\theta}{f''(k^*)} - k^* < 0.\end{aligned} \tag{3}$$

The sign uses Romer's maintained restriction $\rho - n - (1-\theta)g > 0$. Hence a fall in g raises both k^* and c^*, even though the initial jump in c is not determined.

(d) On impact, k and therefore $r = f'(k)$ do not change, while g falls. Consequently,

$$(r-g)_{t_0^+} - (r-g)_{t_0^-} = g_{\text{old}} - g_{\text{new}} > 0. \tag{4}$$

Thus $r - g$ rises immediately.

(e) On a balanced growth path, (2) implies $r^* = f'(k^*) = \rho + \theta g$. Therefore,

$$r^* - g = \rho + (\theta - 1)g, \qquad \frac{\partial (r^* - g)}{\partial g} = \theta - 1. \tag{5}$$

A fall in g raises long-run $r - g$ if $\theta < 1$, leaves it unchanged if $\theta = 1$, and lowers it if $\theta > 1$. The impact answer in part (d) is unambiguous because r cannot jump; the long-run answer is different because r adjusts as k changes.

(f) Let s^* denote the fraction of output saved on the balanced growth path. Since $\dot{k} = 0$, equation (1) gives

$$s^* \equiv \frac{f(k^*) - c^*}{f(k^*)} = \frac{(n+g)k^*}{f(k^*)}. \tag{6}$$

Differentiate the modified-golden-rule condition in (2):

$$f''(k^*)\frac{\partial k^*}{\partial g} = \theta \quad \Longrightarrow \quad \frac{\partial k^*}{\partial g} = \frac{\theta}{f''(k^*)} < 0. \tag{7}$$

Differentiating (6) and then using (7) yields

$$\frac{\partial s^*}{\partial g} = \frac{k^*}{f(k^*)} + (n+g)\frac{f(k^*) - k^* f'(k^*)}{[f(k^*)]^2}\,\frac{\theta}{f''(k^*)}. \tag{8}$$

The first term is positive. The second is negative because $f(k^*) - k^* f'(k^*) > 0$ and $f''(k^*) < 0$. Without additional restrictions on production and preferences, the sign is ambiguous.

(g) With $f(k) = k^\alpha$, the condition $\alpha (k^*)^{\alpha-1} = \rho + \theta g$ implies $(k^*)^{1-\alpha} = \alpha/(\rho + \theta g)$.

Therefore

$$s^* = \frac{(n+g)k^*}{(k^*)^\alpha} = \frac{\alpha(n+g)}{\rho + \theta g}. \tag{9}$$

Differentiating gives

$$\boxed{\frac{\partial s^*}{\partial g} = \frac{\alpha(\rho - \theta n)}{(\rho + \theta g)^2}.}$$

Thus a fall in g lowers the balanced-growth saving rate if $\rho > \theta n$, raises it if $\rho < \theta n$, and leaves it unchanged if $\rho = \theta n$.

The figure below summarizes the shifts in the two zero-motion loci, the resulting balanced-growth points, and local motion under the new system. The black point-arrows are phase-direction markers, not an impact jump in c. No impact jump is drawn because part (c) shows that its direction is not determined without more information about the new saddle path.

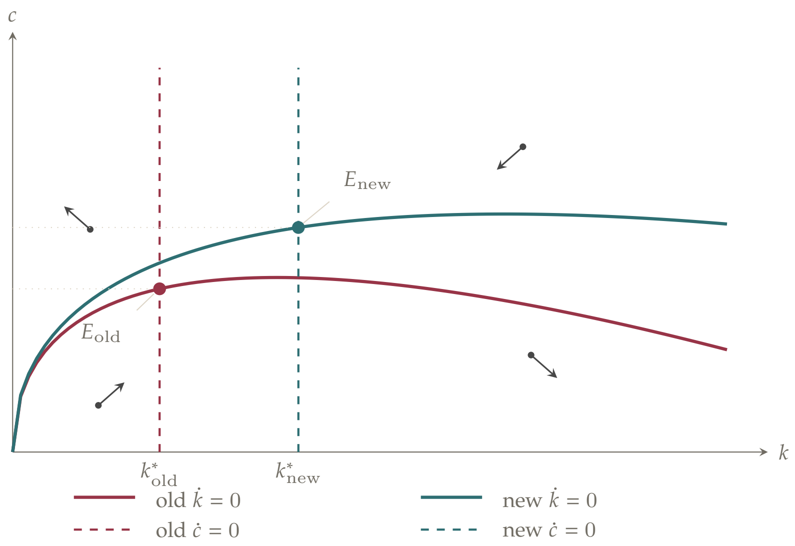

Figure 2.1. A permanent fall in g shifts the $\dot{k} = 0$ locus upward and the $\dot{c} = 0$ locus to the right. Both balanced-growth-path values k^* and c^* rise. Black point-arrows show motion under the new system in each phase region.

NOTE **IMPACT VERSUS LONG RUN.** At the instant of the shock, k and $r = f'(k)$ are predetermined, so $r - g$ must rise. In the long run, capital accumulation changes r, and the sign depends on $\theta - 1$.

NOTE **TWO OPPOSING SAVING-RATE EFFECTS.** A higher g directly raises the investment needed to keep k constant, but it also lowers k^*. These effects have opposite signs in (8), which is why the general answer is ambiguous.

NOTE **COBB–DOUGLAS SHORTCUT.** Do not solve explicitly for k^* before finding s^*. Combining $s^* = (n+g)(k^*)^{1-\alpha}$ with $\alpha(k^*)^{\alpha-1} = \rho + \theta g$ immediately eliminates k^*.

PROBLEM 2.7

Describe how each of the following affects the $\dot{c} = 0$ and $\dot{k} = 0$ curves in Figure 2.5, and thus how they affect the balanced-growth-path values of c and k:

(a) A rise in θ.

(b) A downward shift of the production function.

(c) A change in the rate of depreciation from the value of zero assumed in the text to some positive level.

SOLUTION

With zero depreciation, the two equations of motion are

$$\frac{\dot{c}}{c} = \frac{f'(k) - \rho - \theta g}{\theta}, \qquad \dot{c} = 0 \iff f'(k^*) = \rho + \theta g, \tag{1}$$

$$\dot{k} = f(k) - c - (n+g)k, \qquad \dot{k} = 0 \iff c = f(k) - (n+g)k. \tag{2}$$

Romer's parameter restriction implies $f'(k^*) - (n+g) = \rho - n - (1-\theta)g > 0$. Thus the balanced-growth point lies on the upward-sloping portion of the $\dot{k} = 0$ curve.

(a) Suppose $g > 0$. A rise in θ raises $\rho + \theta g$. Since $f''(k) < 0$, the higher marginal product required by (1) occurs at a lower k:

$$\frac{\partial k^*}{\partial \theta} = \frac{g}{f''(k^*)} < 0.$$

The vertical $\dot{c} = 0$ locus therefore shifts left, while the $\dot{k} = 0$ locus in (2) is unchanged. Because the latter locus is rising over the relevant range, both k^* and c^* fall. If $g = 0$, θ drops out of the zero-motion conditions, so neither balanced-growth value changes.

(b) Interpret a downward shift in the production function in the standard way used for this exercise: at each k, both $f(k)$ and $f'(k)$ are lower. The fall in $f(k)$ shifts the $\dot{k} = 0$ curve downward. The fall in $f'(k)$ means that the unchanged target $\rho + \theta g$ is reached at a lower k, so the $\dot{c} = 0$ locus shifts left. The new balanced-growth point has lower k^* and lower c^*.

(c) Let the depreciation rate be $\delta > 0$. The equations of motion become

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho - \theta g}{\theta}, \qquad \dot{c} = 0 \iff f'(k^*) = \rho + \theta g + \delta, \tag{3}$$

$$\dot{k} = f(k) - c - (n + g + \delta)k, \qquad \dot{k} = 0 \iff c = f(k) - (n + g + \delta)k. \tag{4}$$

The higher required marginal product shifts the $\dot{c} = 0$ locus left. Extra replacement investment, δk, shifts the $\dot{k} = 0$ curve downward, by more at larger k. For completeness, differentiating the two steady-state conditions gives

$$\frac{\partial k^*}{\partial \delta} = \frac{1}{f''(k^*)} < 0, \qquad \frac{\partial c^*}{\partial \delta} = \big[\rho - n - (1-\theta)g\big]\frac{1}{f''(k^*)} - k^* < 0.$$

Hence both balanced-growth values fall.

Figure 2.2 places the three comparative-static experiments side by side. The zero-motion loci and old and new balanced-growth points are accompanied by four local direction markers for the new system; the impact jump in consumption is not needed to answer this problem.

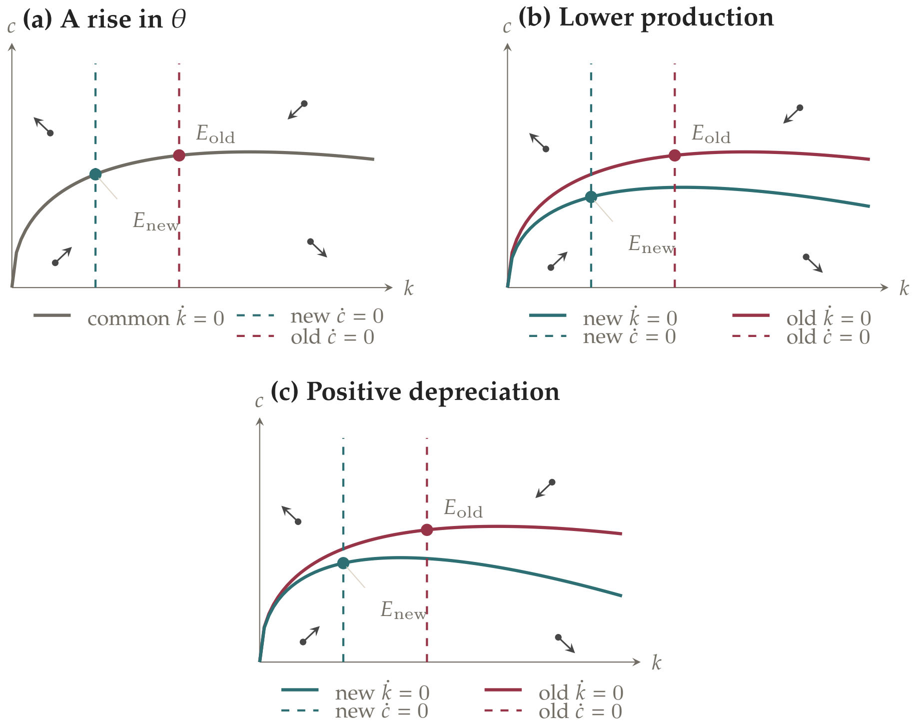

Figure 2.2. How the three changes in Problem 2.7 shift the zero-motion loci. In every panel, the new balanced-growth point lies below and to the left of the old one. Black point-arrows show motion under the new system in each phase region.

Note **The special case $g = 0$.** A larger θ matters for the balanced-growth point through the term θg. When technology is not growing, the change in θ affects transition dynamics but not the two zero-motion loci.

Note **What "downward shift" must mean.** A lower $f(k)$ shifts $\dot{k} = 0$ down, while a lower $f'(k)$ shifts $\dot{c} = 0$ left. A statement about output alone does not pin down the marginal-product shift; this exercise uses both changes.

PROBLEM 2.8

Derive an expression analogous to (2.40) for the case of a positive depreciation rate.

SOLUTION

Let $\delta > 0$. Around the balanced-growth point, define $\widetilde{c} = c - c^*$ and $\widetilde{k} = k - k^*$. The equations of motion and the steady-state condition are

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho - \theta g}{\theta}, \qquad \dot{k} = f(k) - c - (n + g + \delta)k, \qquad f'(k^*) = \rho + \theta g + \delta. \tag{1}$$

Linearizing (1) at (k^*, c^*) gives

$$\begin{pmatrix} \dot{\widetilde{c}} \\ \dot{\widetilde{k}} \end{pmatrix} \simeq \begin{pmatrix} 0 & f''(k^*)c^*/\theta \\ -1 & \rho - n - (1-\theta)g \end{pmatrix} \begin{pmatrix} \widetilde{c} \\ \widetilde{k} \end{pmatrix}. \tag{2}$$

Writing $\beta \equiv \rho - n - (1-\theta)g$, the characteristic polynomial of (2) has one negative root:

$$\mu_1 = \frac{1}{2}\left\{\beta - \left[\beta^2 - \frac{4f''(k^*)c^*}{\theta}\right]^{1/2}\right\} < 0. \tag{3}$$

For Cobb–Douglas production, $f(k) = k^\alpha$, use

$$\alpha(k^*)^{\alpha-1} = \rho + \theta g + \delta, \qquad c^* = (k^*)^\alpha - (n + g + \delta)k^*.$$

Substitution into (3) gives the expression analogous to Romer's equation (2.40):

$$\boxed{\mu_1 = \frac{1}{2}\left\{\beta - \left[\beta^2 + \frac{4}{\theta}\frac{1-\alpha}{\alpha}(\rho + \theta g + \delta)\big[\rho + \theta g + \delta - \alpha(n + g + \delta)\big]\right]^{1/2}\right\}.}$$

Setting $\delta = 0$ returns equation (2.40).

PROBLEM 2.9 *A closed-form solution of the Ramsey model.*

(Smith, 2006) Consider the Ramsey model with Cobb–Douglas production, $y(t) = k(t)^\alpha$, and with the coefficient of relative risk aversion (θ) and capital's share (α) assumed to be equal.

(a) What is k on the balanced growth path (k^*)?

(b) What is c on the balanced growth path (c^*)?

(c) Let $z(t)$ denote the capital-output ratio,

$$k(t)/y(t)$$

and $x(t)$ denote the consumption-capital ratio,

$$c(t)/k(t)$$

Find expressions for $\dot{z}(t)$ and $\dot{x}(t)/x(t)$ in terms of z, x, and the parameters of the model.

(d) Tentatively conjecture that x is constant along the saddle path. Given this conjecture:

(i) Find the path of z given its initial value, $z(0)$.

(ii) Find the path of y given the initial value of k, $k(0)$. Is the speed of convergence to the balanced growth path, $d \ln\left[y(t) - y^*\right]/dt$, constant as the economy moves along the saddle path?

(e) In the conjectured solution, are the equations of motion for c and k, (2.25) and (2.26), satisfied?

Solution

Here $f(k) = k^\alpha$ and $\theta = \alpha$. There is no depreciation, as in the chapter's baseline model.

(a) On the balanced growth path, $\dot{c} = 0$. Hence

$$\alpha(k^*)^{\alpha-1} = \rho + \alpha g \quad \Longrightarrow \quad \boxed{k^* = \left(\frac{\alpha}{\rho + \alpha g}\right)^{1/(1-\alpha)}.} \tag{1}$$

(b) The condition $\dot{k} = 0$ then gives

$$\boxed{\begin{aligned} c^* &= (k^*)^\alpha - (n+g)k^* \\ &= \left(\frac{\alpha}{\rho+\alpha g}\right)^{\alpha/(1-\alpha)} - (n+g)\left(\frac{\alpha}{\rho+\alpha g}\right)^{1/(1-\alpha)}. \end{aligned}} \tag{2}$$

(c) Since $y = k^\alpha$, the capital-output ratio satisfies $z = k/y = k^{1-\alpha}$. Therefore

$$\dot{z} = (1-\alpha)k^{-\alpha}\dot{k}.$$

Substituting $\dot{k} = k^\alpha - c - (n+g)k$, and using $ck^{-\alpha} = (c/k)(k/y) = xz$, yields

$$\boxed{\dot{z} = (1-\alpha)\left[1 - (x+n+g)z\right].} \tag{3}$$

Next, $x = c/k$, so

$$\frac{\dot{x}}{x} = \frac{\dot{c}}{c} - \frac{\dot{k}}{k}.$$

Under $\theta = \alpha$, the Euler equation and capital-accumulation equation imply

$$\frac{\dot{c}}{c} = \frac{\alpha k^{\alpha-1} - \rho - \alpha g}{\alpha} = \frac{1}{z} - \frac{\rho}{\alpha} - g, \qquad \frac{\dot{k}}{k} = \frac{1}{z} - x - n - g.$$

Subtracting cancels the two terms involving z:

$$\boxed{\frac{\dot{x}}{x} = x + n - \frac{\rho}{\alpha}.} \tag{4}$$

(d) If x is constant on the saddle path, then (4) requires

$$x(t) = x^* = \frac{\rho}{\alpha} - n. \tag{5}$$

This is positive under the model's parameter restriction.

(i) Substitute (5) into (3). Since

$$z^* = (k^*)^{1-\alpha} = \frac{\alpha}{\rho + \alpha g},$$

the resulting linear differential equation can be written as

$$\dot{z} = -(1-\alpha)\left(\frac{\rho}{\alpha} + g\right)(z - z^*).$$

Its solution for an arbitrary initial value is

$$\boxed{z(t) = z^* + [z(0) - z^*]\exp\left\{-(1-\alpha)\left(\frac{\rho}{\alpha} + g\right)t\right\}.} \tag{6}$$

(ii) Substituting (6), and using $z = k^{1-\alpha}$ and $y = k^\alpha = z^{\alpha/(1-\alpha)}$, the initial condition $z(0) = k(0)^{1-\alpha}$ gives

$$\boxed{y(t) = \left\{z^* + \left[k(0)^{1-\alpha} - z^*\right]\exp\left[-(1-\alpha)\left(\frac{\rho}{\alpha} + g\right)t\right]\right\}^{\alpha/(1-\alpha)}.} \tag{7}$$

To discuss convergence from either side, use the absolute output gap. Differentiating (7), or equivalently differentiating $y = z^{\alpha/(1-\alpha)}$, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln|y(t) - y^*| = -(\rho + \alpha g)\frac{z(t)^{(2\alpha-1)/(1-\alpha)}[z(t) - z^*]}{z(t)^{\alpha/(1-\alpha)} - (z^*)^{\alpha/(1-\alpha)}}. \tag{8}$$

The right-hand side of (8) depends on $z(t)$, so the speed is not constant in general. It converges to $-(1-\alpha)(\rho/\alpha + g)$ as the economy approaches the balanced growth path. The special case $\alpha = 1/2$ makes $y = z$, and only then is the gap's proportional rate constant along the entire saddle path.

(e) It remains to verify the conjecture, not merely use it. With $x = x^* = \rho/\alpha - n$, the capital-accumulation equation divided by k gives

$$\frac{\dot{k}}{k} = \frac{1}{z} - x^* - n - g = \frac{1}{z} - \frac{\rho}{\alpha} - g. \tag{9}$$

Since $c = x^* k$ and x^* is constant, $\dot{c}/c = \dot{k}/k$. The Euler equation with $\theta = \alpha$ gives exactly the last expression in (9). Thus both (2.25) and (2.26) are satisfied, and the conjectured path is the saddle path.

Note **Why the closed form is possible.** The restriction $\theta = \alpha$ makes the two $1/z$ terms cancel in (4). The consumption-capital ratio can therefore remain constant, reducing the two-dimensional system to one linear equation for z.

Problem 2.10 *Capital taxation in the Ramsey–Cass–Koopmans model.*

Consider a Ramsey–Cass–Koopmans economy that is on its balanced growth path. Suppose that at some time, which we will call time 0, the government switches to a policy of taxing investment income at rate τ. Thus the real interest rate that households face is now given by $r(t) = (1 - \tau) f'(k(t))$. Assume that the government returns the revenue it collects from this tax through lump-sum transfers. Finally, assume that this change in tax policy is unanticipated.

(a) How, if at all, does the tax affect the $\dot{c} = 0$ locus? The $\dot{k} = 0$ locus?

(b) How does the economy respond to the adoption of the tax at time 0? What are the dynamics after time 0?

(c) How do the values of c and k on the new balanced growth path compare with their values on the old balanced growth path?

(d) (Barro, Mankiw, and Sala-i-Martin, 1995) Suppose there are many economies like this one. Workers' preferences are the same in each country, but the tax rates on investment income may vary across countries. Assume that each country is on its balanced growth path.

  (i) Show that the saving rate on the balanced growth path, $\left(y^* - c^*\right)/y^*$, is decreasing in τ.

  (ii) Do citizens in low-τ, high-k^*, high-saving countries have any incentive to invest in low-saving countries? Why or why not?

(e) Does your answer to part (c) imply that a policy of *subsidizing investment* (that is, making $\tau < 0$), and raising the revenue for this subsidy through lump-sum taxes, increases welfare? Why or why not?

(f) How, if at all, do the answers to parts (a) and (b) change if the government does not rebate the revenue from the tax but instead uses it to make government purchases?

Solution

Let $\tau \in (0,1)$. When the revenue is returned through lump-sum transfers, the tax changes the return relevant for household saving but does not absorb any of the economy's output. The equations of motion are

$$\frac{\dot{c}}{c} = \frac{(1-\tau)f'(k) - \rho - \theta g}{\theta}, \tag{1}$$
$$\dot{k} = f(k) - c - (n+g)k. \tag{2}$$

(a) The post-tax $\dot{c} = 0$ condition is

$$(1-\tau)f'(k_\tau^*) = \rho + \theta g.$$

Thus the required pre-tax marginal product is $(\rho + \theta g)/(1-\tau)$, which exceeds its no-tax value. Since $f''(k) < 0$, $k_\tau^* < k_0^*$: the $\dot{c} = 0$ locus shifts left. Equation (2) is unchanged, so the $\dot{k} = 0$ locus remains $c = f(k) - (n+g)k$.

(b) At time 0, k cannot jump and remains at k_0^*. Consumption can jump because the policy was unanticipated. At the old point, the new saddle path lies above the unchanged $\dot{k} = 0$ curve, so c jumps upward. The lower after-tax return makes households substitute toward current consumption.

After the jump, the economy is to the right of the new $\dot{c} = 0$ locus and above the $\dot{k} = 0$ locus. Hence $\dot{c} < 0$ and $\dot{k} < 0$. Consumption and capital then fall together along the new saddle path toward the new balanced-growth point.

(c) We already know $k_\tau^* < k_0^*$. Both points lie on the unchanged $\dot{k} = 0$ curve. Its slope at the no-tax balanced-growth point is

$$f'(k_0^*) - (n+g) = \rho - n - (1-\theta)g > 0,$$

and its slope is even larger at lower k. Thus the relevant part of the curve is increasing, and $c_\tau^* < c_0^*$ as well.

(d) (i) Differentiate the post-tax balanced-growth condition:

$$-f'(k_\tau^*) + (1-\tau)f''(k_\tau^*)\frac{\partial k_\tau^*}{\partial \tau} = 0 \quad \Longrightarrow \quad \frac{\partial k_\tau^*}{\partial \tau} = \frac{f'(k_\tau^*)}{(1-\tau)f''(k_\tau^*)} < 0. \tag{3}$$

On a balanced growth path, saving equals break-even investment, so

$$s^*(\tau) \equiv \frac{f(k_\tau^*) - c_\tau^*}{f(k_\tau^*)} = \frac{(n+g)k_\tau^*}{f(k_\tau^*)}. \tag{4}$$

Define capital's pre-tax income share by $\alpha_K(k) \equiv kf'(k)/f(k)$. Differentiating (4) and using (3) gives

$$\boxed{\frac{\partial s^*}{\partial \tau} = \frac{n+g}{f(k^*_\tau)}\frac{\partial k^*_\tau}{\partial \tau}\left[1-\alpha_K(k^*_\tau)\right] < 0.} \tag{5}$$

The sign follows from $\partial k^*_\tau/\partial\tau < 0$ and $0 < \alpha_K < 1$.

(ii) No. Under the tax system in the problem, an investment in a country with tax rate τ earns the host country's after-tax return. Every country on its balanced growth path satisfies

$$(1-\tau)f'(k^*_\tau) = \rho + \theta g.$$

Thus low-saving, high-tax countries have a higher pre-tax marginal product, but the tax exactly offsets that advantage. After-tax returns are equal, so residents of low-tax countries have no return incentive to move their saving there.

(e) No. With lump-sum taxes available and no externality, the no-tax competitive allocation solves the representative household's social-planner problem. An investment subsidy raises long-run k^* and c^*, but reaching that point requires a short-run consumption sacrifice. At the optimum, the discounted utility cost of this distortion outweighs the later gain. A higher long-run consumption level by itself is not a welfare comparison.

(f) If the revenue finances government purchases that provide neither private utility nor productive capital, then purchases per unit of effective labor are $G(t) = \tau f'(k(t))k(t)$. The Euler equation (1) is unchanged, but capital accumulation becomes

$$\dot{k} = f(k) - c - G - (n+g)k = f(k) - c - \tau f'(k)k - (n+g)k. \tag{6}$$

Therefore the $\dot{c} = 0$ locus shifts left exactly as in part (a), while the $\dot{k} = 0$ locus now shifts downward. The new balanced-growth value of k is the same as with rebates because it is still pinned down by (1). Private consumption is lower than in the rebate case by $G^* = \tau f'(k^*_\tau)k^*_\tau$.

On impact, k again cannot jump, but the new saddle path may pass either above or below the old balanced-growth point. Thus c may jump up or down. After the jump, the economy follows that saddle path toward lower k and lower private c.

Note A TRANSFER IS NOT A RESOURCE COST. Tax revenue returned lump-sum changes household incentives but stays in private budgets. Government purchases absorb output, so only the latter policy shifts the $\dot{k} = 0$ locus.

NOTE **STATE AND CONTROL VARIABLES.** Capital k is inherited from the past and cannot jump. Consumption c is a control variable and can jump at an unanticipated policy change so that the economy lands on the relevant saddle path.

NOTE **MORE CAPITAL IS NOT AUTOMATICALLY BETTER.** The golden rule maximizes sustainable consumption, not discounted lifetime utility. The no-tax Ramsey allocation already balances current sacrifice against future consumption according to household preferences.

Figure 2.3 distinguishes the resource-neutral rebate case from the case in which tax revenue is absorbed by government purchases.

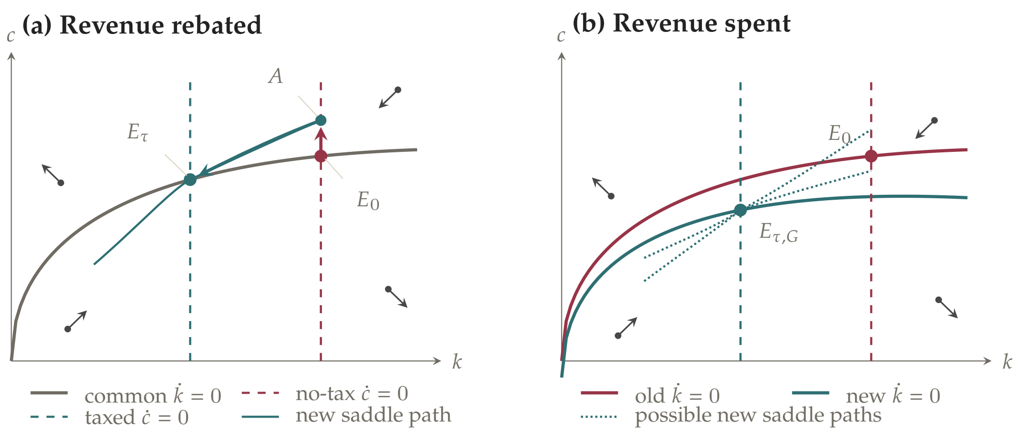

Figure 2.3. A capital-income tax shifts the $\dot{c} = 0$ locus left. With rebates, the $\dot{k} = 0$ locus is unchanged and consumption jumps up. With government purchases, the $\dot{k} = 0$ locus also shifts down, so the direction of the initial consumption jump is ambiguous. Black point-arrows show motion under the post-policy system.

PROBLEM 2.11 *Using the phase diagram to analyze the impact of an anticipated change.*

Consider the policy described in Problem 2.10, but suppose that instead of announcing and implementing the tax at time 0, the government announces at time 0 that at some later time, time t_1, investment income will begin to be taxed at rate τ.

(a) Draw the phase diagram showing the dynamics of c and k after time t_1.

(b) Can c change discontinuously at time t_1? Why or why not?

(c) Draw the phase diagram showing the dynamics of c and k before t_1.

(d) In light of your answers to parts (a), (b), and (c), what must c do at time 0?

(e) Summarize your results by sketching the paths of c and k as functions of time.

Solution

Let $t_0 = 0$ denote the announcement date. Before t_1, the original Euler equation applies; from t_1 onward, the taxed return applies:

$$\frac{\dot{c}}{c} = \begin{cases} \dfrac{f'(k) - \rho - \theta g}{\theta}, & 0 \le t < t_1, \\ \dfrac{(1-\tau)f'(k) - \rho - \theta g}{\theta}, & t \ge t_1. \end{cases} \tag{1}$$

Because the tax revenue is rebated, capital accumulation is governed in both regimes by

$$\dot{k} = f(k) - c - (n+g)k. \tag{2}$$

(a) After t_1, the $\dot{c} = 0$ locus is determined by $(1-\tau)f'(k) = \rho + \theta g$, so it lies to the left of the no-tax locus. The $\dot{k} = 0$ curve is unchanged. The post-tax saddle path leads to a new balanced-growth point with lower k and lower c, exactly as in Problem 2.10.

(b) Consumption cannot jump at t_1. The tax change and its date are known in advance, and there is no instantaneous confiscation of wealth at that date. A jump would create a predictable discontinuity in marginal utility that households could profitably smooth. Hence

$$c(t_1^-) = c(t_1^+). \tag{3}$$

The growth rate $\dot{c}/c$, however, can change discontinuously because the after-tax return changes at t_1.

(c) During $[0, t_1)$, the no-tax part of (1) and the unchanged capital law (2) govern motion. Thus the old, right-hand $\dot{c} = 0$ line is relevant before implementation, even though the tax has already been announced.

(d) At t_1, the economy must be on the post-tax saddle path; by (3), it must arrive there without a jump in c. This terminal requirement pins down the response at the announcement date. Consumption jumps upward at time 0 to a point such as A in Figure 2.4. Capital remains at its old value.

At A, consumption is above the $\dot{k} = 0$ curve, so k begins to fall. Once k moves left of the old $\dot{c} = 0$ line, consumption rises. The economy therefore moves northwest to B, reaching the post-tax saddle path exactly at t_1. When the tax begins, consumption is continuous but its direction reverses: from B, both c and k fall along the new saddle path toward E_τ.

(e) Thus c jumps up at time 0, rises further until t_1, and then falls toward its lower post-tax balanced-growth value. Capital does not jump; it begins falling after the announcement and continues falling toward its new balanced-growth value. The level path of c has a kink at t_1, whereas k and $\dot{k}$ are continuous there because the resource equation is unchanged.

Figure 2.4 combines the pre- and post-implementation phase diagrams. The associated time paths are separated out in Figure 2.5.

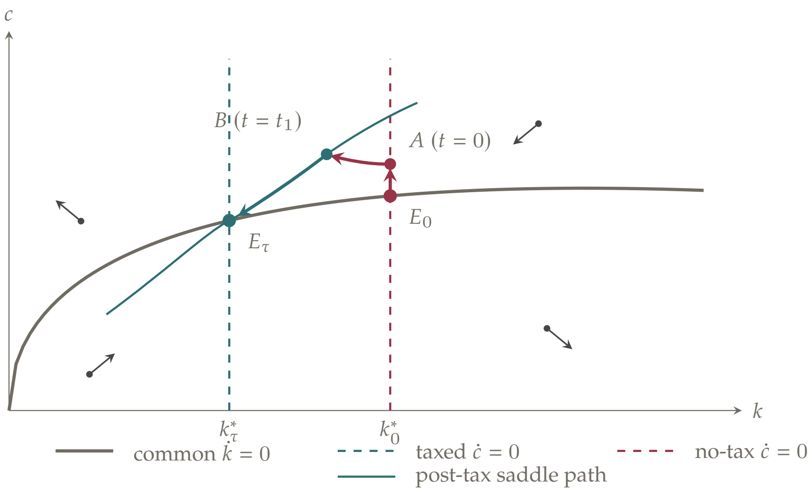

Figure 2.4. An anticipated permanent capital-income tax. Consumption jumps at the announcement date to A, follows the no-tax dynamics to B, and is continuous when the tax begins. Thereafter the economy follows the post-tax saddle path to E_τ. Black point-arrows show post-tax motion in each phase region.

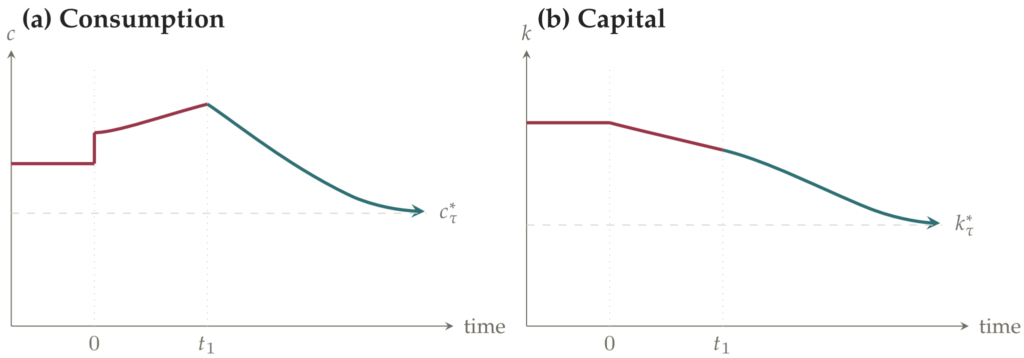

Figure 2.5. Time paths following announcement of the future permanent tax. Consumption jumps at time 0, leaves the impact point with zero slope, and has a kink at t_1. Capital has a kink at time 0 but is smooth at t_1; both paths flatten as they approach the new balanced-growth values.

Note **Announcement dates versus implementation dates.** A forward-looking control variable can jump when genuinely new information arrives at time 0. It cannot jump at the already anticipated implementation date t_1, although its growth rate can change there.

PROBLEM 2.12 *Using the phase diagram to analyze the impact of unanticipated and anticipated temporary changes.*

Analyze the following two variations on Problem 2.11:

(a) At time 0, the government announces that it will tax investment income at rate τ from time 0 until some later date t_1; thereafter investment income will again be untaxed.

(b) At time 0, the government announces that from time t_1 to some later time t_2, it will tax investment income at rate τ; before t_1 and after t_2, investment income will not be taxed.

SOLUTION

In either variation, the Euler equation depends on whether the capital-income tax is currently in force:

$$\frac{\dot{c}}{c} = \begin{cases} \dfrac{f'(k) - \rho - \theta g}{\theta}, & \text{while income is untaxed,} \\ \dfrac{(1-\tau)f'(k) - \rho - \theta g}{\theta}, & \text{while the tax is in force.} \end{cases} \tag{1}$$

Since the revenue is rebated, the resource equation is always

$$\dot{k} = f(k) - c - (n+g)k. \tag{2}$$

Thus the $\dot{k} = 0$ locus never moves. The $\dot{c} = 0$ locus shifts left while the tax is in force and returns to its original position when the tax ends.

(a) The adoption of the tax at time 0 is unanticipated, so consumption can jump then. Its removal at t_1 is known from time 0, so $c(t_1^-) = c(t_1^+)$. Moreover, at t_1 the economy must lie on the original, no-tax saddle path; otherwise the no-tax dynamics after t_1 would not return it to the original balanced-growth point.

These requirements determine the initial jump. Consumption jumps upward from E to A. To see why the jump cannot be downward, suppose instead that the economy remained at or below E. Under the taxed branch of (1) it would have $\dot{c} < 0$, and being below the $\dot{k} = 0$ curve in (2) would give $\dot{k} > 0$; the economy would move southeast, away from the original saddle path.

From A, both c and k initially fall. The economy eventually crosses the $\dot{k} = 0$ curve at Q; thereafter c continues to fall but k begins to rise. The impact point A is the unique one that makes the economy reach point B on the original saddle path exactly at t_1. When the tax is removed, consumption remains continuous and both variables rise from B back to E.

(b) Consumption cannot jump at either t_1 or t_2, because both dates and both policy changes are announced at time 0. It can jump at time 0, when the new information first arrives. The terminal condition is now that the economy be on the original saddle path at t_2.

Consumption jumps upward from E to A at time 0, while capital remains at k_0^*. Until t_1, the no-tax equations apply. At A, $\dot{c} = 0$ initially and $\dot{k} < 0$; after k begins to fall, the economy is left of the no-tax $\dot{c} = 0$ line, so consumption rises. It moves northwest to B.

At t_1, the tax begins and the relevant $\dot{c} = 0$ line shifts left. Consumption is continuous, but $\dot{c}$ switches from positive to negative. Capital continues to fall because the economy is still above $\dot{k} = 0$. After crossing that curve at Q, capital begins to rise while consumption continues to fall. The initial point A is chosen so that this taxed trajectory reaches C, on the original saddle path, exactly at t_2. The tax then ends without a jump in consumption, and the economy moves northeast from C back to E.

Figure 2.6 shows the complete shooting paths for the two temporary-tax experiments; the labels B and C mark the points that must lie on the original saddle path when the tax is removed.

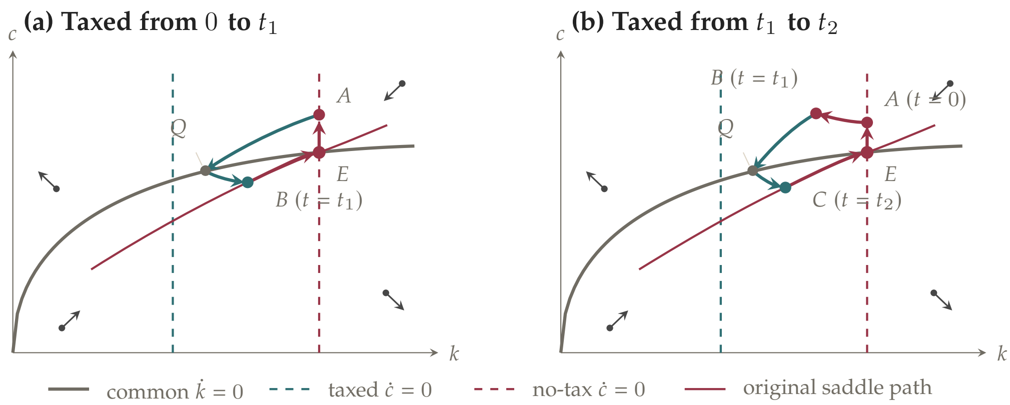

Figure 2.6. Temporary capital-income taxes. In panel (a), implementation at time 0 is unanticipated but removal at t_1 is anticipated. In panel (b), both implementation and removal are anticipated. Consumption jumps only at the announcement date, and the economy must reach the original saddle path when the temporary tax ends. Black point-arrows show motion while the tax is in force.

Note **Where consumption may jump.** The determining event is the arrival of new information, not the moment a tax rate mechanically changes. Consumption may jump at time 0 in both cases, but it is continuous at every previously announced policy date.

NOTE THE TERMINAL SADDLE-PATH CONDITION. Work backward from the date the temporary tax ends. The economy must be on the original saddle path then; this requirement selects the unique impact value of consumption at the announcement date.

PROBLEM 2.13 *An interesting situation in the Ramsey–Cass–Koopmans model.*

(a) Consider the Ramsey–Cass–Koopmans model where k at time 0 (which–as always–the model takes as given) is at the golden-rule level: $k(0) = k^{GR}$. Sketch the paths of c and k.

(b) Consider the same initial situation as in part (a), but in the version of the model that includes government purchases. Assume that G is constant and equal $\bar{G}$. Crucially, $\bar{G}$ is strictly less than $f\left(k^{GR}\right) - \left(n+g\right)k^{GR}$ and strictly greater than $f\left(k^{*}\right) - \left(n+g\right)k^{*}$ (where k^{*} is the level of k on the balanced growth path the economy would have if G were constant and equal to 0). Sketch the paths of c and k.

Hint: This problem is hard but can be answered by thinking things through slowly and carefully.

SOLUTION

Write net output after break-even investment as $h(k) \equiv f(k) - (n+g)k$. With government purchases per unit of effective labor equal to G, the two equations of motion are

$$\frac{\dot{c}}{c} = \frac{f'(k) - \rho - \theta g}{\theta}, \qquad \dot{k} = h(k) - c - G. \tag{1}$$

The Ramsey balanced-growth value and the golden-rule value satisfy

$$f'(k^{*}) = \rho + \theta g, \qquad f'(k^{GR}) = n + g. \tag{2}$$

The chapter assumes $\rho - n - (1-\theta)g > 0$, or $\rho + \theta g > n + g$. Since f' is decreasing,

$$k^{*} < k^{GR}. \tag{3}$$

(a) Here $G = 0$. Capital is predetermined, so the economy begins at $k(0) = k^{GR}$; consumption jumps to the unique value $c(0)$ that places the economy on the stable saddle path.

By (3), the initial capital stock lies to the right of the $\dot{c} = 0$ locus. Hence $\dot{c} < 0$. On the stable arm to the right of k^{*}, consumption is above the $\dot{k} = 0$ curve $c = h(k)$, so $\dot{k} < 0$. Thus both variables fall monotonically:

$$k(t) \downarrow k^{*}, \qquad c(t) \downarrow c^{*} = h(k^{*}).$$

The convergence is asymptotic; neither variable jumps after time 0.

(b) Now set $G = \bar{G}$. The $\dot{c} = 0$ locus is unchanged, while the $\dot{k} = 0$ curve becomes

$$c = h(k) - \bar{G}. \tag{4}$$

The assumptions in the question say

$$h(k^*) - \bar{G} < 0 \quad \text{and} \quad h(k^{GR}) - \bar{G} > 0. \tag{5}$$

Moreover, $h'(k) = f'(k) - (n+g) > 0$ between k^* and k^{GR}. Therefore there is a unique value $\widehat{k} \in (k^*, k^{GR})$ such that

$$h(\widehat{k}) = \bar{G}. \tag{6}$$

No balanced-growth point with positive consumption exists: the unchanged $\dot{c} = 0$ line is at k^*, but (5) would require negative consumption there. The feasible limiting point is instead the boundary $(\widehat{k}, 0)$.

Starting from k^{GR}, consumption again jumps to the unique path leading to that boundary point. Along this path $k > \widehat{k} > k^*$, so $\dot{c} < 0$; the path lies above (4), so $\dot{k} < 0$. Consequently,

$$k(t) \downarrow \widehat{k}, \qquad c(t) \downarrow 0.$$

Holding k fixed at k^{GR} is not an optimum: at any point on the $\dot{k} = 0$ curve to the right of k^*, $\dot{c} < 0$, so households wish to tilt consumption toward the present rather than keep it constant.

Figure 2.7 locates the initial capital stock and the two limiting points in the phase diagram. Figure 2.8 then displays the corresponding time paths, so its panels should be read as paths over time rather than as additional phase regions.

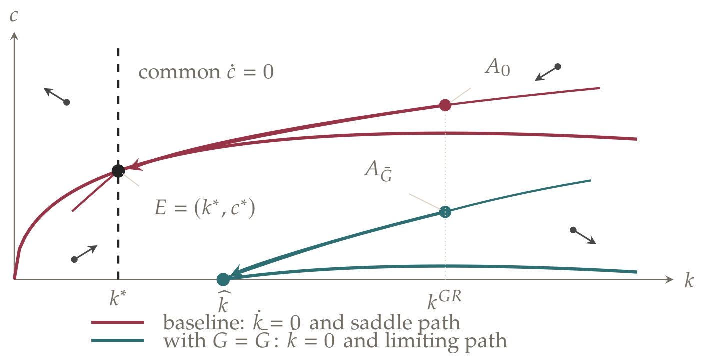

Figure 2.7. Problem 2.13 in the phase diagram. The golden-rule stock is to the right of the Ramsey balanced-growth stock. Constant government purchases shift the $\dot{k} = 0$ curve down so far that it lies below the horizontal axis at k^*; the feasible limiting point is therefore $E_{\bar{G}} = (\widehat{k}, 0)$. Black point-arrows mark the direction of motion in each of the four baseline phase regions.

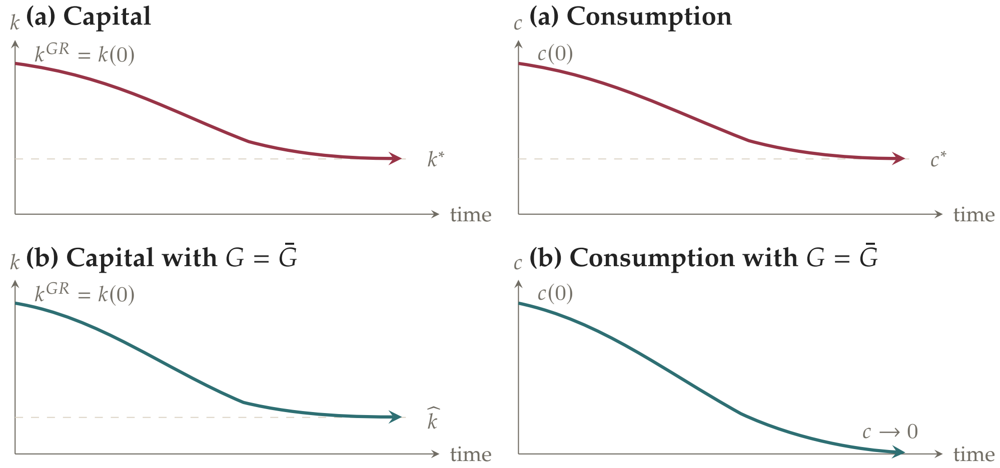

Figure 2.8. Time paths in Problem 2.13. Without government purchases, capital and consumption converge to positive balanced-growth values. Under the purchases specified in part (b), both decline, but consumption approaches zero while capital approaches $\widehat{k} > k^*$.

NOTE **WHY THE GOLDEN RULE IS NOT THE RAMSEY STEADY STATE.** The golden rule maximizes sustainable consumption and uses $f'(k^{GR}) = n + g$. The Ramsey household discounts the future, so its balanced-growth condition is $f'(k^*) = \rho + \theta g$, which places k^* below k^{GR} under the chapter's parameter restriction.

NOTE **A BOUNDARY LIMIT, NOT A POSITIVE-CONSUMPTION STEADY STATE.** In part (b), the two zero-motion loci do not meet at positive consumption. The economy approaches $(\widehat{k}, 0)$ only asymptotically; calling k^* the new steady state would incorrectly require negative consumption.

PROBLEM 2.14

Consider the Diamond model with logarithmic utility and Cobb–Douglas production. Describe how each of the following affects k_{t+1} as a function of k_t:

(a) A rise in n.

(b) A downward shift of the production function (that is, $f(k)$ takes the form Bk^α, and B falls).

(c) A rise in α.

SOLUTION

For logarithmic utility, the young save the fraction $1/(2+\rho)$ of their labor income. If $f(k) = Bk^\alpha$, the real wage per unit of effective labor is

$$w_t = f(k_t) - k_t f'(k_t) = (1-\alpha)Bk_t^\alpha.$$

Since the saving of the L_t young finances K_{t+1}, the transition equation is

$$k_{t+1} = \frac{(1-\alpha)B}{(1+n)(1+g)(2+\rho)} k_t^{\alpha}. \tag{1}$$

(a) A rise in n increases the denominator of (1). Thus k_{t+1} is lower at every positive k_t: the entire transition curve shifts down. The same saving is spread over a larger next generation of workers.

(b) A fall in B also shifts the transition curve down proportionally. At a given k_t, productivity, the wage, and therefore the saving of the young are all lower.

(c) A rise in α changes both labor's share $1-\alpha$ and the term k_t^{α}, so the sign need not be the same at all values of k_t. Differentiating (1) gives

$$\frac{\partial k_{t+1}}{\partial \alpha} = \frac{Bk_t^{\alpha}}{(1+n)(1+g)(2+\rho)} \big[(1-\alpha)\ln k_t - 1\big]. \tag{2}$$

Hence the transition curve moves down where $\ln k_t < 1/(1-\alpha)$, moves up where $\ln k_t > 1/(1-\alpha)$, and is unchanged at

$$k_t = \exp\!\left(\frac{1}{1-\alpha}\right).$$

Economically, a larger α raises the output elasticity of capital but lowers labor's income share. Which force dominates depends on the level and units of k_t.

PROBLEM 2.15 *A discrete-time version of the Solow model.*

Suppose $Y_t = F(K_t, A_t L_t)$, with $F(\cdot)$ having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$, and $K_{t+1} = K_t + sY_t - \delta K_t$.

(a) Find an expression for k_{t+1} as a function of k_t.

(b) Sketch k_{t+1} as a function of k_t. Does the economy have a balanced growth path? If the initial level of k differs from the value on the balanced growth path, does the economy converge to the balanced growth path?

(c) Find an expression for consumption per unit of effective labor on the balanced growth path as a function of the balanced-growth-path value of k. What is the marginal product of capital, $f'(k)$, when k maximizes consumption per unit of effective labor on the balanced growth path?

(d) Assume that the production function is Cobb–Douglas.

(i) What is k_{t+1} as a function of k_t?

(ii) What is k^*, the value of k on the balanced growth path?

(iii) Along the lines of equations (2.65)–(2.67), in the text, linearize the expression in subpart (i) around $k_t = k^*$, and find the rate of convergence of k to k^*.

Solution

(a) Next period's effective labor is $A_{t+1}L_{t+1} = (1+g)(1+n)A_t L_t$. Dividing the capital-accumulation equation by this quantity gives

$$k_{t+1} = \frac{(1-\delta)k_t + sf(k_t)}{(1+n)(1+g)}. \tag{1}$$

This is the discrete-time counterpart of the Solow equation of motion. Notice that exact discrete-time break-even investment contains the product ng, which disappears only under a continuous-time approximation.

(b) Let the right-hand side of (1) be $T(k_t)$. Diminishing returns and the Inada conditions imply

$$T'(k) > 0, \qquad T''(k) < 0, \qquad \lim_{k\to 0} T'(k) = \infty, \qquad \lim_{k\to\infty} T'(k) = \frac{1-\delta}{(1+n)(1+g)} < 1.$$

Thus $T(k)$ crosses the $45°$ line once at a strictly positive value k^*. The fixed-point condition is

$$sf(k^*) = (n+g+ng+\delta)k^*. \tag{2}$$

If $0 < k_t < k^*$, then $k_t < T(k_t) < k^*$; if $k_t > k^*$, then $k^* < T(k_t) < k_t$. Capital therefore converges monotonically to k^* from either side. Once k is constant, all variables per unit of effective labor are constant and the economy is on a balanced growth path.

(c) On a balanced growth path, consumption per unit of effective labor is $c^* = (1-s)f(k^*)$. Solving (2) for s and substituting gives

$$c^* = f(k^*) - (n+g+ng+\delta)k^*. \tag{3}$$

Maximizing the right-hand side of (3) with respect to the balanced-growth value of capital yields

$$f'(k^{GR}) = n+g+ng+\delta. \tag{4}$$

(d) Let $f(k) = k^\alpha$, with $0 < \alpha < 1$.

(i) Equation (1) becomes

$$k_{t+1} = \frac{(1-\delta)k_t + s k_t^{\alpha}}{(1+n)(1+g)}. \tag{5}$$

(ii) Applying the fixed-point condition (2) to Cobb–Douglas production gives

$$(k^*)^{1-\alpha} = \frac{s}{n+g+ng+\delta}.$$

Therefore

$$\boxed{k^* = \left(\frac{s}{n+g+ng+\delta}\right)^{1/(1-\alpha)}.} \tag{6}$$

(iii) A first-order Taylor expansion of (5) around $k_t = k^*$ is

$$k_{t+1} - k^* \simeq \lambda\,(k_t - k^*), \qquad \lambda \equiv \left.\frac{\partial k_{t+1}}{\partial k_t}\right|_{k_t = k^*}. \tag{7}$$

Differentiating (5) and using $s(k^*)^{\alpha-1} = n+g+ng+\delta$ yields

$$\lambda = \alpha + \frac{(1-\alpha)(1-\delta)}{(1+n)(1+g)}, \qquad 0 < \lambda < 1. \tag{8}$$

Iterating (7) gives

$$k_t - k^* \simeq \lambda^t (k_0 - k^*).$$

Hence the fraction of the remaining gap closed each period is

$$1 - \lambda = \frac{(1-\alpha)(n+g+ng+\delta)}{(1+n)(1+g)}. \tag{9}$$

Equivalently, the exact log rate associated with the linearized path is $-\ln\lambda$ per period.

Figure 2.9 illustrates the fixed point and the monotone convergence result in part (b); the cobweb arrows convert a value on the vertical axis into the next period's value on the horizontal axis.

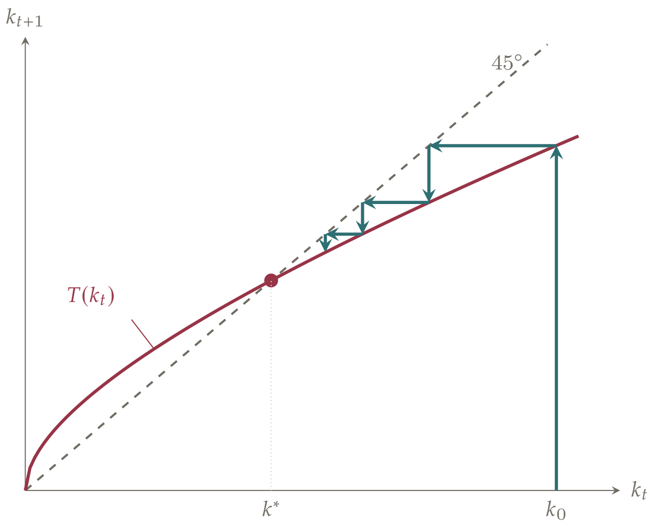

Figure 2.9. The discrete-time Solow transition map. It is increasing and strictly concave, crosses the 45° line once at k^*, and maps any positive initial capital stock monotonically toward that fixed point.

NOTE **DISCRETE TIME HAS AN ng TERM.** Since effective labor grows by the exact factor $(1+n)(1+g) = 1+n+g+ng$, break-even investment is $n+g+ng+\delta$, not $n+g+\delta$.

NOTE **TWO WAYS TO REPORT CONVERGENCE.** $1-\lambda$ is the fraction of the remaining gap closed in one period. The corresponding continuously compounded rate is $-\ln\lambda$; these are close only when convergence per period is slow.

PROBLEM 2.16 *Depreciation in the Diamond model and microeconomic foundations for the Solow model.*

Suppose that in the Diamond model capital depreciates at rate δ, so that $r_t = f'(k_t) - \delta$.

(a) How, if at all, does this change in the model affect equation (2.60) giving k_{t+1} as a function of k_t?

(b) In the special case of logarithmic utility, Cobb–Douglas production, and $\delta = 1$, what is the equation for k_{t+1} as a function of k_t? Compare this with the analogous expression for the discrete-time version of the Solow model with $\delta = 1$ from part (a) of Problem 2.15.

SOLUTION

Let $s(r)$ denote the fraction of labor income that a young household saves when the net real interest rate is r.

(a) The household's intertemporal problem and therefore the function $s(r)$ are unchanged. Depreciation changes the return that must be inserted into that function.

Since the young at t earn $A_t[f(k_t) - k_t f'(k_t)]$, equation (2.60) becomes

$$k_{t+1} = \frac{1}{(1+n)(1+g)} s\left(f'(k_{t+1}) - \delta\right) \left[f(k_t) - k_t f'(k_t)\right]. \tag{1}$$

Compared with (2.60), $s(f'(k_{t+1}))$ is replaced by $s(f'(k_{t+1}) - \delta)$. Thus the mapping generally changes, and whether it moves up or down depends on how saving responds to the interest rate. For CRRA utility, saving rises with the interest rate when $\theta < 1$, falls when $\theta > 1$, and is independent of it when $\theta = 1$.

(b) With logarithmic utility, $s(r) = 1/(2+\rho)$, so depreciation no longer affects the fraction of wage income saved. Cobb–Douglas production gives $f(k) - kf'(k) = (1-\alpha)k^{\alpha}$. Hence, when $\delta = 1$,

$$k_{t+1} = \frac{1-\alpha}{(1+n)(1+g)(2+\rho)} k_t^{\alpha}. \tag{2}$$

In the discrete-time Solow model with full depreciation, Problem 2.15 gives

$$k_{t+1} = \frac{\widetilde{s}}{(1+n)(1+g)} k_t^{\alpha},$$

where $\widetilde{s}$ is saving as a fraction of total output. The two transition equations coincide when

$$\widetilde{s} = \frac{1-\alpha}{2+\rho}. \tag{3}$$

The factor $1-\alpha$ appears because Diamond households save a fraction $1/(2+\rho)$ of labor income, and labor income is the share $1-\alpha$ of Cobb–Douglas output. With $\delta = 1$, no undepreciated capital remains for the old to sell, so this is also the economy's aggregate saving rate.

Note **Two different saving rates.** $s(r)$ is the young household's saving as a share of its wage income; $\widetilde{s}$ is aggregate saving as a share of total output. They coincide only after accounting for labor's income share.

Problem 2.17 *Social security in the Diamond model.*

Consider a Diamond economy where g is zero, production is Cobb–Douglas, and utility is logarithmic.

(a) **Pay-as-you-go social security.** Suppose the government taxes each young individual an amount T and uses the proceeds to pay benefits to old individuals; thus each old person receives $(1+n)T$.

(i) How, if at all, does this change affect equation (2.61) giving k_{t+1} as a function of k_t?

(ii) How, if at all, does this change affect the balanced-growth-path value of k?

(iii) If the economy is initially on a balanced growth path that is dynamically efficient, how does a marginal increase in T affect the welfare of current and future generations? What happens if the initial balanced growth path is dynamically inefficient?

(b) **Fully funded social security.** Suppose the government taxes each young person an amount T and uses the proceeds to purchase capital. Individuals born at t therefore receive $(1 + r_{t+1})T$ when they are old.

(i) How, if at all, does this change affect equation (2.61) giving k_{t+1} as a function of k_t?

(ii) How, if at all, does this change affect the balanced-growth-path value of k?

Solution

Because $g = 0$, A is constant. With Cobb–Douglas production,

$$w_t = (1 - \alpha)k_t^{\alpha}, \qquad r_{t+1} = \alpha k_{t+1}^{\alpha-1}.$$

(a) (i) Let S_t be the young person's private saving. Under pay-as-you-go social security,

$$C_{1t} + S_t = Aw_t - T, \qquad C_{2,t+1} = (1 + r_{t+1})S_t + (1 + n)T. \tag{1}$$

Eliminating S_t from (1) gives the present-value budget constraint

$$C_{1t} + \frac{C_{2,t+1}}{1 + r_{t+1}} = Aw_t - \frac{r_{t+1} - n}{1 + r_{t+1}}T. \tag{2}$$

The last term compares the market return $1 + r_{t+1}$ with the implicit pay-as-you-go return $1 + n$.

With logarithmic utility, first-period consumption is the fraction $(1+\rho)/(2+\rho)$ of the lifetime wealth on the right of (2). Substituting that choice back into the first constraint in (1) yields

$$S_t = \frac{Aw_t}{2 + \rho} - Z(r_{t+1})T,$$
$$Z(r) \equiv 1 - \frac{1 + \rho}{2 + \rho}\frac{r - n}{1 + r} = \frac{1 + r + (1 + \rho)(1 + n)}{(2 + \rho)(1 + r)} > 0. \tag{3}$$

Since $K_{t+1} = L_t S_t$, division by $AL_{t+1} = A(1 + n)L_t$ gives the new transition equation:

$$k_{t+1} = \frac{1}{1+n}\left[\frac{1-\alpha}{2+\rho}k_t^{\alpha} - Z(r_{t+1})\frac{T}{A}\right], \qquad r_{t+1} = \alpha k_{t+1}^{\alpha-1}. \tag{4}$$

Relative to equation (2.61), pay-as-you-go benefits crowd out private saving. The equation is now implicit because $Z(r_{t+1})$ depends on k_{t+1}.

(ii) Since $Z(r) > 0$, the social-security term in (4) lowers k_{t+1} at a given k_t. On the standard unique-equilibrium branch, the transition curve shifts down and its positive fixed point falls. Thus the balanced-growth value of k is lower.

(iii) Consider a small permanent increase $\mathrm{d}T > 0$. The old alive when the policy changes pay no new tax and receive $(1+n)\mathrm{d}T$, so they gain.

If the initial balanced growth path is dynamically efficient, $r^* > n$. The pay-as-you-go component offers the currently young a lower return than private saving, as (2) shows. Their full transitional welfare change also includes the rise in r_{t+1} caused by lower saving, so its exact size is not pinned down by the return comparison alone. The reduction in the eventual k^*, however, moves the economy farther below the golden-rule stock, so future generations on the new balanced growth path are worse off.

If the initial path is dynamically inefficient, $r^* < n$. The implicit pay-as-you-go return then exceeds the market return, and the fall in k^* reduces overaccumulation. A sufficiently small increase in T, one that does not push capital past the golden-rule level, can therefore implement a Pareto improvement: the initial old gain immediately, and the economy's future consumption possibilities rise.

(b) (i) Under a fully funded system, the two within-generation constraints are

$$C_{1t} + S_t = Aw_t - T, \qquad C_{2,t+1} = (1+r_{t+1})(S_t + T).$$

Eliminating private saving gives

$$C_{1t} + \frac{C_{2,t+1}}{1+r_{t+1}} = Aw_t. \tag{5}$$

This is exactly the budget constraint without social security. Therefore desired total saving remains $Aw_t/(2+\rho)$, and private saving is

$$S_t = \frac{Aw_t}{2+\rho} - T. \tag{6}$$

The government invests T for each young person, so total investment is $L_t(S_t +$

T). Using (6),

$$k_{t+1} = \frac{1-\alpha}{(1+n)(2+\rho)} k_t^{\alpha}, \tag{7}$$

exactly equation (2.61) with $g = 0$. Private saving falls one-for-one with government saving.

(ii) Since the transition equation is unchanged, the balanced-growth value of k is unchanged. This conclusion assumes $T \leq Aw_t/(2+\rho)$, or that households can borrow, so that the one-for-one private offset in (6) is feasible.

Note **The key return comparison.** Pay-as-you-go social security earns the population-growth return $1+n$; private capital earns $1+r$. The sign of $r-n$ is therefore both the dynamic-efficiency test and the long-run marginal welfare test.

Note **Why fully funded is neutral here.** The government buys the same asset households would buy and pays the same return. With no binding borrowing constraint, households simply reduce private saving by the amount saved for them.

Note **A small-policy qualification.** When the economy is dynamically inefficient, lowering capital is beneficial only until the golden-rule stock is reached. A large tax can overshoot that point even though a marginal increase is welfare-improving.

Problem 2.18 *The basic overlapping-generations model.*

(Samuelson, 1958; Allais, 1947) Suppose, as in the Diamond model, that L_t two-period-lived individuals are born in period t and that $L_t = (1+n)L_{t-1}$. For simplicity, let utility be logarithmic with no discounting: $U_t = \ln(C_{1t}) + \ln(C_{2t+1})$.

The production side of the economy is simpler than in the Diamond model. Each individual born at time t is endowed with A units of the economy's single good. The good can be either consumed or stored. Each unit stored yields $x > 0$ units of the good in the following period.[1]

Finally, assume that in the initial period, period 0, in addition to the L_0 young individuals each endowed with A units of the good, there are $[1/(1+n)]L_0$ individuals who are alive only in period 0. Each of these "old" individuals is endowed with some amount Z of the good; their utility is simply their consumption in the initial period, C_{20}.

(a) Describe the decentralized equilibrium of this economy.

Hint: Given the overlapping-generations structure, will the members of any gener-

[1] Note that this is the same as the Diamond economy with $g = 0$, $F(K_t, AL_t) = AL_t + xK_t$, and $\delta = 1$. With this production function, since individuals supply 1 unit of labor when they are young, an individual born in t obtains A units of the good. And each unit saved yields $1+r = 1+\partial F(K,AL)/\partial K - \delta = 1 + x - 1 = x$ units of second-period consumption.

ation engage in transactions with members of another generation?

(b) Consider paths where the fraction of agents' endowments that is stored, f_t, is constant over time. What is total consumption (that is, consumption of all the young plus consumption of all the old) per person on such a path as a function of f? If $x < 1+n$, what value of f satisfying $0 \le f \le 1$ maximizes consumption per person? Is the decentralized equilibrium Pareto efficient in this case? If not, how can a social planner raise welfare?

Solution

(a) A young person cannot make an enforceable trade with the next generation, because that generation has not yet been born. The old alive today cannot provide a useful counterparty either: they will not be alive next period to repay the young. Thus there is no intergenerational trade in the decentralized equilibrium.

Let F_t be the amount a member of generation t stores. The two consumption levels are

$$C_{1t} = A - F_t, \qquad C_{2,t+1} = xF_t. \tag{1}$$

Substitution into utility gives

$$U_t = \ln(A - F_t) + \ln(xF_t).$$

The first-order condition is

$$-\frac{1}{A - F_t} + \frac{1}{F_t} = 0,$$

so the unique optimum is

$$F_t = \frac{A}{2}, \qquad C_{1t} = \frac{A}{2}, \qquad C_{2,t+1} = \frac{xA}{2}. \tag{2}$$

The initial old simply consume their endowment Z. Every later generation stores half of its endowment when young.

(b) Suppose the constant storage fraction is f, so each young person consumes $(1-f)A$ and each old person consumes xfA. Total consumption in period t is

$$C_t = L_t(1-f)A + L_{t-1}xfA. \tag{3}$$

Since $L_t = (1+n)L_{t-1}$, consumption per person alive is

$$\frac{C_t}{L_t + L_{t-1}} = \frac{A\big[(1+n)(1-f) + xf\big]}{2+n}. \tag{4}$$

Its derivative with respect to f has the sign of $x-(1+n)$. Therefore, when $x < 1+n$, aggregate consumption per person is maximized at

$$\boxed{f = 0.}$$

The decentralized allocation nevertheless has $f = 1/2$, so it is not Pareto efficient. A planner can leave each young person's consumption at $A/2$, but transfer the other $A/2$ directly to the current old rather than store it. Because there are $1+n$ young people for every old person, the resulting allocation is

$$C_{1t}^{P} = \frac{A}{2}, \qquad C_{2,t+1}^{P} = \frac{(1+n)A}{2} > \frac{xA}{2}. \tag{5}$$

Repeating the transfer every period leaves consumption when young unchanged and raises consumption when old for every generation. The initial old also receive the first transfer, in addition to their endowment Z.

Note **Aggregate efficiency is not enough by itself.** The fact that $f = 0$ maximizes total consumption does not alone prove a Pareto improvement, because the old would consume nothing without a transfer. The planner's intergenerational transfer is what makes every generation weakly better off.

Note **Where the return $1+n$ comes from.** One unit collected from every young person finances $1+n$ units for each old person because the young generation is $1+n$ times as large. This transfer technology dominates storage when $1+n > x$.

Problem 2.19 *Stationary monetary equilibria in the Samuelson OLG model.*

(Samuelson, 1958) Consider the setup described in Problem 2.18. Assume that $x < 1+n$. Suppose that the old individuals in period 0, in addition to being endowed with Z units of the good, are each endowed with M units of a storable, divisible commodity, which we will call money. Money is not a source of utility.

(a) Consider an individual born at t. Suppose the price of the good in units of money is P_t in t and P_{t+1} in $t+1$. Thus the individual can sell units of endowment for P_t units of money and then use that money to buy P_t/P_{t+1} units of the next generation's endowment the following period. What is the individual's behavior as a function of P_t/P_{t+1}?

(b) Show that there is an equilibrium with $P_{t+1} = P_t/(1+n)$ for all $t \geq 0$ and no storage, and thus that the presence of "money" allows the economy to reach the golden-rule level of storage.

(c) Show that there are also equilibria with $P_{t+1} = P_t/x$ for all $t \geq 0$.

(d) Finally, explain why $P_t = \infty$ for all t (that is, money is worthless) is also an equilibrium. Explain why this is the only equilibrium if the economy ends at some date, as in Problem 2.20(b) below.

Hint: Reason backward from the last period.

Solution

Define the gross real return on money by

$$R_t \equiv \frac{P_t}{P_{t+1}}. \tag{1}$$

With logarithmic utility, a young person consumes $A/2$ and saves $A/2$, regardless of the return. The only remaining decision is whether to save through storage or money.

(a) Let F_t denote storage and M_t^d nominal money holdings. The optimal portfolio is

return comparison	F_t	M_t^d/P_t	$C_{2,t+1}$
$R_t < x$	$A/2$	0	$xA/2$
$R_t > x$	0	$A/2$	$R_tA/2$
$R_t = x$	$(1-\mu_t)A/2$	$\mu_tA/2$	$xA/2, \quad \mu_t \in [0,1]$.

(2)

Thus all saving goes to the asset with the higher gross return; when the returns are equal, any mixture is optimal.

(b) The initial old population is $L_0/(1+n)$, so the fixed nominal stock of money is

$$\mathcal{M} = \frac{L_0}{1+n}M.$$

Consider a path with $R_t = 1+n > x$. By (2), every young person sells $A/2$ units of the good for money and stores nothing. Money-market clearing requires

$$L_tP_t\frac{A}{2} = \mathcal{M}, \qquad P_t = \frac{2M}{A(1+n)^{t+1}}. \tag{3}$$

The second equality uses $L_t = L_0(1+n)^t$. It immediately implies

$$P_{t+1} = \frac{P_t}{1+n}.$$

Hence the proposed price path clears the money market in every period and is an equilibrium. Storage is zero, which is the golden-rule value from Problem 2.18 when $x < 1+n$. Money implements the superior intergenerational return $1+n$.

(c) Now impose $R_t = x$, or $P_{t+1} = P_t/x$. Households are indifferent between money and storage. If μ_t is the fraction of their saving held as money, market clearing requires

$$L_t P_t \mu_t \frac{A}{2} = \mathcal{M}, \qquad P_t = \frac{2M}{A\mu_t(1+n)^{t+1}}. \tag{4}$$

Taking the ratio of adjacent prices in (4), the proposed return is obtained exactly when

$$\frac{\mu_{t+1}}{\mu_t} = \frac{x}{1+n} < 1. \tag{5}$$

Choose any feasible $\mu_0 \in (0,1]$ and then use (5). This constructs a continuum of equilibria in which money and storage both earn x, and the fraction of saving held as money declines over time.

(d) If $P_t = \infty$, one unit of money buys $1/P_t = 0$ goods. Money is therefore worthless. If every young generation expects the next generation to reject money, no one offers goods for it; each person instead stores $A/2$, exactly as in Problem 2.18. Zero real money demand equals the zero real value of the outstanding nominal stock, so this is also an equilibrium.

If the economy ends at T, the last young generation lives for only one period and will not exchange goods for money that can never be spent. Knowing this, generation $T-1$ will not acquire money when young. Repeating the same reasoning backward shows that no generation accepts money. Thus the worthless-money equilibrium is the only one in a finite-horizon economy.

Note **Keep the price convention straight.** P_t is units of money per unit of the good. Thus the real value of money is $1/P_t$, and its gross return is P_t/P_{t+1}, not the inverse.

Note **Stationary refers to the allocation.** In the no-storage monetary equilibrium, the real allocation is stationary even though the nominal goods price P_t falls with population growth. More young agents must share the fixed nominal money stock.

Problem 2.20 *The source of dynamic inefficiency.*

(Shell, 1971) There are two ways in which the Diamond and Samuelson models differ from textbook models. First, markets are incomplete: because individuals cannot trade with individuals who have not been born, some possible transactions are ruled out. Second, because time goes on forever, there are an infinite number of agents. This problem asks you to investigate which of these is the source of the possibility of dynamic inefficiency. For simplicity, it focuses on the Samuelson overlapping-generations model (see the previous two problems), again with log utility and no discounting. To simplify further, it assumes $n = 0$ and $0 < x < 1$.

(a) **Incomplete markets.** Suppose we eliminate incomplete markets from the model by allowing all agents to trade in a competitive market "before" the beginning of time. That is, a Walrasian auctioneer calls out prices $Q_0, Q_1, Q_2, \ldots$ for the good at each date. Individuals can then make sales and purchases at these prices given their endowments and their ability to store. The budget constraint of an individual born at t is thus $Q_t C_{1t} + Q_{t+1} C_{2t+1} = Q_t (A - S_t) + Q_{t+1} x S_t$, where S_t (which must satisfy $0 \leq S_t \leq A$) is the amount the individual stores.

(i) Suppose the auctioneer announces $Q_{t+1} = Q_t / x$ for all $t > 0$. Show that in this case individuals are indifferent concerning how much to store, that there is a set of storage decisions such that markets clear at every date, and that this equilibrium is the same as the equilibrium described in part (a) of Problem 2.18.

(ii) Suppose the auctioneer announces prices that fail to satisfy $Q_{t+1} = Q_t / x$ at some date. Show that at the first date that does not satisfy this condition the market for the good cannot clear, and thus that the proposed price path cannot be an equilibrium.

(b) **Infinite duration.** Suppose that the economy ends at some date T. That is, suppose the individuals born at T live only one period (and hence seek to maximize C_{1T}), and that thereafter no individuals are born. Show that the decentralized equilibrium is Pareto efficient.

(c) In light of these answers, is it incomplete markets or infinite duration that is the source of dynamic inefficiency?

Solution

(a) For a person born at t, rewrite the budget constraint as

$$Q_t C_{1t} + Q_{t+1} C_{2,t+1} = Q_t A + (x Q_{t+1} - Q_t) S_t. \tag{1}$$

One unit sold at t and repurchased at $t+1$ earns the gross return Q_t / Q_{t+1}; one unit stored earns x.

(i) If $Q_{t+1} = Q_t / x$, then $Q_t / Q_{t+1} = x$, and the coefficient on S_t in (1) is zero. Storage does not change lifetime wealth, so the individual is indifferent about S_t. Log utility implies

$$C_{1t} = \frac{A}{2}, \qquad C_{2,t+1} = \frac{xA}{2}.$$

To exhibit market clearing, let $\alpha_t \in [0, 1]$ be the fraction of the young person's saving $A/2$ transferred through market trade rather than storage. Then

$$S_t = (1 - \alpha_t)\frac{A}{2}.$$

Generation t supplies $\alpha_t A/2$ goods when young and, because $Q_t/Q_{t+1} = x$, demands $x\alpha_t A/2$ goods when old. With $n = 0$, goods-market clearing in period $t+1$ therefore requires

$$\alpha_{t+1} = x\alpha_t. \tag{2}$$

The initial old simply consume their endowment Z, so they neither demand nor supply goods in the complete market and $\alpha_0 = 0$. Equation (2) then gives $\alpha_t = 0$ for every t. Everyone stores $A/2$, and the allocation is exactly the decentralized equilibrium in Problem 2.18.

(ii) Let t be the first date at which the price condition fails. All earlier market trades are zero by the argument above, so the old at date t plan no net trade.

If $Q_{t+1} < Q_t/x$, then $Q_t/Q_{t+1} > x$: market trade dominates storage. The young at t choose $S_t = 0$ and are net suppliers of the date-t good, but the old have no matching demand.

If $Q_{t+1} > Q_t/x$, then storage dominates market trade. The coefficient on S_t in (1) is positive, so the young choose the upper bound $S_t = A$. To consume while young they must be net demanders of the date-t good, but the old have no matching supply. In either case the date-t goods market cannot clear. Thus the only equilibrium price sequence satisfies $Q_{t+1} = Q_t/x$ at every date.

(b) If the economy ends at T, the generation born at T lives for one period and consumes its entire endowment. A planner cannot take goods from this last generation without making it worse off, because there is no generation at $T+1$ that can compensate it.

Now reason backward. Without a transfer from the young at T, the old at T can consume only the proceeds of the saving they chose at $T-1$. Any attempt to make an earlier old generation better off by replacing storage with transfers must eventually place an uncompensated burden on the last generation. Therefore the infinite chain of Pareto-improving transfers used in Problem 2.18 cannot be started. Given this terminal condition, each generation's private storage choice is optimal, and the finite-horizon decentralized equilibrium is Pareto efficient.

(c) Eliminating incomplete markets in part (a) reproduces the same dynamically inefficient allocation. Eliminating infinite duration in part (b) eliminates the possible Pareto improvement. Therefore the source of dynamic inefficiency is the economy's *infinite duration*, not incomplete markets.

NOTE **WHY THE LAST GENERATION MATTERS.** The planner's improvement is a chain: each young generation supports the current old and is repaid by the next young generation. A finite terminal date breaks the chain because the last young generation has no successor.

PROBLEM 2.21 *Explosive paths in the Samuelson overlapping-generations model.*

(Black, 1974; Brock, 1975; Calvo, 1978) Consider the setup described in Problem 2.19. Assume that x is zero, and assume that utility is constant-relative-risk-aversion with $\theta < 1$ rather than logarithmic. Finally, assume for simplicity that $n = 0$.

(a) What is the behavior of an individual born at t as a function of P_t/P_{t+1}? Show that the amount of his or her endowment that the individual sells for money is an increasing function of P_t/P_{t+1} and approaches zero as this ratio approaches zero.

(b) Suppose $P_0/P_1 < 1$. How much of the good are the individuals born in period 0 planning to buy in period 1 from the individuals born then? What must P_1/P_2 be for the individuals born in period 1 to want to supply this amount?

(c) Iterating this reasoning forward, what is the qualitative behavior of P_t/P_{t+1} over time? Does this represent an equilibrium path for the economy?

(d) Can there be an equilibrium path with $P_0/P_1 > 1$?

SOLUTION

Let $R_t \equiv P_t/P_{t+1} > 0$ be the gross real return on money. Because $x = 0$, money is the only way to transfer consumption to old age.

(a) The lifetime budget constraint, in units of the date-t good, is

$$C_{1t} + \frac{C_{2,t+1}}{R_t} = A. \tag{1}$$

With no discounting, the first-order conditions for CRRA utility imply

$$C_{2,t+1} = R_t^{1/\theta} C_{1t}. \tag{2}$$

Let $h(R_t)$ be the fraction of the endowment sold for money. Substituting (2) into (1), and using $Ah(R_t) = A - C_{1t}$, gives

$$C_{1t} = \frac{A}{1 + R_t^{(1-\theta)/\theta}}, \qquad h(R) = \frac{R^{(1-\theta)/\theta}}{1 + R^{(1-\theta)/\theta}}. \tag{3}$$

Write $a \equiv (1-\theta)/\theta > 0$. Then $h'(R) = aR^{a-1}/(1+R^a)^2 > 0$ and $\lim_{R \to 0} h(R) = 0$. Thus a higher return on money raises the amount sold for money, and that amount approaches zero with the return.

(b) Generation 0 sells $Ah(R_0)$ goods for money at date 0. Its nominal balances are therefore $P_0Ah(R_0)$, which buy

$$\frac{P_0Ah(R_0)}{P_1} = R_0Ah(R_0) \tag{4}$$

units of the good at date 1. Generation 1 supplies $Ah(R_1)$ units. Since $n = 0$, the two generations have the same size, so date-1 market clearing requires

$$h(R_1) = R_0h(R_0). \tag{5}$$

When $R_0 < 1$, the right-hand side is strictly between zero and $h(R_0)$. Since h is strictly increasing, there is a unique solution and it satisfies $0 < R_1 < R_0$. Explicitly,

$$R_1 = \left[\frac{R_0h(R_0)}{1 - R_0h(R_0)}\right]^{\theta/(1-\theta)}.$$

(c) Applying (5) in every period gives

$$1 > R_0 > R_1 > R_2 > \cdots.$$

The sequence has a limit $\overline{R} \geq 0$. Taking limits in (5) gives $h(\overline{R}) = \overline{R}\,h(\overline{R})$. A positive limit would require $\overline{R} = 1$, which is impossible because every $R_t < R_0 < 1$. Hence

$$\boxed{R_t = \frac{P_t}{P_{t+1}} \longrightarrow 0, \qquad h(R_t) \longrightarrow 0.}$$

This is an equilibrium path: choose P_0 to clear the initial money market and set $P_{t+1} = P_t/R_t$. Equation (5) then clears both markets every period. Thus P_{t+1}/P_t explodes, while the real value of money and the quantity of goods traded for it vanish.

(d) Suppose instead that $R_0 > 1$. As long as a next-period return can be found, (5) implies

$$h(R_{t+1}) = R_th(R_t) > h(R_t),$$

so $R_{t+1} > R_t > 1$. Such a sequence cannot converge to a finite limit: the limiting recursion would require $R = 1$ or $h(R) = 0$. Hence it becomes unbounded. But eventually $R_th(R_t) > 1$, whereas $h(R_{t+1}) \leq 1$; market clearing would require the next young generation to sell more than its endowment. Thus

$$\boxed{\text{there is no equilibrium path with } P_0/P_1 > 1.}$$

Chapter 3

Problem 3.1

Consider the model of Section 3.2 with $\theta < 1$.

(a) On the balanced growth path, $\dot{A} = g_A^* A(t)$, where g_A^* is the balanced-growth-path value of g_A. Use this fact and equation (3.6) to derive an expression for $A(t)$ on the balanced growth path in terms of B, a_L, γ, θ, and $L(t)$.

(b) Use your answer to part (a) and the production function, (3.5), to obtain an expression for $Y(t)$ on the balanced growth path. Find the value of a_L that maximizes output on the balanced growth path.

Problem 3.2

Consider two economies (indexed by $i = 1, 2$) described by $Y_i(t) = K_i(t)^\theta$ and $\dot{K}_i(t) = s_i Y_i(t)$, where $\theta > 1$. Suppose that the two economies have the same initial value of K, but that $s_1 > s_2$. Show that Y_1/Y_2 is continually rising.

Problem 3.3

Consider the economy analyzed in Section 3.3. Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:

(a) An increase in n.

(b) An increase in a_K.

(c) An increase in θ.

Problem 3.4

Consider the economy described in Section 3.3, and assume $\beta + \theta < 1$ and $n > 0$. Suppose the economy is initially on its balanced growth path, and that there is a permanent increase in s.

(a) How, if at all, does the change affect the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines? How, if at all, does it affect the location of the economy in (g_A, g_K) space at the time of the change?

(b) What are the dynamics of g_A and g_K after the increase in s? Sketch the path of log output per worker.

(c) Intuitively, how does the effect of the increase in s compare with its effect in the Solow model?

PROBLEM 3.5

Consider the model of Section 3.3 with $\beta + \theta = 1$ and $n = 0$.

(a) Using (3.14) and (3.16), find the value that A/K must have for g_K and g_A to be equal.

(b) Using your result in part (a), find the growth rate of A and K when $g_K = g_A$.

(c) How does an increase in s affect the long-run growth rate of the economy?

(d) What value of a_K maximizes the long-run growth rate of the economy? Intuitively, why is this value not increasing in β, the importance of capital in the R&D sector?

PROBLEM 3.6

Consider the model of Section 3.3 with $\beta + \theta > 1$ and $n > 0$.

(a) Draw the phase diagram for this case.

(b) Show that regardless of the economy's initial conditions, eventually the growth rates of A and K (and hence the growth rate of Y) are increasing continually.

(c) Repeat parts (a) and (b) for the case of $\beta + \theta = 1, n > 0$.

PROBLEM 3.7 *Learning-by-doing.*

Suppose that output is given by equation (3.22), $Y(t) = K(t)^\alpha [A(t)L(t)]^{1-\alpha}$; that L is constant and equal to 1; that $\dot{K}(t) = sY(t)$; and that knowledge accumulation occurs as a side effect of goods production: $\dot{A}(t) = BY(t)$.

(a) Find expressions for $g_A(t)$ and $g_K(t)$ in terms of $A(t)$, $K(t)$, and the parameters.

(b) Sketch the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines in (g_A, g_K) space.

(c) Does the economy converge to a balanced growth path? If so, what are the growth rates of K, A, and Y on the balanced growth path?

(d) How does an increase in s affect long-run growth?

PROBLEM 3.8

Consider the model of Section 3.5. Suppose, however, that households have constant-relative-risk-aversion utility with a coefficient of relative risk aversion of θ. Find the equilibrium level of labor in the R&D sector, L_A.

PROBLEM 3.9

Suppose that policymakers, realizing that monopoly power creates distortions, put controls on the prices that patent-holders in the Romer model can charge for the inputs embodying their ideas. Specifically, suppose they require patent-holders to charge $\delta w(t)/\phi$, where δ satisfies $\phi \leq \delta \leq 1$.

(a) What is the equilibrium growth rate of the economy as a function of δ and the other parameters of the model? Does a reduction in δ increase, decrease, or have no effect on the equilibrium growth rate, or is it not possible to tell?

(b) Explain intuitively why setting $\delta = \phi$, thereby requiring patent-holders to charge marginal cost and so eliminating the monopoly distortion, does not maximize social welfare.

PROBLEM 3.10

(a) Show that (3.48) follows from (3.47).

(b) Derive (3.49).

PROBLEM 3.11 *The balanced growth path of a semi-endogenous version of the Romer model.*

(Jones, 1995) Consider the model of Section 3.5 with two changes. First, existing knowledge contributes less than proportionally to the production of new knowledge, as in Case 1 of the model of Section 3.2: $\dot{A}(t) = BL_A(t)A(t)^{\theta}$, $\theta < 1$. Second, population is growing at rate n rather than constant: $L(t) = L(0)e^{nt}$, $n > 0$. (Consistent with this, assume that utility is given by equation [2.2] with $u(\cdot)$ logarithmic.) The analysis of Section 3.2 implies that such a model will exhibit transition dynamics rather than always immediately being on a balanced growth path. This problem therefore asks you not to analyze the full dynamics of the model, but to focus on its properties when it is on a balanced growth path. Specifically, it looks at situations where the fraction of the labor force engaged in R&D, a_L, is constant, and all variables of the model are growing at constant rates.

Hint: You are welcome to assume rather than derive that on a balanced growth path, the wage and consumption per person grow at the same rate as Y/L.

(a) What are balanced-growth-path values of the growth rates of Y and Y/L and of

r as functions of the balanced-growth-path value of the growth rate of A and parameters of the model?

(b) On the balanced growth path, what is the present value of the profits from the discovery of a new idea at time t as a function of $L(t)$, $A(t)$, $w(t)$, and exogenous parameters?

(c) What is $\dot{A}(t)/A(t)$ on the balanced growth path as a function of a_L and exogenous parameters? What is $L(t)A(t)^{\theta-1}$ on the balanced growth path?

Hint: Consider equations [3.6] and [3.7] in the case of $\gamma = 1$.

(d) What is a_L on the balanced growth path?

(e) Discuss how changes in each of ρ, B, ϕ, n, and θ affect the balanced-growth-path value of a_L. In the cases of ρ, B, and ϕ, are the effects in the same direction as the effects on L_A in the model of Section 3.5?

Problem 3.12 *Learning-by-doing with microeconomic foundations.*

Consider a variant of the model in equations (3.22)–(3.25). Suppose firm i's output is $Y_i(t) = K_i(t)^{\alpha}[A(t)L_i(t)]^{1-\alpha}$, and that $A(t) = BK(t)$. Here K_i and L_i are the amounts of capital and labor used by firm i, and K is the aggregate capital stock. Capital and labor earn their *private* marginal products. As in the model of Section 3.5, the economy is populated by infinitely lived households that own the economy's initial capital stock. The utility of the representative household takes the constant-relative-risk-aversion form in equations (2.2)–(2.3). Population growth is zero.

(a) (i) What are the private marginal products of capital and labor at firm i as functions of $K_i(t)$, $L_i(t)$, $K(t)$, and the parameters of the model?

(ii) Explain why the capital-labor ratio must be the same at all firms, so

$$\frac{K_i(t)}{L_i(t)} = \frac{K(t)}{L(t)}$$

(iii) What are $w(t)$ and $r(t)$ as functions of $K(t)$, L, and the parameters of the model?

(b) What must the growth rate of consumption be in equilibrium? Assume for simplicity that the parameter values are such that the growth rate is strictly positive and less than the interest rate. Sketch an explanation of why the equilibrium growth rate of output equals the equilibrium growth rate of consumption.

Hint: Consider equation [2.22].

(c) Describe how long-run growth is affected by:

(i) A rise in B.

(ii) A rise in ρ.

(iii) A rise in L.

(d) Is the equilibrium growth rate more than, less than, or equal to the socially optimal rate, or is it not possible to tell?

Problem 3.13

(Rebelo, 1991) Assume that there are two sectors, one producing consumption goods and one producing capital goods, and two factors of production: capital and land. Capital is used in both sectors, but land is used only in producing consumption goods. Specifically, the production functions are $C(t) = K_C(t)^{\alpha} T^{1-\alpha}$ and $\dot{K}(t) = BK_K(t)$, where K_C and K_K are the amounts of capital used in the two sectors (so $K_C(t)+K_K(t) = K(t)$) and T is the amount of land, and $0 < \alpha < 1$ and $B > 0$. Factors are paid their marginal products, and capital can move freely between the two sectors. T is normalized to 1 for simplicity.

(a) Let $P_K(t)$ denote the price of capital goods relative to consumption goods at time t. Use the fact that the earnings of capital in units of consumption goods in the two sectors must be equal to derive a condition relating $P_K(t)$, $K_C(t)$, and the parameters α and B. If K_C is growing at rate $g_K(t)$, at what rate must P_K be growing (or falling)? Let $g_P(t)$ denote this growth rate.

(b) The real interest rate in terms of consumption is $B + g_P(t)$.[2] Thus, assuming that households have our standard utility function, (2.22)–(2.23), the growth rate of consumption must be $(B + g_P - \rho)/\theta \equiv g_C$. Assume $\rho < B$.

(i) Use your results in part (a) to express $g_C(t)$ in terms of $g_K(t)$ rather than $g_P(t)$.

(ii) Given the production function for consumption goods, at what rate must K_C be growing for C to be growing at rate $g_C(t)$?

(iii) Combine your answers to (i) and (ii) to solve for $g_K(t)$ and $g_C(t)$ in terms of the underlying parameters.

(c) Suppose that investment income is taxed at rate τ, so that the real interest rate households face is $(1-\tau)(B + g_P)$. How, if at all, does τ affect the equilibrium growth rate of consumption?

[2] To see this, note that capital in the investment sector produces new capital at rate B and changes in value relative to the consumption good at rate g_P. (Because the return to capital is the same in the two sectors, the same must be true of capital in the consumption sector.)

Problem 3.14 *Delays in the transmission of knowledge to poor countries.*

(a) Assume that the world consists of two regions, the North and the South. The North is described by $Y_N(t) = A_N(t)(1 - a_L)L_N$ and $\dot{A}_N(t) = a_L L_N A_N(t)$. The South does not do R&D but simply uses the technology developed in the North; however, the technology used in the South lags the North's by τ years. Thus $Y_S(t) = A_S(t)L_S$ and $A_S(t) = A_N(t - \tau)$. If the growth rate of output per worker in the North is 3 percent per year, and if a_L is close to 0, what must τ be for output per worker in the North to exceed that in the South by a factor of 10?

(b) Suppose instead that both the North and the South are described by the Solow model: $y_i(t) = f(k_i(t))$, where

$$y_i(t) \equiv \frac{Y_i(t)}{L_i(t)} \quad \text{and} \quad k_i(t) \equiv \frac{K_i(t)}{L_i(t)}, \qquad i = N, S$$

As in the Solow model, assume $\dot{K}_i(t) = sY_i(t) - \delta K_i(t)$ and $\dot{L}_i(t) = nL_i(t)$; the two countries are assumed to have the same saving rates and rates of population growth. Finally, $\dot{A}_N(t) = gA_N(t)$ and $A_S(t) = A_N(t - \tau)$.

  (i) Show that the value of k on the balanced growth path, k^*, is the same for the two countries.

  (ii) Does introducing capital change the answer to part (a)? Explain. (Continue to assume $g = 3\%$.)

Problem 3.15

Which of the following possible regression results concerning the elasticity of long-run output with respect to the saving rate would provide the best evidence that differences in saving rates are not important to cross-country income differences? (1) A point estimate of 5 with a standard error of 2; (2) a point estimate of 0.1 with a standard error of 0.01; (3) a point estimate of 0.001 with a standard error of 5; (4) a point estimate of -2 with a standard error of 5. Explain your answer.

Hint: See the discussion of confidence intervals versus t-statistics in Section 3.6.

Chapter 4

Problem 4.1 *The golden-rule level of education.*

Consider the model of Section 4.1 with the assumption that $G(E)$ takes the form $G(E) = e^{\phi E}$.

(a) Find an expression that characterizes the value of E that maximizes the level of output per person on the balanced growth path. Are there cases where this value equals 0? Are there cases where it equals T?

(b) Assuming an interior solution, describe how, if at all, the golden-rule level of E (that is, the level of E you characterized in part (a)) is affected by each of the following changes:

(i) A rise in T.

(ii) A fall in n.

Problem 4.2 *Endogenizing the choice of E.*

(Bils and Klenow, 2000) Suppose that the wage of a worker with education E at time t is $be^{gt}e^{\phi E}$. Consider a worker born at time 0 who will be in school for the first E years of life and will work for the remaining $T - E$ years. Assume that the interest rate is constant and equal to $\bar{r}$.

(a) What is the present discounted value of the worker's lifetime earnings as a function of E, T, b, $\bar{r}$, ϕ, and g?

(b) Find the first-order condition for the value of E that maximizes the expression you found in part (a). Let E^* denote this value of E. (Assume an interior solution.)

(c) Describe how each of the following developments affects E^*:

(i) A rise in T.

(ii) A rise in $\bar{r}$.

(iii) A rise in g.

Problem 4.3

Suppose output in country i is given by $Y_i = A_i Q_i e^{\phi E_i} L_i$. Here E_i is each worker's years of education, Q_i is the quality of education, and the rest of the notation is standard. Higher output per worker raises the quality of education. Specifically, Q_i is given by $B_i (Y_i/L_i)^{\gamma}$, $0 < \gamma < 1$, $B_i > 0$.

Our goal is to decompose the difference in log output per worker between two countries, 1 and 2, into the contributions of education and all other forces. We have data on Y, L, and E in the two countries, and we know the values of the parameters ϕ and γ.

(a) Explain in what way attributing amount $\phi (E_2 - E_1)$ of $\ln (Y_2/L_2) - \ln (Y_1/L_1)$ to education and the remainder to other forces would understate the contribution of education to the difference in log output per worker between the two countries.

(b) What would be a better measure of the contribution of education to the difference in log output per worker?

Problem 4.4

Suppose the production function is $Y = K^{\alpha} \left(e^{\phi E} L\right)^{1-\alpha}$, $0 < \alpha < 1$. E is the amount of education workers receive; the rest of the notation is standard. Assume that there is perfect capital mobility. In particular, K always adjusts so that the marginal product of capital equals the world rate of return, r^*.

(a) Find an expression for the marginal product of capital as a function of K, E, L, and the parameters of the production function.

(b) Use the equation you derived in part (a) to find K as a function of r^*, E, L, and the parameters of the production function.

(c) Use your answer in part (b) to find an expression for $d (\ln Y) / dE$, incorporating the effect of E on Y via K.

(d) Explain intuitively how capital mobility affects the impact of the change in E on output.

Problem 4.5

(Mankiw, Romer, and Weil, 1992) Suppose output is given by

$$Y(t) = K(t)^{\alpha} H(t)^{\beta} [A(t) L(t)]^{1-\alpha-\beta}, \qquad \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < 1.$$

Here L is the number of workers and H is their total amount of skills. The remainder of the notation is standard. The dynamics of the inputs are $\dot{L}(t) = nL(t)$, $\dot{A}(t) = gA(t)$, $\dot{K}(t) = s_K Y(t) - \delta K(t)$, $\dot{H}(t) = s_H Y(t) - \delta H(t)$, where $0 < s_K < 1$, $0 < s_H < 1$, and

$n+g+\delta > 0$. $L(0)$, $A(0)$, $K(0)$, and $H(0)$ are given, and are all strictly positive. Finally, define $y(t) \equiv Y(t)/[A(t)L(t)]$, $k(t) \equiv K(t)/[A(t)L(t)]$, and $h(t) \equiv H(t)/[A(t)L(t)]$.

(a) Derive an expression for $y(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model.

(b) Derive an expression for $\dot{k}(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model. In (k, h) space, sketch the set of points where $\dot{k} = 0$.

(c) Derive an expression for $\dot{h}(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model. In (k, h) space, sketch the set of points where $\dot{h} = 0$.

(d) Does the economy converge to a balanced growth path? Why or why not? If so, what is the growth rate of output per worker on the balanced growth path? If not, in general terms what is the behavior of output per worker over time?

Problem 4.6

Consider the model in Problem 4.5.

(a) What are the balanced-growth-path values of k and h in terms of s_K, s_H, and the other parameters of the model?

(b) Suppose $\alpha = 1/3$ and $\beta = 1/2$. Consider two countries, A and B, and suppose that both s_K and s_H are twice as large in Country A as in Country B and that the countries are otherwise identical. What is the ratio of the balanced-growth-path value of income per worker in Country A to its value in Country B implied by the model?

(c) Consider the same assumptions as in part (b). What is the ratio of the balanced-growth-path value of skills per worker in Country A to its value in Country B implied by the model?

Problem 4.7

(Jones, 2002) Consider the model of Section 4.1 with the assumption that $G(E) = e^{\phi E}$. Suppose, however, that E, rather than being constant, is increasing steadily: $\dot{E}(t) = m$, where $m > 0$. Assume that, despite the steady increase in the amount of education people are getting, the growth rate of the number of workers is constant and equal to n, as in the basic model.

(a) With this change in the model, what is the long-run growth rate of output per worker?

(b) In the United States over the past century, if we measure E as years of schooling, $\phi \simeq 0.1$ and $m \simeq 1/15$. Overall growth of output per worker has been about 2 percent per year. In light of your answer to (a), approximately what fraction of this overall growth has been due to increasing education?

(c) Can $\dot{E}(t)$ continue to equal $m > 0$ forever? Explain.

Problem 4.8

Consider the following model with physical and human capital:

$$\begin{aligned}
Y(t) &= [(1-a_K)K(t)]^{\alpha}[(1-a_H)H(t)]^{1-\alpha} && 0<\alpha, a_K, a_H<1\\
\dot{K}(t) &= sY(t) - \delta_K K(t)\\
\dot{H}(t) &= B[a_K K(t)]^{\gamma}[a_H H(t)]^{\phi}[A(t)L(t)]^{1-\gamma-\phi} - \delta_H H(t) && \gamma, \phi > 0,\ \gamma+\phi<1\\
\dot{L}(t) &= nL(t)\\
\dot{A}(t) &= gA(t),
\end{aligned}$$

where a_K and a_H are the fractions of the stocks of physical and human capital used in the education sector.

This model assumes that human capital is produced in its own sector with its own production function. Bodies (L) are useful only as something to be educated, not as an input into the production of final goods. Similarly, knowledge (A) is useful only as something that can be conveyed to students, not as a direct input to goods production.

(a) Define $k \equiv K/(AL)$ and $h \equiv H/(AL)$. Derive equations for $\dot{k}$ and $\dot{h}$.

(b) Find an equation describing the set of combinations of h and k such that $\dot{k} = 0$. Sketch in (h, k) space. Do the same for $\dot{h} = 0$.

(c) Does this economy have a balanced growth path? If so, is it unique? Is it stable? What are the growth rates of output per person, physical capital per person, and human capital per person on the balanced growth path?

(d) Suppose the economy is initially on a balanced growth path, and that there is a permanent increase in s. How does this change affect the path of output per person over time?

Problem 4.9 *Increasing returns in a model with human capital.*

(Lucas, 1988) Suppose that $Y(t) = K(t)^{\alpha}[(1-a_H)H(t)]^{\beta}$, $\dot{H}(t) = Ba_H H(t)$, and $\dot{K}(t) = sY(t)$. Assume $0 < \alpha < 1$, $0 < \beta < 1$, and $\alpha + \beta > 1$.

(a) What is the growth rate of H?

(b) Does the economy converge to a balanced growth path? If so, what are the growth rates of K and Y on the balanced growth path?

Problem 4.10 *A different form of measurement error.*

Suppose the true relationship between social infrastructure (SI) and log income per person (y) is $y_i = a + bSI_i + e_i$. There are two components of social infrastructure, SI_i^A and SI_i^B (with $SI_i = SI_i^A + SI_i^B$), and we only have data on one of the components, SI_i^A. Both SI_i^A and SI_i^B are uncorrelated with e. We are considering running an OLS regression of y on a constant and SI^A.

(a) Derive an expression of the form, $y_i = a + bSI_i^A + \text{other terms}$.

(b) Use your answer to part (a) to determine whether an OLS regression of y on a constant and SI^A will produce an unbiased estimate of the impact of social infrastructure on income if:

(i) SI^A and SI^B are uncorrelated.

(ii) SI^A and SI^B are positively correlated.

Problem 4.11

Briefly explain whether each of the following statements concerning a cross-country regression of income per person on a measure of social infrastructure is true or false:

(a) "If the regression is estimated by ordinary least squares, it shows the effect of social infrastructure on output per person."

(b) "If the regression is estimated by instrumental variables using variables that are not affected by social infrastructure as instruments, it shows the effect of social infrastructure on output per person."

(c) "If the regression is estimated by ordinary least squares and has a high R^2, this means that there are no important influences on output per person that are omitted from the regression; thus in this case, the coefficient estimate from the regression is likely to be close to the true effect of social infrastructure on output per person."

Problem 4.12 *Convergence regressions.*

(a) **Convergence.** Let y_i denote log output per worker in country i. Suppose all countries have the same balanced-growth-path level of log income per worker, y^*. Suppose also that y_i evolves according to $\mathrm{d}y_i(t)/dt = -\lambda\left[y_i(t) - y^*\right]$.

(i) What is $y_i(t)$ as a function of $y_i(0)$, y^*, λ, and t?

(ii) Suppose that $y_i(t)$ in fact equals the expression you derived in part (i) plus a mean-zero random disturbance that is uncorrelated with $y_i(0)$. Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \varepsilon_i$.

What is the relation between β, the coefficient on $y_i(0)$ in the regression, and λ, the speed of convergence? Given this, how could you estimate λ from an estimate of β?

Hint: For a univariate OLS regression, the coefficient on the right-hand-side variable equals the covariance between the right-hand-side and left-hand-side variables divided by the variance of the right-hand-side variable.

(iii) If β in part (ii) is negative (so that rich countries on average grow less than poor countries), is $\operatorname{Var}(y_i(t))$ necessarily less than $\operatorname{Var}(y_i(0))$, so that the cross-country variance of income is falling? Explain. If β is positive, is $\operatorname{Var}(y_i(t))$ necessarily more than $\operatorname{Var}(y_i(0))$? Explain.

(b) **Conditional convergence.** Suppose $y_i^* = a + bX_i$, and that

$$\frac{\mathrm{d}y_i(t)}{\mathrm{d}t} = -\lambda\left[y_i(t) - y_i^*\right]$$

(i) What is $y_i(t)$ as a function of $y_i(0)$, X_i, λ, and t?

(ii) Suppose that $y_i(0) = y_i^* + u_i$ and that $y_i(t)$ equals the expression you derived in part (i) plus a mean-zero random disturbance, e_i, where X_i, u_i, and e_i are uncorrelated with one another. Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \varepsilon_i$. Suppose one attempts to infer λ from the estimate of β using the formula in part (a)(ii). Will this lead to a correct estimate of λ, an overestimate, or an underestimate?

(iii) Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \gamma X_i + \varepsilon_i$. Under the same assumptions as in part (ii), how could one estimate b, the effect of X on the balanced-growth-path value of y, from estimates of β and γ?

Chapter 5

Problem 5.1

Redo the calculations reported in Table 5.1, 5.2, or 5.3 for any country other than the United States.

Problem 5.2

Redo the calculations reported in Table 5.3 for the following:

(a) Employees' compensation as a share of national income.

(b) The labor force participation rate.

(c) The federal government budget deficit as a share of GDP.

(d) The Standard and Poor's 500 composite stock price index.

(e) The difference in yields between Moody's Baa and Aaa bonds.

(f) The difference in yields between 10-year and 3-month U.S. Treasury securities.

(g) The weighted average exchange rate of the U.S. dollar against major currencies.

Problem 5.3

Let A_0 denote the value of A in period 0, and let the behavior of $\ln A$ be given by equations (5.8)–(5.9).

(a) Express $\ln A_1$, $\ln A_2$, and $\ln A_3$ in terms of $\ln A_0$, ε_{A1}, ε_{A2}, ε_{A3}, $\bar{A}$, and g.

(b) In light of the fact that the expectations of the ε_A's are zero, what are the expectations of $\ln A_1$, $\ln A_2$, and $\ln A_3$ given $\ln A_0$, $\bar{A}$, and g?

Problem 5.4

Suppose the period-t utility function, u_t, is $u_t = \ln c_t + b(1-\ell_t)^{1-\gamma}/(1-\gamma)$, $b > 0$, $\gamma > 0$, rather than (5.7).

(a) Consider the one-period problem analogous to that investigated in (5.12)–(5.15). How, if at all, does labor supply depend on the wage?

(b) Consider the two-period problem analogous to that investigated in (5.16)–(5.21). How does the relative demand for leisure in the two periods depend on the relative wage? How does it depend on the interest rate? Explain intuitively why γ affects the responsiveness of labor supply to wages and the interest rate.

Problem 5.5

Consider the problem investigated in (5.16)–(5.21).

(a) Show that an increase in both w_1 and w_2 that leaves w_1/w_2 unchanged does not affect ℓ_1 or ℓ_2.

(b) Now assume that the household has initial wealth of amount $Z > 0$.

(i) Does (5.23) continue to hold? Why or why not?

(ii) Does the result in (a) continue to hold? Why or why not?

Problem 5.6

Suppose an individual lives for two periods and has utility $\ln C_1 + \ln C_2$.

(a) Suppose the individual has labor income of Y_1 in the first period of life and zero in the second period. Second-period consumption is thus $(1+r)(Y_1 - C_1)$; r, the rate of return, is potentially random.

(i) Find the first-order condition for the individual's choice of C_1.

(ii) Suppose r changes from being certain to being uncertain, without any change in $E[r]$. How, if at all, does C_1 respond to this change?

(b) Suppose the individual has labor income of zero in the first period and Y_2 in the second. Second-period consumption is thus $Y_2 - (1+r)C_1$. Y_2 is certain; again, r may be random.

(i) Find the first-order condition for the individual's choice of C_1.

(ii) Suppose r changes from being certain to being uncertain, without any change in $E[r]$. How, if at all, does C_1 respond to this change?

Problem 5.7

(a) Use an argument analogous to that used to derive equation (5.23) to show that household optimization requires

$$\frac{b}{1-\ell_t} = e^{-\rho}\,\mathrm{E}_t\left[\frac{b\,w_t\,(1+r_{t+1})}{w_{t+1}\,(1-\ell_{t+1})}\right]$$

(b) Show that this condition is implied by (5.23) and (5.26). (Note that [5.26] must hold in every period.)

Problem 5.8 *A simplified real-business-cycle model with additive technology shocks.*

(Blanchard and Fischer, 1989, pp. 329–331) Consider an economy consisting of a constant population of infinitely lived individuals. The representative individual maximizes the expected value of $\sum_{t=0}^{\infty} u(C_t)/(1+\rho)^t$, $\rho > 0$. The instantaneous utility function, $u(C_t)$, is $u(C_t) = C_t - \theta C_t^2$, $\theta > 0$. Assume that C is always in the range where $u'(C)$ is positive.

Output is linear in capital, plus an additive disturbance: $Y_t = AK_t + e_t$. There is no depreciation; thus $K_{t+1} = K_t + Y_t - C_t$, and the interest rate is A. Assume $A = \rho$. Finally, the disturbance follows a first-order autoregressive process: $e_t = \phi e_{t-1} + \varepsilon_t$, where $-1 < \phi < 1$ and where the ε_t's are mean-zero, i.i.d. shocks.

(a) Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1}.

(b) Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma e_t$. Given this guess, what is K_{t+1} as a function of K_t and e_t?

(c) What values must the parameters α, β, and γ have for the first-order condition in part (a) to be satisfied for all values of K_t and e_t?

(d) What are the effects of a one-time shock to ε on the paths of Y, K, and C?

Problem 5.9 *A simplified real-business-cycle model with taste shocks.*

(Blanchard and Fischer, 1989, p. 361) Consider the setup in Problem 5.8. Assume, however, that the technological disturbances (the e's) are absent and that the instantaneous utility function is $u(C_t) = C_t - \theta(C_t + \nu_t)^2$. The ν's are mean-zero, i.i.d. shocks.

(a) Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1}.

(b) Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma \nu_t$. Given this guess, what is K_{t+1} as a function of K_t and ν_t?

(c) What values must the parameters α, β, and γ have for the first-order condition in (a) to be satisfied for all values of K_t and ν_t?

(d) What are the effects of a one-time shock to ν on the paths of Y, K, and C?

Problem 5.10 *The balanced growth path of the model of Section 5.3.*

Consider the model of Section 5.3 without any shocks. Let y^*, k^*, c^*, and G^* denote the values of $Y/(AL)$, $K/(AL)$, $C/(AL)$, and $G/(AL)$ on the balanced growth path; w^* the value of w/A; ℓ^* the value of L/N; and r^* the value of r.

(a) Use equations (5.1)–(5.4), (5.23), and (5.26) and the fact that y^*, k^*, c^*, w^*, ℓ^*, and r^* are constant on the balanced growth path to find six equations in these six variables.

Hint: The fact that c in [5.23] is consumption per person, C/N, and c^* is the balanced-growth-path value of consumption per unit of effective labor, $C/(AL)$, implies that $c = c^*\ell^*A$ on the balanced growth path.

(b) Consider the parameter values assumed in Section 5.7. What are the implied shares of consumption and investment in output on the balanced growth path? What is the implied ratio of capital to annual output on the balanced growth path?

Problem 5.11 *Solving a real-business-cycle model by finding the social optimum.*

Consider the model of Section 5.5. Assume for simplicity that $n = g = \bar{A} = \bar{N} = 0$. Let $V(K_t, A_t)$, the *value function*, be the expected present value from the current period forward of lifetime utility of the representative individual as a function of the capital stock and technology.

(a) Explain intuitively why $V(\cdot)$ must satisfy

$$V(K_t, A_t) = \max_{C_t,\ell_t} \{[\ln C_t + b\ln(1-\ell_t)] + e^{-\rho}E_t[V(K_{t+1}, A_{t+1})]\}.$$

This condition is known as the *Bellman equation*.

Given the log-linear structure of the model, let us guess that $V(\cdot)$ takes the form $V(K_t, A_t) = \beta_0 + \beta_K \ln K_t + \beta_A \ln A_t$, where the values of the β's are to be determined. Substituting this conjectured form and the facts that $K_{t+1} = Y_t - C_t$ and $E_t[\ln A_{t+1}] = \rho_A \ln A_t$ into the Bellman equation yields

$$V(K_t, A_t) = \max_{C_t,\ell_t} \left\{[\ln C_t + b\ln(1-\ell_t)] + e^{-\rho}\left[\beta_0 + \beta_K \ln(Y_t - C_t) + \beta_A\rho_A \ln A_t\right]\right\}.$$

(b) Find the first-order condition for C_t. Show that it implies that C_t/Y_t does not depend on K_t or A_t.

(c) Find the first-order condition for ℓ_t. Use this condition and the result in part (b) to show that ℓ_t does not depend on K_t or A_t.

(d) Substitute the production function and the results in parts (b) and (c) for the optimal C_t and ℓ_t into the equation above for $V(\cdot)$, and show that the resulting expression has the form $V(K_t, A_t) = \beta_0' + \beta_K' \ln K_t + \beta_A' \ln A_t$.

(e) What must β_K and β_A be so that $\beta_K' = \beta_K$ and $\beta_A' = \beta_A$?

(f) What are the implied values of C/Y and ℓ? Are they the same as those found in Section 5.5 for the case of $n = g = 0$?

Problem 5.12

Suppose technology follows some process other than (5.8)–(5.9). Do $s_t = \hat{s}$ and $\ell_t = \hat{\ell}$ for all t continue to solve the model of Section 5.5? Why or why not?

Problem 5.13

Consider the model of Section 5.5. Suppose, however, that the instantaneous utility function, u_t, is given by $u_t = \ln c_t + b(1-\ell_t)^{1-\gamma}/(1-\gamma)$, $b > 0$, $\gamma > 0$, rather than by (5.7) (see Problem 5.4).

(a) Find the first-order condition analogous to equation (5.26) that relates current leisure and consumption, given the wage.

(b) With this change in the model, is the saving rate (s) still constant?

(c) Is leisure per person $(1-\ell)$ still constant?

Problem 5.14

(a) If the $\tilde{A}_t$'s are uniformly 0 and if $\ln Y_t$ evolves according to (5.39), what path does $\ln Y_t$ settle down to?

Hint: Note that we can rewrite [5.39] as

$$\ln Y_t - (n+g)\,t = Q + \alpha\left[\ln Y_{t-1} - (n+g)(t-1)\right] + (1-\alpha)\tilde{A}_t$$

where $Q \equiv \alpha \ln \hat{s} + (1-\alpha)\left(\bar{A} + \ln \hat{\ell} + \bar{N}\right) - \alpha(n+g)$.

(b) Let $\tilde{Y}_t$ denote the difference between $\ln Y_t$ and the path found in (a). With this definition, derive (5.40).

Problem 5.15 *The derivation of the log-linearized equation of motion for capital.*

Consider the equation of motion for capital, $K_{t+1} = K_t + K_t^{\alpha}(A_t L_t)^{1-\alpha} - C_t - G_t - \delta K_t$.

(a) (i) Show that

$$\frac{\partial \ln K_{t+1}}{\partial \ln K_t} = \frac{1 + r_t}{K_{t+1}/K_t}$$

(ii) Show that this implies that $\partial \ln K_{t+1}/\partial \ln K_t$ evaluated at the balanced growth path is $(1 + r^*)/e^{n+g}$.

(b) Show that

$$\tilde{K}_{t+1} \simeq \lambda_1 \tilde{K}_t + \lambda_2 \left(\tilde{A}_t + \tilde{L}_t\right) + \lambda_3 \tilde{G}_t + (1 - \lambda_1 - \lambda_2 - \lambda_3)\,\tilde{C}_t,$$

where

$$\lambda_1 \equiv \frac{(1 + r^*)}{e^{n+g}}, \quad \lambda_2 \equiv \frac{(1 - \alpha)(r^* + \delta)}{\alpha\, e^{n+g}}, \quad \lambda_3 \equiv -\frac{(r^* + \delta)(G/Y)^*}{\alpha\, e^{n+g}}$$

and where $(G/Y)^*$ denotes the ratio of G to Y on the balanced growth path without shocks. (Hints: Since the production function is Cobb–Douglas, $Y^* = (r^* + \delta) K^*/\alpha$. On the balanced growth path, $K_{t+1} = e^{n+g} K_t$, which implies that $C^* = Y^* - G^* - \delta K^* - (e^{n+g} - 1) K^*$.)

(c) Use the result in (b) and equations (5.43)–(5.44) to derive (5.53), where $b_{KK} = \lambda_1 + \lambda_2 a_{LK} + (1 - \lambda_1 - \lambda_2 - \lambda_3)\, a_{CK}$, $b_{KA} = \lambda_2 (1 + a_{LA}) + (1 - \lambda_1 - \lambda_2 - \lambda_3)\, a_{CA}$, and $b_{KG} = \lambda_2 a_{LG} + \lambda_3 + (1 - \lambda_1 - \lambda_2 - \lambda_3)\, a_{CG}$.

Problem 5.16

Redo the regression reported in equation (5.55):

(a) Incorporating more recent data.

(b) Incorporating more recent data, and using $M1$ rather than $M2$.

(c) Including eight lags of the change in log money rather than four.

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